

B O O K I.

(*Ch. I.*) Every systematic science, the humblest and the noblest alike, seems to admit of two distinct kinds of proficiency; one of which may be properly called scientific knowledge of the subject, while the other is a kind of educational acquaintance with it. For an educated man should be able to form a fair offhand judgment as to the goodness or badness of the method used by a professor in his exposition. To be educated is in fact to be able to do this; and even the man of universal education we deem to be such in virtue of his having this ability. It will, however, of course, be understood that we only ascribe universal education to one who in his own individual person is thus critical in all or nearly all branches of knowledge, and not to one who has a like ability merely in some special subject. For it is possible for a man to have this competence in some one branch of knowledge without having it in all.¹

It is plain then that, as in other sciences, so in that which enquires into nature, there must be certain canons, by reference to which a hearer shall be able to criticise the method of a professed exposition, quite independently of the question whether the statements made be true or false. Ought we, for instance, to give an illustration of what I mean, to begin by discussing the specific characters of each kind of animal—man, lion, ox, and the like—and by describing separately the distinctive features of each, or ought we rather to deal first with the characters which all these animals have in common, and thus form a basis for the consideration of them separately? For genera² that are quite distinct yet oftentimes resemble each other in many of their phenomena; in sleep, for instance, in respiration, in growth, in decay, in death,

and in other similar affections and conditions, which may be passed over for the present, as we are not yet prepared to treat of them with clearness and precision. Now it is plain that if we discuss each kind of animal separately, we shall frequently be obliged to repeat the same statements over and over again ; for horse and dog and man present, each and all, every one of the phenomena just enumerated. A discussion therefore of these several animals separately would necessarily involve frequent repetitions as to characters, themselves identical, but recurring in animals specifically distinct. (Very possibly also there may be other characters which, though they present specific differences, yet come under one and the same category. For instance, flying, walking, swimming, creeping, are plainly specifically distinct, but yet are all forms of animal progression.) We must, then, have some clear understanding as to the manner in which our investigation is to be conducted ; whether, I mean, we are first to deal with the common or generical characters, and afterwards to take into consideration special peculiarities ; or whether we are to start straight off with the ultimate species. For as yet no definite rule has been laid down in this matter. So also there is a like uncertainty as to another point now to be mentioned. Ought the writer who deals with the works of nature to follow the plan adopted by the mathematicians in their astronomical demonstrations, and after considering the phenomena presented by animals, and their several parts, proceed subsequently to treat of the causes and the reason why ; or ought he to follow some other method ? And when these questions are answered, there yet remains another. The causes concerned in the generation of the works of nature are, as we see, more than one. There is the final cause and there is the motor cause. Now we must decide which of these two causes³ comes first, which second. Plainly, however, that cause is the first which we call the final one. For this is the Reason, and the Reason forms the starting-point, alike in works of art and in works of nature. For consider how the physician or how the builder sets about his work. He starts by forming for himself a definite picture, in the one case mental, in the other actual, of his end—the physician of health, the builder of a house—and this he holds forward as the reason and explanation of each subsequent step that he takes, and of his acting in this or that way as the case may be. Now in the works of nature the good end

and the final cause is still more dominant than in works of art, such as these, and necessity is a much less constant factor in their production; though it is to this that almost all writers would seek to refer their origin, though they do not distinguish the various senses in which the term necessity is used. For there is absolute necessity, manifested in eternal phenomena; and there is hypothetical necessity, manifested in everything that is generated by nature as in everything that is produced by art, be it a house or what it may. For if a house or other such final object is to be realised, it is necessary that such and such material shall exist; and it is necessary that first this and then that shall be produced, and first this and then that set in motion, and so on in continuous succession, until the end and final result is reached, for the sake of which each prior thing is produced and exists. As with these productions of art, so also is it with the productions of nature. The mode of necessity, however, and the mode of ratiocination are different in natural science [and in art] from what they are in the theoretical sciences;⁴ of which we have spoken elsewhere. For in the latter the starting-point is that which is; in the former that which is to be. For it is that which is yet to be—health, let us say, or a man—which, owing to its being of such and such characters, necessitates the pre-existence or previous production of this and that antecedent; and not this or that antecedent which, because it exists or is generated, makes it necessary that health or a man shall come into existence. Nor is it possible to trace back the series of necessary antecedents to a starting-point, of which you can say that, existing itself from eternity, it has determined their existence as its consequent. These, however, again are matters that have been dealt with in another treatise. There too it was stated in what cases absolute and hypothetical necessity exist; in what cases also the proposition expressing hypothetical necessity is simply convertible, and what cause it is that determines this convertibility.⁵

Another matter which must not be passed over without consideration is, whether the proper subject of our exposition is that with which the ancient writers concerned themselves, namely, what is the process of formation of each animal; or whether it is not rather, what are the characters of a given creature when formed. For there is no small difference between these two views.⁶ The best course appears to be that we should follow the method

already mentioned, and begin with the phenomena presented by each group of animals, and, when this is done, proceed afterwards to state the causes of those phenomena, and to deal with their evolution. For elsewhere, as for instance in house building, this is the true sequence. The plan of the house, or the house, has this and that features; and because it has this and that features, therefore is its construction carried out in this or that manner. For the process of evolution is for the sake of the thing finally evolved, and not this for the sake of the process. Empedocles, then, was in error when he said that many of the characters presented by animals were merely the results of incidental occurrences during their development; for instance, that the backbone was divided as it is into vertebræ, because it happened to be broken owing to the contorted position of the foetus in the womb.⁷ In so saying he overlooked the fact that propagation implies a creative seed endowed with certain formative properties. Secondly, he neglected another fact, namely, that the parent animal pre-exists, not only in idea, but actually in time. For man is generated from man; and thus it is the possession of certain characters by the parent that determines the development of like characters in the child. The same statement⁸ [namely, that the process of production occurs for the sake of the thing produced] holds good also for the operations of art, and even for those which are apparently spontaneous. For the same result as is commonly produced only by art may sometimes occur spontaneously. Spontaneity, for instance, may do the work of the physician and bring about the restoration of health. As a general rule, however, the products of art are such as cannot possibly be produced spontaneously; for they require the pre-existence of an efficient cause homogeneous with themselves, such as the statuary's art which must necessarily precede the statue; art indeed consisting in the conception of the result to be produced before its realisation in the material. As with spontaneity,⁹ so with chance; for this also occasionally produces the same result as art, and by the same process of evolution.

The fittest mode, then, of treatment is to say, a man has such and such parts, because the conception of a man includes their presence, and because they are necessary conditions of his existence, or at any rate of his perfection; and these conditions in their turn imply other prior conditions. Thus we should say,

because man is an animal with such and such characters, therefore is the process of his development necessarily such as it is; and therefore is it accomplished in such and such an order, this part being formed first, that next, and so on in succession; and after a like fashion should we explain the evolution of all other works of nature.

Now that with which the ancient writers, who first philosophised about Nature, busied themselves, was the material principle and the material cause. They enquired what this is, and what its character; how the universe is generated out of it, and by what motor influence, whether, for instance, by antagonism or friendship, whether by intelligence or spontaneous action,¹⁰ the substratum of matter being assumed to have certain inseparable properties; fire, for instance, to have a hot nature, earth a cold one; the former to be light, the latter heavy. For even the genesis of the universe is thus explained by them. After a like fashion do they deal also with the development of plants and of animals. They say, for instance, that the water contained in the body causes by its currents¹¹ the formation of the stomach and the other receptacles of food or of excretion; and that the breath by its passage breaks open the outlets of the nostrils; air and water being the materials of which bodies are made; for all represent nature as composed of such or similar substances.

But if men and animals and their several parts are natural phenomena, then the natural philosopher must take into consideration [not merely the ultimate substances of which they are made, but also] flesh, blood, bone, and all the other homogeneous parts;¹² nor only these, but also the heterogeneous parts, such as face, hand, foot, and the like; and must examine how each of these comes to be what it is, and in virtue of what force. For to say what are the ultimate substances out of which an animal is formed, to state, for instance, that it is made of fire and water, is no more sufficient, than would be a similar account in the case of an inanimate object, such as a couch or the like. For we should not be content with saying that the couch was made of bronze or wood or whatever it might be, but should try to describe its design or mode of composition in preference to the material; or, if we did deal with the material, it would at any rate be with the concretion of material and form. For a couch is such and such a form of this or that matter, or such and such a matter

with this or that form ; so that its shape and structure must be included in our description. For the formal nature is of much greater importance than the material nature.

Does, then, configuration and colour constitute the essence of the various animals and of their several parts? For if so, what Democritus says will be strictly correct. For such appears to have been his notion. At any rate he says that it is evident to every one what form it is that makes the man ; as if man were constituted by a certain recognisable shape and colour. And yet a dead body has exactly the same configuration as a living one ; but for all that is not a man. So also no hand of bronze or wood or of any but the appropriate materials can possibly be a hand in more than name. For like a physician in a painting, or like a flute in a sculpture, in spite of its name, it will be unable to do the office which that name implies. Precisely in the same way no part of a dead body, such I mean as its eye or its foot, is really an eye or a foot. To say, then, that shape and colour constitute the animal is an inadequate statement, and is much the same as if a woodcarver were to insist that the hand he had cut out was really a hand. Yet the physiologists, when they give an account of the development and causes of the animal form, speak very much like such a craftsman. What, however, I would ask, are the forces by which the hand or the body was fashioned into its shape? The woodcarver will perhaps say, by the axe or the auger ; the physiologist, by air and by earth. Of these two answers the artificer's is the better, but it is nevertheless insufficient.¹³ For it is not enough for him to say that by the stroke of his tool this part was formed into a concavity, that into a flat surface ; but he must state the reasons why he struck his blow in such a way as to effect this, and what his final object was ; namely, that the piece of wood should develop eventually into this or that shape. It is plain, then, that the teaching of the old physiologists is inadequate, and that the true method is to state what the definitive characters are that distinguish the animal as a whole ; to explain what it is both in substance and in form, and to deal after the same fashion with its several organs ; in fact, to proceed in exactly the same way as we should do, were we giving a complete description of a couch.

If now this something that constitutes the form of the living being be the soul,¹⁴ or part of the soul, or something that without

the soul cannot exist; as would seem to be the case, seeing at any rate that when the soul departs, what is left is no longer a living animal, and that none of the parts remain what they were before, excepting in mere configuration, like the animals that in the fable are turned into stone; if, I say, this be so, then it will come within the province of the natural philosopher to inform himself concerning the soul, and to treat of it, either in its entirety, or, at any rate, of that part of it which constitutes the essential character of an animal; and it will be his duty to say what this soul or this part of a soul is; and to discuss the attributes that attach to this essential character. Again, nature is spoken of in two senses, and the nature of a thing is either its matter or its essence; nature as essence including both the motor cause and the final cause.¹⁵ Now it is in the latter of these two senses that either the whole soul or some part of it constitutes the nature of an animal; and inasmuch as it is the presence of the soul that enables matter to constitute the animal nature, much more than it is the presence of matter which so enables the soul, the enquirer into nature is bound on every ground to treat of the soul rather than of the matter. For though the wood of which they are made constitutes the couch and the tripod, it only does so because it is capable of receiving such and such a form.

So far as has yet been said it remains a debatable question, whether it is the whole soul or only some part of it, the consideration of which comes within the province of natural science. Now if it be of the whole soul that this should treat, then there is no place for any other philosophy beside it. For as it belongs in all cases to one and the same science to deal with correlated subjects—one and the same science, for instance, deals with sensation and with the objects of sense—and as therefore the intelligent soul and the objects of intellect, being correlated, must belong to one and the same science, it follows that natural science will have to include the whole universe in its province. But perhaps it is not the whole soul, nor all its parts collectively, that constitutes the source of motion; but there may be one part, identical with that in plants, which is the source of [change of quantity or] growth, another which is the source of feeling or change of quality, while still another, and this not the intellectual part, is the source of [change of place or] locomotion.¹⁶ I say not the intellectual part; for other animals than man have the power of locomotion, but

in none but him is there intellect. Thus then it is plain that it is not of the whole soul that we have to treat. For it is not the whole soul that constitutes the animal nature, but only some part or parts of it.¹⁷ Moreover, it is impossible that any abstraction¹⁸ can form a subject of natural science, seeing that everything that Nature makes is means to an end. For just as human creations are the products of art, so living objects are manifestly the products of an analogous cause or principle, not external but internal,¹⁹ derived like the hot and the cold [and the other material elements of our bodies] from the enviring universe.²⁰ And that the heaven, if it had an origin,²¹ was evolved by such a cause, there is even more reason to believe, than that mortal animals so originated. For order and arrangement and constancy are much more plainly manifest in the celestial bodies than in our own frame ; while change and chance are characteristic of the perishable things of earth. Yet there are some who, while they allow that animals were all generated by nature, nevertheless hold that the heaven was constructed to be what it is by chance and spontaneity ; the heaven, in which not the faintest sign of hap-hazard or of disorder is discernible !²² Again, whenever there is plainly some final end, to which a motion tends, should nothing stand in the way, we always say that such final end is the aim or purpose of the motion ; and from this it is evident that there must be a something or other really existing, corresponding to what we call by the name of Nature. For a given germ does not give rise to any chance living being, nor spring from any chance one ; but each germ springs from a definite parent and gives rise to a definite progeny. And thus it is the germ that is the ruling influence and fabricator of the offspring ; for the offspring is that which in the course of nature will spring from it. At the same time [in the order of thought] the offspring is anterior to the germ ; for germ and perfected progeny are related as the developmental process and the result, [and result is in thought anterior to evolution]. Anterior, however, to both germ and product is the organism from which the germ was derived. For every germ connotes two organisms, the parent and the progeny. For germ or seed is both the seed of the organism from which it came, of the horse, for instance, from which it was derived, and the seed of the organism that will eventually arise from it, of the mule, for example, which is developed from the seed of the horse. The

same seed then is the seed both of the horse and of the mule, though in different ways. Moreover, the seed is potentially that which will spring from it, and the relation of potentiality to actuality we know [to be that of consequent to antecedent].²³

There are then two causes, namely, necessity and the final end. For many things are produced, simply as the results of necessity. It may, however, be asked, of what mode of necessity²⁴ are we speaking when we say this. For it can be of neither of those two modes which are set forth in the philosophical treatises.²⁵ There is, however, the third mode, in such things at any rate as are generated. For instance, we say that food is necessary; because an animal cannot possibly do without it. This third mode is what may be called hypothetical necessity. Here is another example of it. If a piece of wood is to be split with an axe, the axe must of necessity be hard; and, if hard, must of necessity be made of iron, or bronze, or the like. Now exactly in the same way the body, which like the axe is an instrument—for both the body as a whole and its several parts individually have definite operations for which they are made—just in the same way, I say, the body, if it is to do its work, must of necessity be of such and such a character, and made of such and such materials.

There are then, as before said, two modes of causation, and both of these must, so far as possible, be taken into account in explaining the works of nature.²⁶ At any rate it is plain that an attempt must be made to include them both; and that those who fail in this tell us in reality nothing about nature. For the main factor in the nature of an animal is much more the final cause than the necessary material. There are indeed passages in which even Empedocles hits upon this, and following the guidance of fact, finds himself constrained to speak of the final cause as constituting the essence and real nature of things. Such, for instance, is the case when he explains what is a bone. For he does not merely describe its material, and say it is this one element, or those two or three elements, or a compound of all the elements, but states the law or plan of their combination, and takes this to constitute the bone. As with a bone, so manifestly is it with the flesh and all other similar parts.

The reason why our predecessors failed in hitting upon this method of treatment was, that they were not in possession of the notion of formal cause, nor of any definition of essence.²⁷ The

first who came near it was Democritus, and he was far from adopting it as a necessary method in natural science, but was merely brought to it, spite of himself, by constraint of facts. In the time of Socrates a nearer approach was made to the method. But at this period men gave up enquiring into the works of nature, and philosophers diverted their attention to political science and to the virtues which benefit mankind.

Of the method itself the following is an example. In dealing with respiration we must show that it takes place for such or such a final object; and we must also show that this and that part of the process is necessitated by this and that other stage of it. By necessity we shall sometimes mean hypothetical necessity, the necessity, that is, that the requisite antecedents shall be there, if the final end is to be reached; and sometimes absolute necessity, such necessity as that between substances and their inherent properties and characters. Thus it is necessary [if we are to live] that there shall be alternating discharges and returns of heat from and to the body, and a necessary condition for this is the inflow of air. Here at once we have a necessity [in the former of the two senses]. But the reduction of the internal heat by refrigeration produces [as a necessary result] the outflow of the air, while *vice versâ* the reduction of the cold by the internal heat produces an inflow. [This is a necessity in the second of the two senses.]²⁸

In the foregoing we have an example of the method which we must adopt, and also an example of the kind of phenomena, the causes of which we have to investigate.

(*Ch. 2.*) Some¹ writers propose to reach the definitions of the ultimate forms of animal life by bipartite division. But this method is often difficult, and often impracticable.

Sometimes the final differentia of the subdivision is sufficient by itself and the antecedent differentiæ are mere surplusage. Thus in the series Footed, Two-footed, Cloven-footed,² the last term is all-expressive by itself, and to append the higher terms is only an idle iteration.

Again it is not permissible to break up a natural group, Birds for instance, by putting its members under different bifurcations, as is done in the published dichotomies, where some birds are ranked with Animals of the water, and others with Animals of the air. The group Birds and the group Fishes happen to be named,

while other natural groups have no popular names; for instance the groups that we may call Sanguineous and Exsanguineous are not known popularly by any designations. If such natural groups are not to be broken up, the method of Dichotomy cannot be employed, for it necessarily involves such breaking up and dislocation. The group of the Many-footed, for instance, would, under this method, have to be dismembered, and some of its kinds distributed among land animals, others among water animals.³

(*Ch.* 3.) Again privative terms inevitably form one branch of dichotomous division, as we see in the proposed dichotomies. But privative terms in their character of privatives admit of no subdivision. For there can be no specific forms of what is non-existent, of Featherless for instance or of Footless, as there are of Feathered and of Footed. Yet a generic differentia must be subdivisible; for otherwise what is there that makes it generic rather than specific? There are to be found generic, that is specifically subdivisible, differentia; Feathered for instance and Footed. For feathers are divisible into Barbed and Unbarbed, and feet into Manycleft, like those of a bird, or Two-cleft, like those of an ox, or Uncloven and Undivided, like those of a horse. Now even with differentia capable of this specific subdivision it is difficult enough so to make the classification, as that each animal shall be comprehended in some one subdivision and in not more than one; but far more difficult, nay impossible, is it to do this, if we start with a dichotomy into two contradictories. (Suppose for instance we start with the two contradictories, Feathered and Unfeathered;¹ we shall find that the ant, the glow-worm and some other animals fall under both divisions.) For each differentia must be presented by some species. There must be some species, therefore, under the privative heading. Now specifically distinct animals cannot present in their essence a common undifferentiated element, but any apparently common element must really be differentiated. (Bird and Man for instance are both Two-footed, but their two-footedness is diverse and differentiated. So any two sanguineous groups must have some difference in their blood, if their blood is part of their essence.) From this it follows that a privative term, being insusceptible of differentiation, cannot be a generic differentia; for, if it were, there would be a common undifferentiated element in two different groups.²

Again, if the species are ultimate indivisible groups, that is, are groups with indivisible differentia, and if no differentia be common to several groups, the number of differentia must be equal to the number of species. If a differentia though not divisible could yet be common to several groups, then it is plain that in virtue of that common differentia specifically distinct animals would fall into the same division. It is necessary then that none of the differentia that belong to the ultimate or indivisible groups shall be common; for otherwise, as already said, specifically distinct animals will come into one and the same division. But this would violate one of the requisite conditions, which are as follows. No ultimate group must be included in more than a single division; different groups must not be included in the same division; and every group must be found in some division. It is plain then that we cannot get at the ultimate specific forms of the animal, or any other, kingdom by bifurcate division. If we could, the number of ultimate differentia would equal the number of ultimate animal forms, [which is inconceivable]. For assume an order of beings whose prime differentia are White and Black. Each of these branches will bifurcate, and their branches again, and so on till we reach the ultimate differentia, whose number will be four or some other multiple of two, and will also be the number of the ultimate species comprehended in the order.

(A species is constituted by the combination of differentia and matter. For no animal, nor portion of an animal, is purely material or purely immaterial, in other words purely corporeal or purely incorporeal, as repeatedly observed.)

Further the differentia must be elements of the essence, and not merely properties or attributes. Thus if Figure is the term to be divided, it must not be divided into figures whose angles are equal to two right angles, and figures whose angles are together greater than two right angles. For it is only an attribute of a triangle and not part of its essence that its angles are equal to two right angles.

Again the bifurcations must be contradictories, like White and Black, Straight and Bent; and if we characterize one branch by either term, we must characterize the other by its opposite, and not, for example, characterize one branch by a colour, the other by a mode of progression, swimming for instance.

Furthermore living beings [cannot be divided] by the functions common to body and soul, by Flying, for instance, and Walking, as we see them divided in the current dichotomies. For some groups, Ants for instance, fall under both divisions, some ants flying while others do not. Similarly the division into Wild and Tame [must be rejected]; as it also would involve the disruption of a species into different groups. For in almost all species in which some members are tame, there are other members that are wild. Such for example is the case with Men, Horses, Oxen, Dogs in India,³ Pigs, Goats, Sheep; groups which, if double, ought to have what they have not, namely, different appellations; and which, if single, prove that Wildness and Tameness do not amount to specific differences. And whatever single element we take as a basis of division the same difficulty will occur.

The method then that we must adopt is to attempt to recognise the natural groups, following the indications afforded by the instincts of mankind, which led them for instance to form the class of Birds and the class of Fishes, each of which groups combines a multitude of differentia, and is not defined by a single one as in dichotomy. The method of dichotomy is either impossible (for it would put a single group under different divisions or dissimilar groups under the same division); or it only furnishes a single ultimate differentia for each species, which either alone or with its series of antecedents has to constitute the whole essence.

If, again, a new differential character be introduced at any stage into the division, the necessary result is that the unity of the whole, and the continuity of the division, become merely a unity and continuity of agglomeration, like the unity and continuity of a series of sentences coupled together by conjunctive particles. For instance, suppose we have the bifurcation Feathered and Featherless, and then divide Feathered into Wild and Tame, or into White and Black. Tame and White have no essential relation to Feathered, but are the commencement of an independent bifurcation, and are foreign to the series at the end of which they are introduced.

As we said then, we must define at the outset by a multiplicity of differentia. If we do so, privative terms will be available, which are unavailable to the dichotomist.

The impossibility of reaching the definition of any of the ultimate forms by dichotomy of the larger group, as some propose, is manifest also from the following considerations. It is impossible that a single differentia, either by itself or with its antecedents, shall express the whole essence of a species. (In saying a single differentia by itself I mean such an isolated differentia as Cloven-footed; in saying a single differentia with its antecedents I mean such a series as Footed, Two-footed, Cloven-footed. The very continuity of such a series of successive differentia in a division is intended to show that it is their combination that expresses the character of the resulting unit, or ultimate group. But one is misled by the usages of language into imagining that it is merely the final term of the series, Cloven-footed for instance, that constitutes the whole differentia, and that the antecedent terms, Footed, Two-footed, are superfluous. Now it is evident that such a series cannot consist of many terms. For if one divides and subdivides, one soon reaches the final differential term; but for all that, one has not got to the ultimate division, that is, to the species.) No single differentia, I repeat, either by itself or with its antecedents, can possibly express the essence of a species. Suppose, for example, Man to be the animal to be defined; the single differentia will be Cloven-footed, either by itself or with its antecedents, Footed and Two-footed. Now if man was nothing more than a Cloven-footed animal, this single differentia would duly represent his essence. But seeing that this is not the case, more differentia than this one will necessarily be required to define him; and these cannot come under one division; for each single branch of a dichotomy ends in a single differentia, and cannot possibly include several differentia belonging to one and the same animal.

It is impossible then to reach any of the ultimate animal forms by dichotomous division.

(*Ch.* 4.) It deserves inquiry why a single name denoting a single group was not invented by mankind, as an appellation to comprehend the two groups of Water animals and Winged animals. For there are certain attributes common to the two¹ and [differentiating them from] the rest of the animal kingdom. However, the present nomenclature is just. Groups that only differ in degree, and in the more or less of an identical element that they possess, are aggregated under a single class; groups

whose attributes are not identical but analogous are separated. For instance bird differs from bird by gradation, or by excess and defect; some birds have long feathers, others short ones, but all are feathered. Bird and Fish are more remote and only agree in having analogous organs; for what in the bird is feather, in the fish is scale. Such analogies can scarcely however serve universally as indications for the formation of groups, for almost all animals present analogies in their corresponding parts.

The individuals comprised within a species, such as Socrates and Coriscus, are the real existences; but inasmuch as these individuals possess one common specific form, it will suffice to state the universal attributes of the species, that is, the attributes common to all its individuals, once for all, as otherwise there will be endless reiteration, as has already been pointed out.

But as regards the larger groups—such as Birds—which comprehend many species, there may be a question. For on the one hand it may be urged that as the ultimate species represent the real existences, it will be well, if practicable, to examine these ultimate species separately, just as we examine the species Man separately; to examine, that is, not the whole class Birds collectively, but the Ostrich, the Crane, and the other indivisible groups or species belonging to the class.

On the other hand, however, this course would involve repeated mention of the same attribute, as the same attribute is common to many species, and so far would be irrational and tedious. Perhaps, then, it will be best to treat generically the universal attributes of the groups that have a common nature and contain closely allied subordinate forms, whether they are groups recognised by a true instinct of mankind, such as Birds and Fishes, or groups not popularly known by a common appellation, but withal composed of closely allied subordinate groups; and only to deal individually with the attributes of a single species, when such species—man, for instance, and any other such, if such there be—stands apart from others, and does not constitute with them a larger natural group.

It is generally similarity in the shape of particular organs, or of the whole body, that has determined the formation of the larger groups. It is in virtue of such a similarity that Birds, Fishes, Cephalopoda, and Testacea have been made to form each a

separate class. In their respective subdivisions the organs are not merely analogous, as the bone of man and the spine of fish, but are identical, with the exception of certain corporeal differentiations in respect of largeness smallness, hardness softness, smoothness roughness, and other oppositions of this kind, and, in one word, in respect of degree.

We have now touched upon the canons for criticising the method of natural science, and have considered what is the most systematic and easy course of investigation ; we have also dealt with division, and the mode of conducting it so as best to attain the ends of science, and have shown how dichotomy is either impracticable or inefficacious for its professed purposes.

Having laid this foundation, we proceed to the next topic, and by way of introduction we observe, that (*Ch.* 5) some members of the universe are ungenerated, imperishable, and eternal, while others are subject to generation and decay. The former are excellent beyond compare and divine, but less accessible to knowledge. The evidence that might throw light on them, and on the problems which we long to solve respecting them, is furnished but scantily by sensation ; whereas respecting perishable plants and animals we have abundant information, living as we do in their midst, and ample data may be collected concerning all their various kinds, if only we are willing to take sufficient pains. Both departments, however, have their special charm. The scanty conceptions to which we can attain of celestial things give us, from their excellence, more pleasure than all our knowledge of the world in which we live ; just as a half glimpse of persons that we love is more delightful than a leisurely view of other things, whatever their number and dimensions. On the other hand, in certitude and in completeness our knowledge of terrestrial things has the advantage. Moreover, their greater nearness and affinity to us balances somewhat the loftier interest of the heavenly things that are the objects of the higher philosophy. Having already treated of the celestial world, as far as our conjectures could reach, we proceed to treat of animals, without omitting, to the best of our ability, any member of the kingdom, however ignoble. For if some have no graces to charm the sense, yet even these, by disclosing to intellectual perception the artistic spirit that designed them, give immense pleasure to all who can trace links of causation, and are inclined to philosophy. Indeed, it would be strange if mimic

representations of them were attractive, because they disclose the mimetic skill of the painter or sculptor, and the original realities themselves were not more interesting, to all at any rate who have eyes to discern the reason that presided over their formation. We therefore must not recoil with childish aversion from the examination of the humbler animals. Every realm of nature is marvellous: and as Heraclitus, when the strangers who came to visit him found him warming himself at the furnace in the kitchen, is reported to have bidden them not to be afraid to enter, as even in that kitchen divinities were present, so we should venture on the study of every kind of animal without distaste; for each and all will reveal to us something natural and something beautiful. Absence of hap-hazard and conduciveness of everything to an end are to be found in Nature's works in the highest degree, and the resultant end of her generations and combinations is a form of the beautiful.

If any person thinks the examination of the rest of the animal kingdom an unworthy task, he must hold in like disesteem the study of man. For no one can look at the primordia of the human frame—blood, flesh, bones, vessels, and the like—without much repugnance.¹ Moreover, in every inquiry, the examination of material elements and instruments is not to be regarded as final, but as ancillary to the conception of the total form. Thus the true object of architecture is not bricks mortar or timber, but the house; and so the principal object of natural philosophy is not the material elements, but their composition, and the totality of the form to which they are subservient, and independently of which they have no existence.

The course of exposition must be first to state the attributes common to whole groups of animals, and then to attempt to give their explanation. Many groups, as already noticed, present absolutely common attributes, that is to say, absolutely identical affections, and absolutely identical organs, feet, feathers, scales, and the like; while in other groups the affections and organs are only so far identical as that they are analogous. For instance, some groups have lungs, others have no lung, but an organ analogous to a lung in its place; some have blood, others have no blood, but a fluid analogous to blood, and with the same office. To treat of the common attributes in connexion with each individual group would involve, as already suggested, useless iteration. So much for this topic.

As every instrument and every bodily member subserves some partial end, that is to say, some special action, so the whole body must be destined to minister to some plenary sphere of action. Thus the saw is made for sawing, for this is its function, and not sawing for the saw. Similarly the body too must somehow or other be made for the soul, and each part of it for some subordinate function, to which it is adapted.

We have, then, first to describe the common functions, common, that is, to the whole animal kingdom, or to certain large groups, or to the members of a species. In other words, we have to describe the attributes common to all animals, or to assemblages, like the class of Birds, of closely allied groups differentiated by gradation, or to groups like Man not differentiated into subordinate groups. In the first case the common attributes may be called analogous, in the second generic, in the third specific.

When a function is ancillary to another, a like relation manifestly obtains between the organs which discharge these functions; and similarly if one function is the end of another, their respective organs will stand to each other in the same relation. Other phenomena there are that are [neither ancillary to higher ones, nor the end of lower ones, but] the inevitable results of the existing conditions [under which the animal is living].

Instances of what I mean by functions and affections are Reproduction, Growth, Copulation, Waking, Sleep, Locomotion, and other similar vital actions. Instances of what I mean by parts are Nose, Eye, Face, and other so-called members or limbs, and also the more elementary parts of which these are made. So much for the method to be pursued. Let us now try to set forth the causes of all vital phenomena, whether universal or particular, and in so doing let us follow that order of exposition, which conforms, as we have indicated, to the order of nature.

B O O K I I.

(*Ch.* 1.) The nature and the number of the parts of which animals are severally composed are matters which have already been set forth in detail in the book of *Researches about Animals*.¹ We have now to inquire what are the causes that in each case have determined this composition ; a subject reserved by us for separate consideration.

Now there are three degrees of composition ; and of these the first in order, as all will allow, is composition out of what some call the elements, such ² as air, earth, water, fire. Perhaps however it would be more accurate to say composition out of the elementary forces ;³ nor indeed out of all of these, but out of a limited number of them, as defined in previous treatises. For fluid and solid, hot and cold, form the material of all composite substances ;⁴ and all other differences are secondary to these, such differences, that is, as heaviness or lightness, density or rarity, roughness or smoothness, and any other such properties of matter as there may be. The second degree of composition is that by which the homogeneous parts of animals, such as bone, flesh, and the like, are constituted out of the primary substances. The third and last stage is the composition which forms the heterogeneous parts, such as face, hand, and the rest.⁵

Now the order of actual development and the order of logical existence are always the inverse of each other. For that which is antecedent in the true order of nature is posterior in the order of development, and that is genetically last which is logically first. (That this is so is manifest by induction. For a house does not exist for the sake of bricks and stones, but these materials for the

sake of the house ; and the same is the case with the materials of other bodies. For generation is a process from a something to a something ; that which is generated having a cause in which it originates and a cause in which it ends. The originating cause is the primary efficient cause, which is something already endowed with tangible existence, while the final cause is some definite [and predictable] form ; for man generates man, and plant generates plant, and in each case out of the material which environs both.)⁶ In order of time then the material and the generative process must necessarily be anterior to the being that is generated ; but in logical order the definitive character and form of each being precedes the material. This is evident if one only tries to define the process of formation. For the definition of house-building includes and presupposes that of the house ; but the definition of the house does not include nor presuppose that of house-building ; and the same is true of all other productions. So that it must necessarily be that the elementary material exists for the sake of the homogeneous parts, seeing that these are genetically posterior to it, just as the heterogeneous parts are posterior genetically to them.⁷ For these heterogeneous parts have reached the end and goal, having the third degree of composition, in which degree generation or development often attains its final term.⁸

Animals then are composed of homogeneous parts, and are also composed of heterogeneous parts. The former however exist for the sake of the latter.⁹ For the active functions and operations of the body are carried on by these ; that is, by the heterogeneous parts, such as the eye, the nostril, the whole face, the fingers, the hand, and the whole arm. But inasmuch as there is a great variety in the functions and motions not only of aggregate animals but also of the individual organs, it is necessary that the substances out of which these are composed shall present a diversity of properties. For some purposes softness is advantageous, for others hardness ; some parts must be capable of extension, others of flexion. Such properties then are distributed separately to the different homogeneous parts, one being soft another hard, one fluid another solid, one viscous another brittle ; whereas each of the heterogeneous parts presents a combination of multifarious properties. For the hand, to take an example, requires one property to enable it to effect pressure, and another and different

property for simple prehension.¹⁰ For this reason the active or executive parts of the body are compounded out of bones, sinews, flesh, and the like, but not these latter out of the former.

So far, then, as has yet been stated, the relations between these two orders of parts are determined by a final cause. We have however to inquire whether necessity may not also have a share in the matter; and it must be admitted that these mutual relations could not from the very beginning have possibly been other than they are. For heterogeneous parts can be made up out of homogeneous parts, either from a plurality of them, or from a single one, as is the case with some of the viscera, which varying in configuration are yet, to speak broadly, formed from a single homogeneous substance; but that homogeneous substances should be formed out of a combination of heterogeneous parts is clearly an impossibility. For these causes, then, some parts of animals are simple and homogeneous, while others are composite and heterogeneous; and dividing the parts into the active or executive and the sensitive, each one of the former is, as before said, heterogeneous, and each one of the latter homogeneous. For it is in homogeneous parts alone that sensation can occur, as the following considerations show.

Each sense is confined to a single kind or order of sensibles, and its organ must therefore be such as to admit the action of that kind, and [of no other. It must therefore have the properties of that kind and of no other; for] that which is endowed with a property *in posse* is acted upon by that which has the like property *in esse*.¹¹ The one-ness of its sensibles implies then the one-ness or homogeneity of the sense-organ. Thus it is that while no physiologists ever dream of saying of the hand or face or other such part that one is earth, another water, another fire, they couple each separate sense-organ with a separate element, asserting this one to be air and that other to be fire.¹²

Sensation then is confined to the simple or homogeneous parts. But, as might reasonably be expected, the organ of touch, though still homogeneous, is yet the least simple of all the sense-organs. For touch more than any other sense appears to be correlated to several distinct kinds of objects, and to recognise more than one category of contrasts, heat and cold, for instance, solidity and fluidity, and other similar oppositions.¹³ Accordingly the organ which deals with these varied objects is of all the sense-organs

the most corporeal,¹⁴ being either the flesh, or the substance which in some animals takes the place of flesh.

Now as there cannot possibly be an animal without sensation, it follows as a necessary consequence that every animal must have some homogeneous parts ; for these alone are capable of sensation, the heterogeneous parts serving for the active functions. Again as the sensory faculty, the motor faculty, and the nutritive faculty, are all lodged in one and the same part of the body, as was stated in a former treatise,¹⁵ it is necessary that the part which is the primary seat of these principles shall on the one hand, in its character of general sensory recipient, be one of the simple parts ; and on the other hand shall, in its motor and active character, be one of the heterogeneous parts. For this reason it is the heart which in sanguineous animals constitutes this central part, and in bloodless animals it is that which takes the place of a heart. For the heart, like the other viscera, is one of the homogeneous parts ; for, if cut up, its pieces are homogeneous in substance with each other. But it is at the same time heterogeneous in virtue of its definite configuration.¹⁶ And the same is true of the other so-called viscera, which are indeed formed from the same material as the heart. For all these viscera have a sanguineous character owing to their being situated upon vascular ducts and branches. For just as a stream of water deposits mud, so the various viscera, the heart excepted,¹⁷ are, as it were, deposits from the stream of blood in the vessels. And as to the heart, the very starting-point of the vessels, and the actual seat of the force by which the blood is first fabricated, it is but what one would naturally expect, that out of the selfsame nutriment of which it is the recipient its own proper substance shall be formed.¹⁸ Such then are the reasons why the viscera are of sanguineous aspect ; and why in one point of view they are homogeneous, in another heterogeneous.

(*Ch. 2.*) Of the homogeneous parts of animals, some are soft and fluid, others hard and solid ; and of the former some are fluid permanently, others only so long as they are in the living body.

Among the fluids¹ are blood, serum, lard, suet, marrow, semen, bile, milk when present, flesh, and their various analogues. For the parts enumerated are not to be found in all animals, some animals only having parts analogous to them. Of the hard and solid homogeneous parts bone, fish-spine,² sinew, blood-vessel, are

examples. The last of these points to a sub-division that may be made in the class of homogeneous parts.³ For in some of them the whole and a portion of the whole are synonymous. Thus a portion of vessel is still called vessel. While in others there is not this identity of name; and so far these parts agree with the heterogeneous parts; for a portion of a face is never called face.

The first question to be asked is what are the causes to which these homogeneous parts owe their existence? The causes are various; and this whether the parts be solid or fluid. Thus one set of homogeneous parts represent the material out of which the heterogeneous parts are formed; for each separate organ is constructed of bones, sinews, flesh, and the like; which are either essential elements in its formation, or contribute to the proper discharge of its function. A second set are the nutriment of the first, and are invariably fluid, for all growth occurs at the expense of fluid matter;⁴ while a third set are the residue of the second. Such for instance are the fæces and, in animals that have a bladder, the urine; the former being the dregs of the solid nutriment, the latter of the fluid.

Even the individual homogeneous parts present variations, which are intended in each case to render them more serviceable for their purpose. The variations of the blood may be selected to illustrate this. For different bloods differ in their degrees of thinness or thickness, of clearness or turbidity, of coldness or heat; and this whether we compare the bloods from different parts of the same individual or the bloods of different animals. For, in the individual, all the differences just enumerated distinguish the blood of the upper and of the lower halves of the body; and, dealing with classes, one section of animals is sanguineous, while the other has no blood, but only something resembling it in its place. As regards the results of such differences, the thicker and the hotter blood is, the more conducive is it to strength, while in proportion to its thinness and its coldness is its suitability for sensation and intelligence. A like distinction exists also in the fluids which are analogous to blood. This explains how it is that bees⁵ and other similar creatures are of a more intelligent nature than many sanguineous animals; and that, of sanguineous animals, those are the most intelligent whose blood is thin and cold. Noblest of all are those whose blood is hot, and at the same time thin and clear. For

such are suited alike for the development of courage and of wisdom. Accordingly the upper parts are superior in these respects to the lower, the male superior to the female, and the right side to the left.⁶ As with the blood so also with the other parts, homogeneous and heterogeneous alike. For here also such variations as occur must be held either to be related to the essential constitution and mode of life of the several animals, or, in other cases, to be merely matters of slightly better or slightly worse. Two animals for instance may have eyes. But in one these eyes may be of fluid consistency, while in the other they are hard; and in one there may be eyelids, in the other no such appendages. In such a case, the fluid consistency and the presence of eyelids, which are intended to add to the accuracy of vision, are differences of degree.⁷

As to why all animals must of necessity have blood or something of a similar character, and what the nature of blood may be, these are matters which can only be considered, when we have first discussed hot and cold. For the natural properties of many substances are referable to these two elementary principles; and it is a matter of frequent dispute what animals or what parts of animals are hot and what cold. For some maintain that water animals are hotter than such as live on land,⁸ asserting that their natural heat counterbalances the coldness of their medium; and again that bloodless animals are hotter than those with blood, and females than males. Parmenides, for instance, and some others declare that women are hotter than men,⁹ and that it is the warmth and abundance of their blood which causes their menstrual flow, while Empedocles maintains the opposite opinion. Again, comparing the blood and the bile, some speak of the former as hot and of the latter as cold, while others invert the description. If there be this endless disputing about hot and cold, which of all things that affect our senses are the most distinct, what are we to think as to our other sensory impressions?

The explanation of the difficulty appears to be that the term "hotter" is used in several senses; so that different statements, though in verbal contradiction with each other, may yet all be more or less true. There ought then to be some clear understanding as to the sense in which natural substances are to be termed hot or cold, solid or fluid. For it appears manifest, that these are properties on which even life and death¹⁰ are largely dependent,

and that they are moreover the causes of sleep and waking, of maturity and old age, of health and disease; while no similar influence belongs to roughness and smoothness, to heaviness and lightness, nor in short to any other such properties of matter. That this should be so is but in accordance with rational expectation. For hot and cold, solid and fluid, as was stated in a former treatise,¹¹ are the foundations of the physical elements.

Is then the term hot used in one sense or in many? To answer this we must ascertain what special effect is attributed to a hotter substance, and if there be several such, how many these may be. A body then is in one sense said to be hotter than another, if it impart a greater amount of heat to an object in contact with it. In a second sense, that is said to be hotter which causes the keener sensation when touched, and especially if the sensation be attended with pain. This criterion however would seem sometimes to be a false one; for occasionally it is the idiosyncrasy of the individual that causes the sensation to be painful.¹² Again, of two masses of one and the same substance, the larger is said to have more heat than the smaller. Again, of two things, that is the hotter, which the more readily melts a fusible substance, or sets on fire an inflammable one. Again, of two bodies, that is said to be the hotter which takes the longer time in cooling, as also we call that which is rapidly heated hotter than that which is long about it; as though the rapidity implied proximity and this again similarity of nature, while the want of rapidity implied distance and this again dissimilarity of nature. The term hotter is used then in all the various senses that have been mentioned, and perhaps in still more. Now it is impossible for one body to be hotter than another in all these different fashions. Boiling water for instance, though it is more scalding than flame, yet has no power of burning or melting combustible or fusible matter, while flame has. So again this boiling water is hotter than a small fire, and yet gets cold much more rapidly and completely. For in fact fire never becomes cold; whereas water invariably does so. Boiling water again is hotter to the touch than oil; yet it gets cold and coagulates more rapidly than this other fluid.¹³ Blood again is hotter to the touch than either water or oil, and yet coagulates before them. Iron again and stones and other similar bodies are much longer in getting heated than water, but when once heated burn other substances with

a much greater intensity. Another distinction is this. In some of the bodies which are called hot the heat is derived from without, while in others it belongs to the bodies themselves; and it makes a most important difference, whether the heat has the former or the latter origin. For to call that one of two bodies the hotter, which is possessed of heat, we may almost say, accidentally and not of its own essence, is very much the same thing as if, finding that some man in a fever was a musician, one were to say musicians are hotter than healthy men. Of that which is hot *per se* and that which is hot *per accidens*, the former is the slower to cool, while not rarely the latter is the hotter to the touch. The former again is the more burning of the two—flame for instance as compared with boiling water—while the latter, as the boiling water, which is hot *per accidens*, is the more heating to the touch. From all this it is clear that it is no simple matter to decide which of two bodies is the hotter. For the first may be the hotter in one sense, the second the hotter in another. Indeed in some of these cases it is impossible to say simply even whether a thing is hot or not. For the actual substratum may not itself be hot, but may be receptive of heat as an attribute; as, supposing hot water or hot iron had single names, would be the case with the substratum of the substances denoted by such names. It is after this manner that blood is hot.¹⁴ In such cases, in those, that is, in which the substratum owes its heat to an external influence, it is plain that cold is not a mere privation, but an actual existence.¹⁵

There is no knowing but that even fire may be another of these cases. For the substratum of fire may be smoke or charcoal, and, though the former of these is always hot, smoke being an uprising vapour, yet the latter becomes cold when its flame is extinguished, as also would oil and pinewood under similar circumstances. But even substances that have been burnt nearly all possess some heat, cinders, for example, and ashes, the dejections also of animals, and, among the excretions, bile; because some residue of heat has been left in them after their combustion. It is in another sense that pinewood and fat substances are hot; namely, because they rapidly assume the actuality of fire.

Heat appears to cause both coagulation and melting.¹⁶ Now such things as are formed merely of water are solidified by cold, while such as are formed of nothing but earth are solidified by

fire. Hot substances again are solidified by cold, and, when they consist chiefly of earth, the process of solidification is rapid, and the resulting substance is insoluble; but, when their main constituent is water, the solid matter is again soluble. What kinds of substances however admit of being solidified, and what are the causes, of solidification, are questions that have already been dealt with more precisely in another treatise.¹⁷

In conclusion, then, seeing that the terms hot and hotter are used in many different senses, and that no one substance can be hotter than others in all these senses, we must, when we attribute this character to an object, add such further statements as that this substance is hotter *per se*, though that other is often hotter *per accidens*; or again that this substance is potentially hot, that other actually so; or again, that this substance is hotter in the sense of causing a greater feeling of heat when touched, while that other is hotter in the sense of producing flame and burning. The term hot being used in all these various senses, it plainly follows that the term cold will also be used with like ambiguity.

So much then as to the signification of the terms hot and cold, hotter and colder.¹⁸

(Ch. 3.) In natural sequence we have next to treat of solid and fluid. These terms are used in various senses. Sometimes for instance they denote things that are potentially, at other times things that are actually, solid or fluid. Ice for example, or any other solidified fluid, is spoken of as being actually and accidentally solid, while potentially and essentially it is fluid. Similarly earth and ashes and the like, when mixed with water, are actually and accidentally fluid, but potentially and essentially are solid. Now separate the two constituents in this last instance; and you have on the one hand the watery part, capable of passing by imbibition into other substances, and this is both potentially and actually fluid; and on the other hand you have the earthy part, and this is both potentially and actually solid. It is to bodies that are solid in this double manner that the term "solid" is most properly and absolutely applicable. So also the opposite term "fluid" is strictly and absolutely applicable to that only which is both potentially and actually fluid. The same remark applies also to hot bodies and to cold.

These distinctions, then, being laid down, it is plain that blood

is essentially hot in so far as that heat is connoted in its name; just as if boiling water were denoted by a single term, boiling would be connoted in that term. But the substratum of blood, that which it is [in substance] while it is blood [in form] is not hot. Blood then in a certain sense is essentially hot, and in another sense is not so. For heat is included in the definition of blood, just as whiteness is included in the definition of a white man, and so far therefore blood is essentially hot. But so far as blood becomes hot from some external influence, it is not hot essentially.¹

As with hot and cold, so also is it with solid and fluid. We can therefore understand how some substances are hot and fluid so long as they remain in the living body, but become perceptibly cold and coagulate so soon as they are separated from it; while others are hot and consistent while in the body, but when withdrawn undergo a change to the opposite condition, and become cold and fluid. Of the former blood is an example; of the latter bile; for while blood solidifies when thus separated, yellow bile under the same circumstances becomes more fluid.² We must attribute to such substances the possession of opposite properties in a greater or less degree.

In what sense, then, the blood is hot and in what sense fluid, and how far it partakes of the opposite properties, has now been fairly explained. Now since everything that grows must take nourishment, and nutriment in all cases consists of fluid and solid substances, and since it is by the force of heat that these are concocted³ and changed, it follows that all living things, animals and plants alike, must on this account if on no other have a natural source of heat. This natural heat moreover must belong to many parts,⁴ seeing that the organs by which the various elaborations of the food are effected are many in number. For first of all there is the mouth and the parts inside the mouth, on which the first share in the duty clearly devolves, in such animals at least as live on food which requires disintegration. The mouth, however, does not actually concoct the food, but merely facilitates concoction;⁵ for the subdivision of the food into small bits facilitates the action of heat upon it. After the mouth come the upper and the lower abdominal cavities,⁶ and here it is that concoction is effected by the aid of natural heat. Again, just as there is a channel for the admission of the

unconcocted food into the stomach, namely the mouth, and in some animals the so-called œsophagus, which is continuous with the mouth and reaches to the stomach, so must there also be other and more numerous channels by which the concocted food or nutriment shall pass out of the stomach and intestines into the body at large, and to which these cavities shall serve as a kind of manger.⁷ For plants get their food from the earth by means of their roots; and this food is already elaborated when taken in, which is the reason that plants produce no excrement,⁸ the earth and its heat serving them in the stead of a stomach. But animals, with scarcely an exception, and notably all such animals as are capable of locomotion, are provided with a stomachal sac,⁹ which is as it were an internal substitute for the earth. They must therefore have some instrument which shall correspond to the roots of plants,¹⁰ with which they may absorb their food from this sac, so that the proper end of the successive stages of concoction may at last be attained. The mouth then, its duty done, passes over the food to the stomach, and there must necessarily be something to receive it in turn from this. This something is furnished by the blood-vessels,¹¹ which run throughout the whole extent of the mesentery from its lowest part right up to the stomach. A description of these will be found in the treatises on Anatomy and Natural History.¹² Now as there is a receptacle for the entire matter taken as food, and also a receptacle for its excremental residue, and again a third receptacle, namely the vessels, which serve as such for the blood, it is plain that this blood must be the final nutritive material in such animals as have it; while in bloodless animals the same is the case with the fluid which represents the blood.¹³ This explains why the blood diminishes in quantity when no food is taken, and increases when much is consumed,¹⁴ and also why it becomes healthy and unhealthy according as the food is of the one or the other character. These facts, then, and others of a like kind, make it quite plain that the purpose of the blood in sanguineous animals is to subserve the nutrition of the body. They also explain why no more sensation is produced by touching the blood, than by touching one of the excretions or the food, whereas when the flesh is touched sensation is produced. For the blood is not continuous nor united by growth with the flesh, but simply lies loose in its receptacle, that is in the heart and

vessels.¹⁵ The manner in which the parts grow at the expense of the blood, and indeed the whole question of nutrition, will find a more suitable place for exposition in the treatise on generation and development, and in other writings.¹⁶ For our present purpose all that need be said is that the blood exists for the sake of nutrition, that is the nutrition of the parts; and with this much let us therefore content ourselves.

(*Ch.* 4.) What are called fibres¹ are found in the blood of some animals but not of all. There are none for instance in the blood of deer and of roes; and for this reason the blood of such animals as these never coagulates.² For that portion of the blood which consists mainly of water is not liable to coagulation, this process occurring only in the earthy part, that is in the fibres, during the evaporation of the moisture.

Some at any rate of the animals with watery blood have a keener intellect than those whose blood is of an earthier nature. This is due not so much to the coldness of their blood as to its thinness and purity; neither of which qualities belongs to the earthy matter. For the thinner and purer its fluid is, the more active is an animal's sensibility. Thus it is that some exsanguineous animals, notwithstanding their want of blood, are yet more intelligent than some among the sanguineous kinds. Such for instance, as already said, is the case with bees³ and ants, and whatever other animals there may be of a like nature. At the same time too great an excess of water makes animals timorous.⁴ For fear chills the body; so that in animals whose heart contains so watery a mixture the way is, as it were, prepared for the operation of this emotion. For water is congealed by cold. This also explains why bloodless animals are, as a general rule, more timorous than such as have blood, so that they remain motionless, when frightened, and discharge their excretions, and in some instances change colour.⁵ Such animals on the other hand as have thick and abundant fibres in their blood are of a more earthy nature, and of a choleric temperament, and liable to bursts of passion. For anger is productive of heat; and solids when they have been made hot give off more heat than fluids. The fibres therefore, being earthy and solid,⁶ are turned into so many hot embers in the blood, like the embers in a vapor-bath, and cause ebullition in the fits of passion.⁷

This explains why bulls and boars are so choleric and so

passionate. For their blood is exceedingly rich in fibres;⁸ and the bull's at any rate coagulates more rapidly than that of any other animal.⁹ If these fibres, that is to say if the earthy constituents of which we are speaking, are taken out of the blood, the fluid that remains behind will no longer coagulate; just as the watery residue of mud will not coagulate after removal of the earth. But if the fibres are left the fluid coagulates, as also does mud, under the influence of cold. For when the heat is expelled by the cold, the fluid, as has been already stated, passes off with it by evaporation, and the residue is dried up and solidified, not by heat but by cold.¹⁰ So long however as the blood is in the body, it is kept fluid by animal heat.

The character of the blood affects both the temperament and the sensory faculties of animals in many ways. This is indeed what might reasonably be expected, seeing that the blood is the material of which the whole body is made. For nutriment supplies the material, and the blood is the ultimate nutriment. It makes then a considerable difference whether the blood be hot or cold, thin or thick, turbid or clear.

The watery part of the blood is serum; and it is watery, either owing to its not being yet concocted, or owing to its having become corrupted; so that one part of the serum is the resultant of a necessary process, while another part is material intended to serve for the formation of the blood.¹¹

(*Ch.* 5.) The differences between lard and suet correspond to differences of blood.¹ For both are concocted blood; being that surplus of it, which, though well concocted and highly nutritious, has not been expended in forming the fleshy substance of the animal, but remains over owing to the superabundance of food. That such is the composition of these substances is evident from their sheeny appearance. For a sheeny look in fluids comes from their being compounded of air and of fire.² It follows from what has been said that no ex-sanguineous animals have either lard or suet; for they have no blood.³ Among sanguineous animals those whose blood is dense are likely to have suet rather than lard. For suet is of an earthy nature, that is to say, it contains but a small proportion of water and is chiefly composed of earth; and this it is that makes it coagulate, just as the fibrous matter of blood coagulates, or broths which contain such fibrous matter.⁴ Thus it is that in those horned animals that have no front teeth

in the upper jaw the fat consists of suet. For the very fact that they have horns and huckle-bones⁵ shows that their composition is rich in earthy matter; for all such appurtenances are solid and earthy in character. On the other hand in those hornless animals that have front teeth in both jaws, and whose feet are divided into toes, there is no suet, but in its place lard;⁶ and this, not being of an earthy character, neither coagulates nor dries up into a friable mass.

Both lard and suet when present in moderate amount are beneficial; for they contribute to health and strength, while they are no hindrance to sensation. But when they are present in great excess, they are injurious and destructive. For were the whole body formed of them it would perish. For an animal is an animal in virtue of its sensory part, that is in virtue of its flesh, or of the substance analogous to flesh.⁷ But the blood, as before stated, is not sensitive; as therefore is neither lard nor suet, seeing that they are nothing but concocted blood. Were then the whole body composed of these substances, it would be utterly without sensation. Such animals, again, as are excessively fat age rapidly. For their blood is used up in forming fat, and so they have but little of it left; and when there is but little blood the way is already open for decay.⁸ For decay may be said to be deficiency of blood, the scantiness of which renders it liable, like all bodies of small bulk, to be injuriously affected by any chance excess of heat or cold. For the same reason fat animals are less prolific than others. For that part of the blood which should go to form semen and seed, is used up in the production of lard and suet, which are nothing but concocted blood; so that in these animals there is either no reproductive excretion at all, or only a scanty amount.⁹

So much then of blood and serum, and of lard and suet. Each of these has been described, and the purposes told for which they severally exist.

(*Ch. 6.*) The marrow also is of the nature of blood, and not as some¹ think the germinal force of the semen. That this is the case is quite evident in very young animals. For in the embryo the marrow of the bones has a blood-like appearance, which is but consistent with the fact that the parts are all constructed out of blood, and that it is on blood that the embryo is nourished.² But, as the young animal grows up

and ripens into maturity, the marrow changes its colour, just as do the external parts and the viscera.³ For the viscera also in very young animals have each and all a blood-like look, owing to the large amount of this fluid which they contain.

The consistency of the marrow agrees with that of the fat. For when the fat consists of lard, then the marrow also is oily and lard-like; but when the concocted blood is converted into suet, and does not assume the form of lard, then the marrow also has a suety character. In those animals, therefore, that have horns and are without upper front teeth, the marrow has the character of suet; while it takes the form of lard in those that have front teeth in both jaws, and that also have the foot divided into toes. What has been said hardly applies to the spinal marrow. For it is necessary that this shall be continuous and extend without break through the whole back-bone, inasmuch as this bone consists of separate vertebræ. But were the spinal marrow either of soft fat or of suet, it could not hold together in such a continuous mass as it does, but would either be too fluid or too frangible.⁴

There are some animals that can hardly be said to have any marrow. These are those whose bones are strong and solid, as is the case with the lion. For in this animal the marrow is so utterly insignificant, that the bones look as though they had none at all.⁵ However, as it is necessary that animals shall have bones or something analogous to them, such as the fish-spines of water-animals,⁶ it is also a matter of necessity that some of these bones shall contain marrow; for the substance contained within the bones is the nutriment out of which these are formed. Now the universal nutriment, as already stated, is blood; and the blood within the bone, owing to the heat which is developed in it from its being thus surrounded, undergoes concoction, and self-concocted⁷ blood is suet or lard; so that it is perfectly intelligible how the marrow within the bone comes to have the character of these substances. So also it is easy to understand, why, in those animals that have strong and solid bones, some of these should be entirely void of marrow, while the rest contain but little of it; for here the nutriment is all spent in forming the bones.

Those animals that have fish-spines in place of bones have no other marrow than that of the chine.⁸ For in the first place

they have naturally but a small amount of blood; and secondly the only hollow fish-spine is that of the chine. In this then marrow is formed; this being the only spine in which there is space for it, and moreover being the only one which owing to its division into parts requires a connecting bond. This too is the reason why the marrow of the chine, as already mentioned, is somewhat different from that of other bones. For, having to act the part of a clasp, it must be of glue-like tenacity, and at the same time sinewy⁹ so as to admit of stretching.

Such then are the reasons for the existence of marrow, in those animals that have any, and such its nature. It is the surplus of the sanguineous nutriment apportioned to the bones and fish-spines, which has undergone concoction owing to its being enclosed within them.

(*Ch. 7.*) From the marrow we pass on in natural sequence to the brain. For there are some¹ who think that the brain itself consists of marrow, and that it forms the commencement of that substance, because they see that the spinal marrow is continuous with it. In reality the two may be said to be utterly opposite to each other in character. For of all the parts of the body there is none so cold as the brain; whereas the marrow is of a hot nature, as is plainly shown by its sheeny appearance,² and by its fatness. Indeed this is the very reason why the brain and spinal marrow are continuous with each other. For, wherever the action of any part is in excess, nature so contrives as to set by it another part with an excess of contrary action, so that the excesses of the two may counterbalance each other. Now that the marrow is hot is clearly shown by many indications. The coldness of the brain is also manifest enough. For in the first place it is cold even to the touch; and, secondly, of all the fluids of the body it is the one that has the least blood; for in fact it has no blood at all in its proper substance.³ This makes it the most consistent of all the fluids.⁴ This brain is not an excretion, nor yet is it one of the parts which are anatomically continuous with each other; but it has a character peculiar to itself, as might indeed be expected. That it has no continuity with the organs of sense is plain from simple inspection, and is still more clearly shown by the fact, that, when it is touched, no sensation is produced;⁵ in which respect it resembles the blood of animals and their excrement.⁶ The purpose of its

presence in animals is no less than the preservation of the whole body. How it effects this will be seen from what follows. Some writers assert that the soul is fire or some such force.⁷ This however is but a rough and inaccurate assertion; and it would perhaps be better to say that the soul is incorporate in some substance of a fiery character. The reason for this being so is that of all substances there is none so suitable for ministering to the operations of the soul as that which is possessed of heat. For the functions of the soul are nutrition and motion, and it is by heat that these are most readily effected. To say however that the soul is fire is much the same thing as to confound the auger or the saw with the carpenter or his craft, simply because the work is wrought by the two in conjunction. So far then this much is plain, that all animals must necessarily have a certain amount of heat. But as all influences require to be counterbalanced, so that they may be reduced to moderation and brought to the mean (for in the mean, and not in either extreme, lies the true and rational position), nature has contrived the brain as a counterpoise to the region of the heart with its contained heat, and has given it to animals to moderate the latter, compounding it of earth and water.⁸ For this reason it is, that every sanguineous animal has a brain; whereas no bloodless creature has such an organ, unless indeed it be, as the Poulp, by analogy.⁹ For where there is no blood, there in consequence there is but little heat. The brain then tempers the heat and seething of the heart. In order, however, that it may not itself be absolutely without heat, but may have a moderate amount, branches run from both blood-vessels, that is to say from the great vessel and from what is called the aorta,¹⁰ and end in the membrane which surrounds the brain;¹¹ while at the same time, in order to prevent any injury from the heat, these encompassing vessels, instead of being few and large, are numerous and small, and their blood scanty and clear, instead of being abundant and thick.¹² We can now understand why defluxions have their origin in the head, and occur whenever the parts about the brain have more than a due proportion of coldness. For when the nutriment steams upwards through the blood-vessels, its refuse portion is chilled by the influence of this region, and forms defluxions of phlegm and serum. We must suppose, to compare small things with great, that the like happens here as occurs in the production

of showers. For when vapor steams up from the earth under the influence of heat and is carried into the upper regions, so soon as it reaches the cold air that is above the earth, it condenses again into water owing to the refrigeration, and falls back to the earth as rain.¹³ These however are matters which may be suitably considered in the Principles of Diseases, so far as natural philosophy has anything to say to them.¹⁴

It is the brain again,—or, in animals that have no brain, the part analogous to it,—which is the cause of sleep. For either by chilling the blood that streams upwards after food, or by some other similar influences, it produces heaviness in the region in which it lies (which is the reason that drowsy persons hang the head), and causes the heat to escape downwards in company with the blood. It is the accumulation of this in excess in the lower part that produces complete sleep, taking away the power of standing upright from such animals as are able to assume that posture; and from the rest the power of holding up the head. These however are matters which have been separately considered in the treatises on Sensation and on Sleep.¹⁵

That the brain is a compound of earth and water¹⁶ is shown by what occurs when it is boiled. For, when so treated, it turns hard and solid, inasmuch as the water is evaporated by the heat, and leaves the earthy part behind.¹⁷ Just the same occurs when pulse and other fruits are boiled. For these also are hardened by the process, because the water which enters into their composition is driven off and leaves the earth, which is their main constituent, behind.

Of all animals, man has the largest brain in proportion to his size; and it is larger in men than in women.¹⁸ This is because the region of the heart and of the lung is hotter and richer in blood in man than in any other animal; and in men than in women.¹⁹ This again explains why man, alone of animals, stands erect. For the heat, overcoming any opposite inclination, makes growth take its own line of direction, which is from the centre of the body upwards.²⁰ It is then as a counterpoise to his excessive heat that in man's brain there is this superabundant fluidity and coldness; and it is again owing to this superabundance that the cranial bone which some call the Bregma²¹ is the last to become solidified;²² so long does evaporation continue to occur through it under the influence of heat.²³ Man is the only

sanguineous animal in which this takes place.²⁴ Man, again, has more sutures in his skull than any other animal,²⁵ and the male more than the female.²⁶ The explanation is again to be found in the greater size of the brain, which demands free ventilation, proportionate to its bulk. For if the brain be either too fluid or too solid, it will not perform its office, but in the one case will freeze the body, and in the other will not cool it at all; and thus will cause disease, madness, and death. For the cardiac heat and the centre of life is most delicate in its sympathies, and is immediately sensitive to the slightest change or affection of the blood on the outer surface of the brain.²⁷

The fluids which are present in the animal body at the time of birth have now nearly all been considered. Amongst those that appear only at a later period are the residua of the food, which include the deposits of the belly and also those of the bladder. Besides these there is the semen and the milk, one or the other of which makes its appearance in appropriate animals. Of these fluids, the excremental residua of the food may be suitably discussed by themselves, when we come to examine and consider the subject of nutrition.²⁸ Then will be the proper time to explain in what animals they are found, and what are the reasons for their presence. Similarly all questions concerning the semen and the milk may be dealt with in the treatise on generation,²⁹ for the former of these fluids is the very starting-point of the generative process, and the latter has no other ground of existence than generative purposes.

(*Ch.* 8.) We have now to consider the remaining homogeneous parts, and will begin with flesh, and with the substance that, in animals that have no flesh, takes its place. The reason for so beginning is that flesh forms the very basis of animals, and is the essential constituent of their body. Its right to this precedence can also be demonstrated logically. For an animal is by our definition something that has sensibility, and chief of all that has the primary sensibility, which is that of Touch;¹ and it is the flesh, or analogous substance, which is the organ of this sense. And it is the organ, either in the same way as the eye is the organ of sight, that is it constitutes the primary organ of the sense; or it is the organ and the medium through which the object acts combined, that is it answers to the eye with some or other transparent medium attached to it.² Now in the case of

the other senses it was impossible for nature to unite the medium with the sense-organ, nor would such a junction have served any purpose; but in the case of touch she was compelled by necessity to do so. For of all the sense-organs that of touch is the only one that has a corporeal medium, or at any rate its medium is more corporeal than any other.

Regarding then the flesh in its sensory character, it is plain that all the other parts exist on its account. By the other parts I mean the bones, the skin, the sinews, and the blood-vessels, and, again, the hair and the various kinds of nails, and anything else there may be of a like character. Thus the bones are a contrivance to give security to the soft parts, to which purpose they are adapted by their hardness; and in animals that have no bones the same office is fulfilled by some analogous substance, as by cartilage in some fishes, and by fish-spine in others.

Now in some animals this supporting substance is situated within the body, while in some of the bloodless species it is placed on the outside. The latter is the case in all the Crustacea,³ as the Carcini and the Carabi; it is the case also in the Testacea, as for instance in the several species known by the general name of oysters. For in all these animals the fleshy substance is within, and the earthy matter, which holds the soft parts together and keeps them from injury, is on the outside. For the shell not only enables the soft parts to hold together, but also, as the animal is bloodless and so has but little natural warmth, surrounds it, as a chaufferette does the embers, and keeps in the smouldering heat. Similar to this seems to be the arrangement in another and distinct genus of animal, namely in the Tortoises, including the Chelone and the several kinds of Emys.⁴ But in Insects and in Cephalopods the plan is entirely different, there being moreover a contrast between these two themselves. For in neither of these does there appear to be any bony or earthy part, worthy of notice, distinctly separated from the rest of the body. Thus in the Cephalopods the mass of the body consists of a soft flesh-like substance, or rather of a substance which is intermediate to flesh and sinew, and not so readily destructible as actual flesh. I call this substance intermediate to flesh and sinew, because it is soft like the former, while it admits of stretching like the latter.⁵ Its cleavage however is such that it splits not longitudinally, like sinew, but into circular segments,

this being the most advantageous condition, so far as strength is concerned. These animals have also a part inside them corresponding to the spinous bones of fishes. For instance in the Cuttle-fishes there is what is known as the *os sepiæ*, and in the Calamaries there is the so-called *gladius*. In the Poulps on the other hand there is no such internal part, because the body or, as it is termed in them, the head,⁶ forms but a short sac, whereas it is of considerable length in the other two; and it was this length which led nature to assign to them their hard support, so as to ensure their straightness and inflexibility; just as she has assigned to sanguineous animals their bones or their fish-spines as the case may be. To come now to Insects. In these the arrangement is quite different from that of the Cephalopods; quite different also from that which obtains in sanguineous animals, as indeed has been already stated. For in an insect there is no distinction into soft and hard parts, but the whole body is hard, the hardness however being of such a character as to be more flesh-like than bone, and more earthy and bone-like than flesh. The purpose of this is to make the body of the insect less liable to get broken into pieces.⁷

(*Ch. 9.*) There is a resemblance between the osseous and the vascular systems; for each has a central part in which it begins, and each forms a continuous whole. For no bone in the body exists as a separate individuality in itself, but each is either a portion of what may be considered a continuous whole, or at any rate is linked with the rest by contact and by attachments; so that nature may use adjoining bones either as though they were actually continuous and formed a single bone, or, for purposes of flexure, as though they were two and distinct. And similarly no blood-vessel has in itself a separate individuality; but they all form parts of one whole. For an isolated bone, if such there were, would in the first place be unable to perform the office, for the sake of which bones exist; for, were it discontinuous and separated from the rest by a gap, it would be perfectly unable to produce either flexure or extension; nor only so, but it would actually be injurious, acting like a thorn or an arrow lodged in the flesh. Similarly if a vessel were isolated, and not continuous with the vascular centre, it would be unable to retain the blood within it in a proper state. For it is the warmth

derived from this centre that hinders the blood from coagulating ;¹ indeed the blood, when withdrawn from its influence, actually becomes putrid. Now the centre or origin of the blood-vessels is the heart, and the centre or origin of the bones, in all animals that have bones, is what is called the chine. With this all the other bones of the body are in continuity ; for it is the chine that keeps the body extended and straight. But since it is absolutely necessary that the body of an animal shall bend during locomotion, this chine, while it is one in virtue of the continuity of its parts, yet by its division into vertebræ is made to consist of many segments. It is from this chine that the bones of the limbs proceed, and with it they are continuous, in those animals that have limbs ; and these bones form joints at the place where the limbs admit of flexure ; being fastened together by sinews, and having their extremities adapted to each other, either by the one being hollowed and the other rounded,² or by both being hollowed and including between them a huckle-bone, as a connecting bolt, so as to allow of flexure and extension. For without some such arrangement these movements would be utterly impossible, or at any rate would be performed with great difficulty. There are some joints, again, in which the lower end of the one bone and the upper end of the other are alike in shape. In these cases the bones are bound together by sinews, and cartilaginous pieces are interposed in the joint, to serve as a kind of padding, and to prevent the two extremities from grating against each other.

Round about the bones, and attached to them by thin fibrous bands, grow the fleshy parts, for the sake of which the bones themselves exist. For just as an artist, when he is moulding an animal out of clay or other soft substance, takes first some solid body as a basis, and round this moulds the clay ; so also has nature acted in fashioning the animal body out of flesh. Thus we find all the fleshy parts, with one exception, supported by bones, which serve, when the parts are organs of motion, to facilitate flexure, and, when the parts are motionless, act as a protection. The ribs, for example, which enclose the chest are intended to ensure the safety of the heart and neighbouring viscera. The exception of which mention was made is the belly. The walls of this are in all animals devoid of bones ; in order that there may be no hindrance to the expansion which necessarily occurs in this

part after a meal, nor, in females, any interference with the growth of the embryo, which is lodged here.³

Now the bones of viviparous animals, of such that is as are not merely externally but also internally viviparous,⁴ vary but very little from each other in point of strength, which in all of them is considerable. For the vivipara in their bodily proportions are far above other animals, and many of them occasionally grow to an enormous size, as is the case in Libya and in hot and dry countries generally. But the greater the bulk of an animal, the stronger, the bigger, and the harder, are the supports which it requires; and comparing the big animals with each other, this requirement will be most marked in those that live a life of rapine. Thus it is that the bones of males are harder than those of females; and the bones of flesh-eaters, that get their food by fighting, are harder than those of herbivora. Of this the Lion is an example; for so hard are its bones, that, when struck, they give off sparks, as though they were flints. It may be mentioned also that the Dolphin, inasmuch as it is viviparous, is provided with bones and not merely with fish-spines.⁵

In those sanguineous animals, on the other hand, that are oviparous, the bones present successive slight variations of character. Thus in Birds there are bones, but these are not so strong as the bones of the Vivipara.⁶ Then come the Oviparous fishes, where there is no bone, but merely fish-spine. In the Serpents too the bones have the character of fish-spine, excepting in the very large species,⁷ where the solid foundation of the body requires to be stronger, in order that the animal itself may be strong, the same reason prevailing as in the case of the Vivipara. Lastly in the Selachia, as they are called, the fish-spines are replaced by cartilage. For it is necessary that the movements of these animals shall be of an undulating character; and this again requires the solid framework of the body to be pliable, and not rigid in its motions. Moreover in these Selachia⁸ nature has used all the earthy matter on the skin; and she is unable to distribute to many different parts one and the same superfluity of material.⁹ Even in viviparous animals many of the bones are cartilaginous. This happens in those parts where it is to the advantage of the surrounding flesh that its solid base shall be soft and mucilaginous. Such, for instance, is the case with the ears and nostrils; for in projecting parts, such as these, rigid substances would soon get

broken. Cartilage and bone are indeed fundamentally the same thing, the differences between them being merely matters of degree.¹⁰ This is manifested in the fact that neither cartilage nor bone, when once cut off, grows again.¹¹ Now the cartilages of these land animals are without marrow, that is without any distinctly separate marrow. For the marrow, which in bones is distinctly separate, is here mixed up with the whole mass, and gives a soft and mucilaginous consistence to the cartilage. But in the *Selachia* the chine, though it is cartilaginous, yet contains marrow ; for here it stands in the stead of a bone.

Very nearly resembling the bones to the touch¹² are such parts as nails, hoofs, claws, horns, and the beaks of birds, all of which are intended to serve as means of defence. For the organs which are made out of these substances, and which are called by the same names as the substances themselves, the organ hoof for instance and the organ horn, are contrivances to ensure the preservation of the animals to which they severally belong. In this class too must be reckoned the teeth, which in some animals have but a single function, namely the mastication of the food, while in others they have an additional office, namely to serve as instruments of offence ; as is the case with all animals that have sharp interfitting teeth or that have tusks. All these parts are necessarily of a solid and earthy character ; for the value of a weapon depends on such properties. Their earthy character explains how it is that all such parts are much more developed in viviparous quadrupeds than in man. For there is always more earth in the composition of these animals than in that of the human body. However, not only all these parts but such others as are nearly connected with them, skin for instance, bladder, membrane, hairs, feathers, and their analogues, and any other similar parts that there may be, will be considered farther on with the heterogeneous parts. There we shall examine into the causes which produce them, and into the objects of their presence severally in the bodies of animals. For, as with the heterogeneous parts, so with these ; it is from a consideration of their functions that alone we can derive any knowledge of them. The reason for dealing with them at all in this part of the treatise, and classifying them with the homogeneous parts, is that under one and the same name are confounded the entire organs and the substances of which they are composed.¹³ But of all these

substances flesh and bone form the basis. Semen and milk were also passed over, when we were considering the homogeneous fluids. For the treatise on generation will afford a more suitable place for their examination, seeing that the former of the two is the very foundation of the thing generated, while the latter is its nourishment.

(*Ch.* 10.) Let us now¹ make, as it were, a fresh beginning, and consider the heterogeneous parts, taking those first which are the first in importance.² For in all animals, at least in all the perfect kinds,³ there are two parts more essential than the rest, namely the part which serves for the ingestion of food, and the part which serves for the discharge of its residue.³ For without food growth and even existence is impossible. Intervening again between these two parts there is invariably a third, in which is lodged the vital principle. As for plants, though they also are included by us among things that have life, yet are they without any part for the discharge of residue.⁴ For the food which they absorb from the ground is already concocted, and they give off as its equivalent their seeds and fruits. Plants, again, inasmuch as they are without locomotion, present no great variety in their heterogeneous parts. For, where the functions are but few, few also are the organs required to effect them.⁵ The configuration of plants is a matter then for separate consideration. Animals, however, that not only live but feel, present a much greater multiformity of parts, and this diversity is greater in some animals than in others, being most varied in those, to whose share has fallen not mere life but life of high degree. Now such an animal is man. For of all living beings with which we are acquainted man alone partakes of the divine, or at any rate partakes of it in a fuller measure than the rest.⁶ For this reason, then, and also because his external parts and their forms are more familiar to us than those of other animals, we must speak of man first; and this the more fitly, because in him alone do the natural parts hold their natural position; his upper part being turned towards that which is upper in the universe. For, of all animals, man alone stands erect.⁷

In man, then, the head is destitute of flesh; this being the necessary consequence of what has already been stated concerning the brain. There are, indeed, some⁸ who hold that the life of man would be longer than it is, were his head more abundantly

furnished with flesh; and they account for the absence of this substance by saying that it is intended to add to the perfection of sensation. For the brain they assert to be the organ of sensation; and sensation, they say, cannot penetrate to parts that are too thickly covered with flesh. But neither part of this statement is true. On the contrary, were the region of the brain thickly covered with flesh, the very purpose for which animals are provided with a brain would be directly contravened. For the brain would itself be heated to excess and so quite unable to cool any other part; and, as to the latter half of their statement, the brain cannot be the cause of any of the sensations, seeing that it is itself as utterly without feeling as any one of the excretions. These writers see that certain of the senses are located in the head, and are unable to discern any reason for this; they see also that the brain is the most peculiar of all the animal organs; and out of these facts they form an argument, by which they link sensation and brain together.⁹ It has however already been clearly set forth in the treatise on Sensation, that it is the region of the heart that constitutes the sensory centre. There also it was stated that two of the senses, namely touch and taste, are manifestly directly dependent on the heart;¹⁰ and that as regards the other three, namely hearing, sight, and the centrally placed sense of smell, it is the character of their sense-organs¹¹ which causes them to be lodged as a rule in the head. Vision is so placed in all animals.¹² But such is not invariably the case with hearing or with smell. For fishes and the like hear and smell, and yet have no visible organs for these senses in the head;¹³ a fact which demonstrates the accuracy of the opinion here maintained. Now that vision, whenever it exists, should be in the neighbourhood of the brain is but what one would rationally expect. For the brain is fluid¹⁴ and cold, and vision is of the character of water,¹⁵ water being of all transparent substances the one most easily confined.¹⁶ Moreover it cannot but necessarily be that the more precise senses will have their precision rendered still greater if ministered to by parts that have the purest¹⁷ blood. For the motion of the heat of blood destroys sensory activity. For these reasons the organs of the precise senses are lodged in the head.

It is not only the fore part of the head that is destitute of flesh, but the hind part also. For, in all animals that have a

head, it is this head which more than any other part requires to be held up. But, were the head heavily laden with flesh, this would be impossible; for nothing so burdened can be held upright. This fact is an additional proof that the absence of flesh from the head has no reference to brain sensation. For there is no brain in the hinder part of the head,¹⁸ and yet this is as much without flesh as is the front.

In some animals hearing as well as vision is lodged in the region of the head. Nor is this without a rational explanation. For what is called the empty space is full of air, and the organ of hearing is, as we say, formed of air. Now there are channels¹⁹ which lead from the eyes to the blood-vessels that surround the brain; and similarly there is a channel which leads back again from each ear and connects it with the hinder part of the head. But no part that is without blood is endowed with sensation, as neither is the blood itself, but only certain parts formed of blood.²⁰ So that none of the bloodless parts of sanguineous animals are endowed with sensation, nor their blood itself; for no such part is sensitive in any animal.

The brain in all animals that have one is placed in the front part of the head; because the direction in which sensation acts is in front; and because the heart, from which sensation proceeds, is in the front part of the body; and lastly because the instruments of sensation are the blood-containing parts, and the cavity in the posterior part of the skull is destitute of blood-vessels.

As to the position of the sense-organs, they have been arranged by nature in the following suitable manner. The organs of hearing are so placed as to divide the circumference of the head into two equal halves; for they have to hear not only sounds which are directly in a line with themselves, but sounds from all quarters. The organs of vision are placed in front, because sight is exercised only in a straight line, and moving as we do in a forward direction it is necessary that we should see before us, in the direction of our motion. Lastly, the organs of smell are placed with good reason between the eyes. For as the body consists of two parts, a right half and a left, so also each organ of sense is double. In the case of touch this is not apparent, the reason being that the primary organ of this sense is not the flesh or analogous part, but lies internally.²¹ In the case of taste, which is merely a modification of touch²² and which

is placed in the tongue, the fact is more apparent than in the case of touch, but still not so manifest as in the case of the other senses. However even in taste the fact is evident enough; for in some animals the tongue is plainly forked.²³ The double character of the sensations is however much more conspicuous in the other organs of sense. For there are two ears and two eyes, and the nostrils, though combined together, are also two. Were these latter otherwise disposed and separated from each other as are the ears, neither they nor the nose in which they are placed would be able to perform their office. For in such animals as have nostrils olfaction is effected by means of inspiration,²⁴ and the organ of inspiration is placed in front and in the middle line. This is the reason why nature has brought the two nostrils together and placed them as the central of the three sense-organs, setting them symmetrically on either side of a perpendicular line, where they benefit by the inspiratory motion.²⁵ In other animals than man the arrangement of these sense-organs is also such as is adapted in each case to the special requirements. (*Ch.* 11.) For instance in quadrupeds the ears project from the head and are set to all appearance above the eyes. Not that they are in reality above the eyes; but they seem to be so, because the animal does not stand upright, but has its head hung downwards. This being the usual attitude of the animal when in motion, it is of advantage that its ears shall be high up and very moveable.¹ For they can then be turned in all directions, and readily take in sounds from all quarters. (*Ch.* 12.) In birds, on the other hand, there are no ears, but only the auditory passages.¹ This is because their skin is hard and because they have feathers instead of hairs, so that they have not got the proper material for the formation of ears.² Exactly the same is the case with such oviparous quadrupeds as are clad with scaly plates, and the same explanation applies to them. There is also one of the viviparous quadrupeds, namely the seal, that has no ears but only the auditory passages.³ The explanation of this is that the seal though a quadruped is a quadruped of stunted formation.⁴

(*Ch.* 13.) Men, and Birds, and Quadrupeds, viviparous and oviparous alike, have their eyes protected by lids. In the vivipara there are two of these; and both are used by these animals not only in closing the eye, but also in the act of blinking; whereas

the oviparous quadrupeds, and¹ the heavy-bodied birds as well as some others, use only the lower lid to close the eye;² while birds blink by means of a membrane connected with the canthus. The reason for the eyes being thus protected is that nature has made them of fluid consistency, in order to ensure keenness of vision. For had they been covered with hard skin, they would, it is true, have been less liable to get injured by anything falling into them from without, but they would not have been sharp-sighted. It is then to ensure keenness of vision that the skin over the pupil is fine and delicate; while the lids are superadded as a protection from injury. It is as a still further safeguard, that all these animals blink, and man most of all; this action (which is not performed from deliberate intention but from a natural instinct)³ serving to keep objects from falling into the eyes; and being more frequent in man than in the rest of these animals, because of the greater delicacy of his skin. These lids are made of a roll of skin; and it is because they are made of skin and contain no flesh that neither they, nor the similarly constructed prepuce, unite again when once cut.⁴

As to the oviparous quadrupeds, and such birds as resemble them in closing the eye with the lower lid, it is the hardness of the skin of their heads which makes them do so.⁵ For such birds as have heavy bodies are not made for flight; and so the materials which would otherwise have gone to increase the size of the feathers are diverted thence, and used to augment the thickness of the skin.⁶ Birds therefore of this kind close the eye with the lower lid; whereas pigeons and the like use both upper and lower lids for the purpose. As birds are covered with feathers, so oviparous quadrupeds are covered with scaly plates; and these in all their forms are harder than hairs, so that the skin⁷ also to which they belong is harder than the skin of hairy animals. In these animals, then, the skin on the head is hard, and so does not allow of the formation of an upper eyelid,⁸ whereas lower down the integument is of a flesh-like character, so that the lower lid can be thin and extensible.

The act of blinking is performed by the heavy-bodied birds⁹ by means of the membrane already mentioned, and not by this lower lid. For in blinking rapid motion is required, and such is the motion of this membrane, whereas that of the lower lid is slow. It is from the inner canthus, that is from the one nearest

to the nostrils, that the membrane comes. For it is better to have one starting-point for nictitation than two; and in these birds this starting-point is the junction of eye and nostrils, an anterior starting-point being preferable to a lateral one.¹⁰ Oviparous quadrupeds do not blink in like manner as the birds; for, living as they do on the ground, they are free from the necessity of having eyes of fluid consistency and of keen sight, whereas these are essential requisites for such birds as have to use their eyes at long distances. This too explains why birds with talons, that have to search for prey by eye from aloft, and therefore soar to greater heights than other birds, are sharp-sighted; while common fowls and the like, that live on the ground and are not made for flight, have no such keenness of vision. For there is nothing in their mode of life which imperatively requires it.

Fishes and Insects and the hard-skinned Crustacea present certain differences in their eyes, but so far resemble each other as that none of them have eyelids.¹¹ As for the hard-skinned Crustacea it is utterly out of the question that they should have any; for an eyelid, to be of use, requires the action of the skin to be rapid. These animals then have no eyelids and, in default of this protection, their eyes are hard, just as though the lid were attached to the surface of the eye, and the animal saw through it. Inasmuch however as such hardness must necessarily blunt the sharpness of vision, nature has endowed the eyes of Insects, and still more those of Crustacea,¹² with great mobility (just as she has given some quadrupeds very moveable ears), in order that they may be able to turn to the light and catch its rays, and so see more plainly. Fishes however have eyes of a fluid consistency. For all animals that move much about have to use their vision at considerable distances. If now they live on land, the air in which they move is transparent enough. But the water in which fishes live is a hindrance to sharp sight, though it has this advantage over the air, that it does not contain so many objects to knock against the eyes. The risk of collision being thus small, nature, who makes nothing in vain, has given no eyelids to fishes, while to counterbalance the opacity of the water she has made their eyes of fluid consistency.

(*Ch.* 14.) All animals that have hairs on the body have lashes on the eyelids; but birds and animals with scale-like plates, being

hairless, have none. The Libyan ostrich, indeed, forms an exception; for, though a bird, it is furnished with eyelashes.¹ This exception, however, will be explained hereafter. Of hairy animals, man alone has lashes on both lids. For in quadrupeds there is a greater abundance of hair on the back than on the under side of the body; whereas in man the contrary is the case, and the hair is more abundant on the front surface than on the back. The reason for this is that hair is intended to serve as a protection to its possessor. Now, in quadrupeds, owing to their inclined attitude, the under or anterior surface does not require so much protection as the back, and is therefore left bald, in spite of its being the nobler of the two sides. But in man, owing to his upright attitude, the anterior and posterior surfaces of the body are on an equality as regards need of protection. Nature therefore has assigned the protective covering to the nobler of the two surfaces;² for invariably she brings about the best arrangement of such as are possible. This then is the reason that there is no lower eyelash in any quadruped; though in some a few scattered hairs sprout out under the lower lid.³ This also is the reason that they never have hair in the axillæ, nor on the pubes, as man has. Their hair, then, instead of being collected in these parts, is either thickly set over the whole dorsal surface, as is the case for instance in dogs, or, sometimes, forms a mane, as in horses and the like, or as in the male lion, where the mane is still more flowing and ample. So, again, whenever there is a tail of any length, nature decks it with hair, with long hair if the stem of the tail be short, as in horses, with short hair if the stem be long, regard also being had to the condition of the rest of the body.⁴ For nature invariably distributes her material, by subtracting from one part and giving to another. Thus when she has covered the general surface of an animal's body with an excess of hair, she leaves a deficiency in the region of the back. This for instance is the case with bears.⁵

No animal has so much hair on the head as man. This in the first place is the necessary result of the fluid character of his brain, and of the presence of so many sutures in his skull. For wherever there is the most fluid and the most heat, there also must necessarily occur the greatest outgrowth.⁶ But, secondly, the thickness of the hair in this part has a final cause, being intended to protect the head, by preserving it from

excess of either heat or cold. And as the brain of man is larger and more fluid than that of any other animal, it requires a proportionately greater amount of protection. For the more fluid a substance is, the more readily does it get excessively heated or excessively chilled, while substances of an opposite character are less liable to such injurious affections.

This however is a digression, into which we have been led by the close connection of hair and eyelashes; the causes to which these latter owe their existence being the real matter in hand. We must therefore put off all further details concerning the hair till the proper occasions arrive and then return to the full consideration of the subject.⁷

(*Ch.* 15.) Both eyebrows and eyelashes exist for the protection of the eyes; the former that they may shelter them, like the eaves of a house, from any fluids that trickle down the head;¹ the latter to act like the palisades which are sometimes placed in front of enclosures, and to keep out any objects which might otherwise get in. The brows are placed over the junction of two bones,² which is the reason that in old age they often become so bushy³ as to require cutting. The lashes are set at the terminations of small blood-vessels. For the vessels come to an end where the skin itself terminates; and, in all places where these endings occur, the exudation of moisture of a corporeal character necessitates the growth of hairs,⁴ unless there be some operation of nature which interferes, by diverting the moisture to another purpose.⁵

(*Ch.* 16.) In the generality of viviparous quadrupeds, there is no great variety in the forms of the organ of smell. In those of them however whose jaws project forwards and taper to a narrow end, so as to form what is called a snout, the nostrils are placed in this projection, there being no other available plan; while, in the rest, there is a more definite demarcation between nostrils and jaws. But in no animal is this part so peculiar as in the elephant, where it attains an extraordinary size and strength. For the elephant uses its nostril as a hand; this being the instrument with which it conveys food, fluid and solid alike, to its mouth. With it, too, it tears up trees,¹ coiling it round their trunks. In fact it applies it generally to the purposes of a hand. For the elephant has the double character of a land animal, and of one that lives in swamps. Seeing then that it

has to get food in the water,² and yet must necessarily breathe, inasmuch as it is a land animal and has blood; seeing, also, that its excessive weight prevents it from passing rapidly from water to land, as some other sanguineous vivipara that breathe can do, it becomes necessary that it shall be suited alike for life in the water and for life on dry land. Just then as divers are sometimes provided with instruments for respiration, through which they can draw air from above the water, and thus may remain for a long time under the sea,³ so also have elephants been furnished by nature with their lengthened nostril; and, whenever they have to traverse the water, they lift this up above the surface and breathe through it. For the elephant's proboscis, as already said, is a nostril. Now it would have been impossible for this nostril to have the form of a proboscis, had it been hard and incapable of bending. For its very length would then have prevented the animal from supplying itself with food, being as great an impediment as the horns of certain oxen, that are said⁴ to be obliged to walk backwards while they are grazing. It is therefore soft and flexible, and, being such, is made, in addition to its own proper functions, to serve the office of the fore-feet; nature in this following her wonted plan of using one and the same part for several purposes.⁵ For in polydactylous quadrupeds the fore-feet are not intended merely to support the body, but also to serve as hands. But in elephants, though they must be reckoned polydactylous, as their foot has neither cloven nor solid hoof,⁶ the fore-feet, owing to the great size and weight of the body, are reduced to the condition of mere supports; and indeed their slow motion and⁷ unfitness for bending make them useless for any other purpose. A nostril, then, is given to the elephant for respiration, as to every other animal that has a lung, and is lengthened out and endowed with its power of coiling, because the animal has to remain for considerable periods of time in the water, and is unable to pass thence to dry ground with any rapidity. But as the feet are shorn of their full office, this same part is also, as before said, made by nature to supply their place, and give such help as otherwise would be rendered by them.

As to other sanguineous animals, the Birds, the Serpents and the oviparous quadrupeds, in all of them there are the nostril-holes, placed in front of the mouth; but in none are there any

distinctly formed nostrils, nothing in fact which can be called nostrils except from a functional point of view. A bird at any rate has nothing which can properly be called a nose. For its so-called beak is a substitute for jaws. The reason for this is to be found in the natural conformation of birds. For they are winged bipeds; and this makes it necessary that their head and neck shall be of light weight; just as it makes it necessary that their breast-bone shall be narrowed. The beak therefore with which they are provided is formed of a bone-like substance, in order that it may serve as a weapon as well as for nutritive purposes, but is made of narrow dimensions to suit the small size of the head. In this beak are placed the olfactory passages. But there are no nostrils; for such could not possibly be placed there.

As for those animals that have no respiration,⁸ it has already been explained why it is that they are without nostrils, and perceive odours either through gills, or through a blow-hole,⁹ or, if they are insects, by the hypozoma;¹⁰ and how their power of smelling depends, like their motions, upon the innate spirit¹¹ of their bodies, which in all of them is implanted by nature and not introduced from without.

Under the nostrils are the lips, in such sanguineous animals, that is, as have teeth. For in birds, as already has been said, the purposes of nutrition and defence are fulfilled by a bone-like beak, which forms a compound substitute for teeth and lips. For supposing that one were to cut off a man's lips, unite his upper teeth together, and similarly his under ones, and then were to lengthen out the two separate pieces thus formed, flattening them on either side and making them project forwards, supposing I say this to be done, we should at once have a bird-like beak.

The use of the lips in all animals except man is to preserve and guard the teeth; and thus it is that the distinctness with which the lips are formed is in direct proportion to the degree of nicety and perfection with which the teeth are fashioned. In man the lips are soft and flesh-like and capable of separating from each other. Their purpose, as in other animals, is to guard the teeth, but they are more especially intended to serve a higher office, contributing in common with other parts to man's faculty of speech. For just as nature has made man's tongue unlike

that of other animals, and, in accordance with what I have said is her not uncommon practice,¹² has used it for two distinct operations, namely for the perception of savours and for speech, so also has she acted with regard to the lips, and made them serve both for speech and for the protection of the teeth. For vocal speech consists of combinations of the letters, and most of these it would be impossible to pronounce, were the lips not moist, nor the tongue such as it is. For some letters are formed by closures of the lips and others by applications of the tongue. But the differences of these movements, their nature and their number, are questions, the discussion of which belongs to the writers on articulation. It was necessary however for us to follow up the several parts in question at once to the function now assigned to them, and to show that they are of a character well suited for its performance. Therefore it is then that they are made of flesh. And the flesh of man is softer than that of any other animal, the reason¹³ for this being, that of all animals man has the most delicate sense of touch.¹⁴

(*Ch.* 17.) The tongue is placed under the vaulted roof of the mouth. In land animals it presents but little diversity. But in other¹ animals it is variable, and this whether we compare them as a class with such as live on land, or compare their several species with each other. It is in man that the tongue attains its greatest degree of freedom, of softness, and of breadth; the object of this being to render it suitable for its double function. For its softness fits it for the perception of savours, a sense which is more delicate in man than in any other animal, softness being most impressionable by touch, of which sense taste is but a variety.² This same softness again, together with its breadth, adapts it for the articulation of letters and for speech. For these qualities, combined with its freedom from attachment, are those which suit it best for advancing and retiring in every direction. That this is so is plain, if we consider the case of those who are tongue-tied in however slight a degree. For their speech is indistinct and lisping; that is to say they have not got the full power of uttering letters. In being broad is comprised the possibility of becoming narrow; for in the great the small is included, but not the great in the small.

What has been said explains why, even among birds, those that are most capable of pronouncing letters are such as have

the broadest tongues;³ and why the viviparous and sanguineous quadrupeds, where the tongue is hard and thick and not free in its motions, have a very limited vocal articulation. Some birds have a considerable variety of notes. These are the smaller kinds.⁴ But it is the birds with talons that have the broader tongues and consequently the greater aptitude for speech. All birds use their tongues to communicate with each other. But some do this in a greater degree than the rest; so that in some cases it even seems as though actual instruction were imparted from one to another by its agency.⁵ These however are matters which have already been discussed in the *Researches concerning Animals*.⁶

As to those oviparous and sanguineous animals that live not in the air but on the earth, their tongue in most cases is tied down and hard, and is therefore useless for vocal purposes; in the serpents however and in the lizards it is long and forked, so as to be suited for the perception of savours.⁷ So long indeed is this part in serpents, that though small while in the mouth it can be protruded to a great distance. In these same animals it is forked and has a fine and hair-like extremity, because of their extreme liking for dainty food. For by this arrangement they derive a two-fold pleasure from savours, their gustatory sensation being as it were doubled.

Even some bloodless animals have an organ that serves for the perception of savours; and in sanguineous animals such an organ is invariably present. For even in such of these as would seem to an ordinary observer to have nothing of the kind, some of the fishes for example, there is a kind of shabby representative of a tongue,⁸ much like what exists in river crocodiles. In most of these cases the apparent absence of the part can be rationally explained on some or other ground. For in the first place the interior of the mouth in animals of this character is invariably spinous. Secondly in water animals there is but short space of time for the perception of savours, and as the use of this sense is thus of short duration, shortened also is the separate part which subserves it. The reason for their food being so rapidly transmitted to the stomach is that they cannot possibly spend any time in sucking out the juices; for were they to attempt to do so, the water would make its way in during the process. Unless therefore one pulls their mouth very widely open indeed,

the projection of this part is quite invisible. The region exposed by thus opening the mouth is spinous; for it is formed by the close apposition of the gills, which are of a spinous character.

In crocodiles the immobility of the lower jaw also contributes in some measure to stunt the development of the tongue. For the crocodile's tongue is adherent to the lower jaw;⁹ and its upper and lower jaws are as it were inverted, for in other animals it is the upper jaw which is the immoveable one. The tongue, then, of this animal does not adhere to the upper jaw, because that would interfere with the ingestion of food, but adheres to the lower jaw, because this is, as it were, the upper one which has changed its place.¹⁰ Moreover it is the crocodile's lot though a land animal to live the life of a fish, and this again necessarily involves an indistinct formation of the part in question.¹¹

The roof of the mouth resembles flesh, even in many of the fishes; and in some of the river species, as for instance in the fishes known as Cyprini,¹² is so very flesh-like and soft as to be taken by careless observers for a tongue. The tongue of fishes, however, though it exists as a separate part, is never distinctly visible like this, as has been already explained. Again, as the gustatory sensibility is intended to serve animals in the selection of food, it is not diffused equally over the whole surface of the tongue-like organ, but is placed chiefly in the tip;¹³ and for this reason it is the tip which is the only part of the tongue separated in fishes from the rest of the mouth. As all animals are sensible to the pleasure derivable from food, they all feel a desire for it. For the object of desire is the pleasant. The part however in which food produces the sensation is not precisely alike in all of them, but while in some it is free from attachments, in others, where it is not required for vocal purposes, it is united with the base of the mouth; in some again it is hard, in others soft or flesh-like. Thus even the Crustacea,¹⁴ the Carabi for instance and the like, and the Cephalopods, such as the Sepias and the Poulps, have some such part inside the mouth. As for the Insects, some of them have the part which serves as tongue inside the mouth, as is the case with ants, and as is also the case with many Testacea, while in others it is placed externally. In this latter case it resembles a sting or piercer, and is hollow and spongy, so as to serve at one and the same time for

the tasting and for the sucking up of nutriment. This is plainly to be seen in flies and bees and all such animals, and likewise in some of the Testacea. In the *Purpuræ*,¹⁵ for instance, so strong is this part that it enables them to bore holes through the hard covering of shell-fish, of the spiral snails, for example, that are used as bait to catch them. So also gad-flies and cattle-flies¹⁶ can pierce through the skin of man, and some of them even through the skins of other animals. Such, then, in these animals is the nature of the tongue, which is thus as it were the counterpart of the elephant's nostril. For as in the elephant the nostril is used as a weapon, so in these animals the tongue serves as a piercer or sting.¹⁷

In all other animals the tongue agrees with the description already given.

BOOK III.

(*Ch.* 1.) We have next to consider the teeth, and with these the mouth, that is the cavity which they enclose and form. The teeth have one invariable office, namely the reduction of the food ; but besides this general function they have other special ones, and these differ in different groups. Thus in some animals the teeth serve as weapons ; but this with a distinction. For there are offensive weapons and there are defensive weapons ; and while in some animals, as the wild carnivora, the teeth answer both purposes, in many others, both wild and domesticated, they serve only for defence.¹ In man the teeth are admirably constructed for their general function, the front ones being sharp, so as to cut the food into bits, and the hinder ones broad and flat, so as to grind it to a pulp ; while between these and separating them are the dog-teeth, which, in accordance with the rule that the mean partakes of both extremes, share in the characters of those on either side, being broad in one part but sharp in another.² Similar distinctions of shape are presented by the teeth of other animals, with the exception of those whose teeth are one and all of the sharp kind. In man, however, the number and the character of the teeth have been mainly determined by the requirements of speech. For the front teeth of man contribute in many ways to the formation of letter-sounds.

In some animals, however, the teeth, as already said, serve merely for the reduction of food. When, besides this, they serve as offensive and defensive weapons, they may either be formed into tusks, as for instance is the case in swine, or may be sharp-pointed and interlock with those of the opposite jaw, in which

case the animal is said to be saw-toothed. The explanation of this latter arrangement is as follows. The strength of such an animal is in its teeth, and the efficacy of these depends on their sharpness. In order, then, to prevent their getting blunted by mutual friction, such of them as serve for weapons fit into each other's interspaces, and are so kept in proper condition. No animal that is saw-toothed is at the same time furnished with tusks.³ For nature never makes anything superfluous or in vain. She gives, therefore, tusks to such animals as strike in fighting, and serrated teeth to such as bite. Sows, for instance, have no tusks, and accordingly sows bite instead of striking.

A general principle must here be noted, which will be found applicable not only in this instance but in many others that will occur later on. Nature allots each weapon, offensive or defensive, to those animals alone that can use it; or, if not to them alone, to them in a more marked degree; and she allots it in its most perfect state to those that can use it best; and this whether it be a sting, or a spur, or horns, or tusks, or what it may of a like kind.

Thus as males are stronger and more choleric than females, it is in males alone that such parts as those just mentioned are found, or at any rate it is in males that they are found in the highest degree of perfection.⁴ For though females are of course provided with such parts as are no less necessary to them than to males, the parts for instance which subserve nutrition, they have even these in an inferior degree,⁵ and the parts which answer no such necessary purpose they do not possess at all. This explains why stags have horns, while does have none; why the horns of cows are different from those of bulls, and, similarly, the horns of ewes from those of rams. It explains also why the females are often without spurs in species where the males are provided with them, and accounts for similar facts relating to other such parts.⁶

All fishes have teeth of the serrated form, with the single exception of the fish known as the *Scarus*.⁷ In many of them there are teeth even on the tongue and on the roof of the mouth.⁸ The reason for this is that, living as they do in the water, they cannot but allow this fluid to pass into the mouth with the food. The fluid, thus admitted, they must necessarily discharge again without delay. For were they not to do so, but to retain it for

a time while triturating the food, the water would run into their digestive cavities. Their teeth therefore are all of the sharp kind, so as to serve for comminution of their food, and at the same time are numerous, and set in many parts, that by their very abundance, in the absence of any grinding faculty, they may mince the food into small bits.⁹ Their teeth are also curved, because these are almost the only weapons which they possess.¹⁰

In all these offices of the teeth the mouth also takes its part ; but besides these functions it is subservient to respiration, in all such animals as breathe and are cooled by external agency. For nature, as already said,¹¹ uses the parts which are common to all animals for many special purposes, and this of her own accord.¹² Thus the mouth has one universal function in all animals alike, namely its alimentary office ; but in some, besides this, the special duty of serving as a weapon is attached to it ; in others that of ministering to speech ; and again in many, though not in all, the office of respiration. All these functions are thrown by nature upon one single organ, the construction of which she varies so as to suit the variations of office. Therefore it is that in some animals the mouth is contracted, while in others it is of wide dimensions. The contracted form belongs to such animals as use the mouth merely for nutritive, respiratory, and vocal purposes ; whereas in such as use it as a means of defence it has a wide gape. This is its invariable form in such animals as are saw-toothed. For seeing that their mode of warfare consists in biting, it is advantageous to them that their mouth shall have a wide opening ; for the wider it opens, the greater will be the extent of the bite, and the more numerous will be the teeth called into play.

What has just been said applies to fishes as well as to other animals ; and thus in such of them as are carnivorous, and made for biting, the mouth has a wide gape ; whereas in the rest it is small, being placed at the extremity of a tapering snout. For this form is suited for their purposes, while the other would be useless.

In birds the mouth consists of what is called the beak, which in them is a substitute for lips and teeth. This beak presents variations in harmony with the functions and protective purposes which it serves. Thus in those birds that are called Crooked-clawed¹³ it is invariably hooked, inasmuch as these birds are carnivorous, and eat no kind of vegetable food whatsoever. For

this form renders it serviceable to them in obtaining the mastery over their prey, and is better suited for deeds of violence than any other. Moreover, as their weapons of offence consist of this beak and of their claws, these latter also are more crooked in them than in the generality of birds. Similarly in each other kind of bird the beak is suited to the mode of life. Thus, in woodpeckers¹⁴ it is hard and strong, as also in crows and birds of crowlike habit, while in the smaller birds it is delicate, so as to be of use in collecting seeds and picking up minute animals. In such birds, again, as eat herbage, and such as live on the edges of marshes—those, for example, that swim and have webbed feet—the bill is broad, or adapted in some other way to the mode of life. For a broad bill enables a bird to dig into the ground with ease, just as, among quadrupeds, does the broad snout of the pig, an animal which, like the birds in question, lives on roots. Moreover, in these root-eating birds and in some others of like habits of life, the tips of the bill end in hard points, which gives them additional facility in dealing with herbaceous food.

The several parts which are set on the head have now, pretty nearly all, been considered. In man, however, the part which lies between the head and the neck is called the face, this name (in Greek *prosōpon*) being, it would seem, derived from the function of the part. For as man is the only animal that stands upright, he is also the only one that looks directly in front (in Greek *prosō*); and the only one whose voice is emitted in that direction.¹⁵

(*Ch. 2.*) We have now to treat of horns; for these also, when present, are appendages of the head. They exist in none but viviparous animals; though in some ovipara certain parts are metaphorically spoken of as horns, in virtue of a certain resemblance.¹ To none of such parts however does the proper office of a horn belong; for they are never used, as are the horns of vivipara, for purposes which require strength, whether it be in self-protection or in offensive strife. So also no polydactylous animal² is furnished with horns. For horns are weapons of strife, and these polydactylous animals possess other means of security. For to some of them nature has given claws, to others teeth suited for combat, and to the rest some or other adequate defensive appliance. There are horns, however, in most of the cloven-hoofed animals, and in some³ of those that have a solid hoof, serving them as an offensive weapon, and in some cases

also as a defensive one. There are horns also in all animals that have not been provided by nature with some other means of security ; such means for instance as speed, which has been given to horses ; or excessive bulk of body, which is sufficient in itself to protect an animal from being destroyed by others and which has been given to camels, and in a still greater measure to elephants. Other animals again are protected by the possession of tusks ; and among these are the swine, though they have a cloven hoof.⁴

All animals again whose horns are but useless appendages have been provided by nature with some additional means of security. Thus deer are endowed with speed ; for the large size and great branching of their horns makes these a source of detriment rather than of profit to their possessors.⁵ Similarly endowed are the Bubalus⁶ and gazelle ;⁷ for though these animals will stand up against some enemies and defend themselves with their horns, yet they run away from such as are fierce and valiant. The Bonasus again, whose horns curve inwards towards each other [and are therefore of no use as weapons], is provided with a means of protection in the discharge of its excrement ; and of this it avails itself when frightened. There are some other animals besides the Bonasus that have a similar mode of defence.⁸ In no case however does nature ever give more than one adequate means of protection to one and the same animal.⁹

Most of the animals that have horns are cloven-hoofed ; but the Indian ass, as they call it, is also reported to be horned, though its hoof is solid.¹⁰

Again as the body, so far as regards its organs of motion,¹¹ consists of two distinct parts, the right and the left, so also and for like reasons the horns of animals are, in the great majority of cases, two in number. Still there are some that have but a single horn ; the Oryx¹² for instance, and the so-called Indian ass ; in the former of which the hoof is cloven, while in the latter it is solid. In such animals the horn is set in the centre of the head ; for as the middle belongs equally to both extremes, this arrangement is the one that comes nearest to each side having its own horn.

Again, it would appear consistent with reason that the single horn should go with the solid rather than with the cloven hoof. For hoof, as also claw, is of the same nature as horn ; so that the two naturally undergo division simultaneously and in the

same animals. Again, since the division of the cloven hoof depends on deficiency of material, it is but rationally consistent, that nature when she gave an animal an excess of material for the hoofs, which thus became solid, should have taken away something from the upper parts and so made the animal to have but one horn.¹³

Rightly too did she act when she chose the head whereon to set the horns. And *Æsop's Momus*¹⁴ is beside the mark, when he finds fault with the bull for not having its horns upon its shoulders. For from this position, says he, they would have delivered their blow with the greatest force, whereas on the head they occupy the weakest part of the whole body. Momus was but dull-sighted in making this hostile criticism. For had the horns been set on the shoulders, or had they been set on any other part than they are, the encumbrance of their weight would have been increased, not only without any compensating gain whatsoever, but with the disadvantage of impeding many bodily operations. For the point whence the blows could be delivered with the greatest force was not the only matter to be considered, but the point also whence they could be delivered with the widest range. But as the bull has no hands and cannot possibly have its horns on its feet or on its knees, where they would prevent flexion, there remains no other site for them but the head; and this therefore they necessarily occupy. In this position, moreover, they are much less in the way of the movements of the body than they would be elsewhere.

Deer are the only animals in which the horns are solid throughout, and are also the only animals that cast them.¹⁵ This casting is not simply advantageous to the deer from the increased lightness which it produces, but, seeing how heavy the horns are, is a matter of actual necessity.

In all other animals the horns are hollow for a certain distance, and the end alone is solid, this being the part of use in a blow. At the same time, to prevent even the hollow part from being weak, the horn does not grow out of the skin, but has a solid piece from the bones fitted into its cavity. For this arrangement is not only that which makes the horns of the greatest service in fighting, but that which causes them to be as little of an impediment as possible in the other actions of life.

Such then are the reasons for which horns exist; and such the reasons why they are present in some animals, absent from others.

Let us now consider how the necessary results of the material nature are made available by rational nature for a final cause.¹⁶

In the first place, then, the larger the bulk of animals, the greater is the amount of corporeal and earthy matter which they contain. Thus no very small animal is known to have horns, the smallest horned animal that we are acquainted with being the gazelle.¹⁷ But in all our speculations concerning nature, what we have to consider is the general rule; for that is natural which applies either universally or generally.¹⁸ And thus when we say that the largest animals have most earthy matter, we say so because such is the general rule. Now this earthy matter is used in the animal body to form bone. But in the larger animals there is an excess of it, and this excess is turned by nature to useful account, being converted into weapons of defence. Part of it necessarily flows to the upper portion of the body, and this is allotted by her in some cases to the formation of tusks and teeth, in others to the formation of horns. Thus it is that no animal that has horns has also front teeth in both jaws, those in the upper jaw being deficient.¹⁹ For nature by subtracting from the teeth adds to the horns; the nutriment which in most animals goes to the former being here spent on the augmentation of the latter. Does, it is true, have no horns and yet are equally deficient with the males as regards the teeth. The reason, however, for this is that they, as much as the males, are naturally horn-bearing animals; but they have been stripped of their horns, because these would not only be useless to them but actually baneful;²⁰ whereas the greater strength of the males causes these organs, though equally useless, to be much less of an impediment. In other animals, where this material is not secreted from the body in the shape of horns, it is used to increase the size of the teeth; in some cases of all the teeth, in others merely of the tusks, which thus become so long as to resemble horns projecting from the jaws.²¹

(*Ch. 3.*) So much, then, of the parts which appertain to the head. Below the head lies the neck, in such animals as have one. This is the case only with those that have the parts to which a neck is subservient. These parts are the larynx¹ and what is called the œsophagus. Of these the former, or larynx, exists for the sake of respiration, being the instrument by which such animals as breathe inhale and discharge the air. Therefore it is that, when there is no lung, there is also no neck. Of this

condition the Fishes are an example. The other part, or œsophagus, is the channel through which food is conveyed to the stomach; so that all animals that are without a neck are also without a distinct œsophagus.² Such a part is in fact not required of necessity for nutritive purposes; for it has no action whatsoever on the food. Indeed there is nothing to prevent the stomach from being placed directly after the mouth. This however is quite impossible in the case of the lung. For there must be some sort of tube common to the two divisions of the lung, by which the breath may be apportioned to their respective bronchi, and thence pass into the air pipes; and such an arrangement will be the best for giving perfection to inspiration and respiration. The organ then concerned in respiration must of necessity be of some length; and this again necessitates there being an œsophagus to unite mouth and stomach.³ This œsophagus is of a flesh-like character, and yet admits of extension like a sinew.⁴ This latter property is given to it, that it may stretch when food is introduced; while the flesh-like character is intended to make it soft and yielding, and to prevent it from being rasped by particles as they pass downwards, and so suffering damage. On the other hand the windpipe and the so-called larynx are constructed out of a cartilaginous substance. For they have to serve not only for respiration, but also for vocal purposes; and an instrument that is to produce sounds must necessarily be not only smooth but firm. The windpipe lies in front of the œsophagus, although this position causes it to be some hindrance to the latter in the act of deglutition.⁵ For if a morsel of food, fluid or solid, slips into it by accident, choking and difficult breathing and violent cough ensue. This must be a matter of astonishment to any of those who assert that it is by the windpipe that an animal imbibes fluid.⁶ For the consequences just mentioned occur invariably, whenever a particle of food slips in, and are quite obvious. Indeed on many grounds it is ridiculous to say that this is the channel through which animals imbibe fluid. For there is no passage leading from the lung to the stomach, such as the œsophagus which we see leading thither from the mouth. Moreover, when any cause produces sickness and vomiting, it is plain enough whence the fluid is discharged. It is manifest also that fluid, when swallowed, does not pass directly into the bladder

and collect there, but goes first into the stomach. For, when red wine is taken, the dejections of the stomach are seen to be coloured by its dregs ; and such discoloration has been even seen on many occasions inside the stomach itself, in cases where there have been wounds opening into that organ. However it is perhaps silly to be minutely particular in dealing with silly statements such as this.

The windpipe then, owing to its position in front of the œsophagus, is exposed, as we have said, to annoyance from the food. To obviate this, however, nature has contrived the epiglottis. This part is not found in all viviparous animals, but only in such of them as have a lung ; nor in all of these, but only in such as at the same time have their skin covered with hairs, and not either with scaly plates or with feathers. In such scaly and feathered animals there is no epiglottis, but its office is supplied by the larynx,⁷ which closes and opens, just as in the other case the epiglottis falls down and rises up ; rising up during the ingress or egress of breath, and falling down during the ingestion of food, so as to prevent any particle from slipping into the trachea. Should there be the slightest want of accuracy in this movement, or should an inspiration be made during the ingestion of food, choking and coughing ensue, as already has been noticed. So admirably contrived, however, is the movement both of the epiglottis and of the tongue, that, while the food is being ground to a pulp in the mouth, the tongue very rarely gets caught between the teeth ; and, while the food is passing over the epiglottis, seldom does a particle of it slip into the windpipe.

The animals which have been mentioned as having no epiglottis owe this deficiency to the dryness of their flesh and to the hardness of their skin. For an epiglottis made of such materials would not admit of easy motion.⁸ In fact it would take a longer time to shut down an epiglottis made of the peculiar flesh of these animals, and shaped like that of those with hairy skins, than to bring the edges of the trachea itself into contact with each other.

Thus much then as to the reasons why some animals have an epiglottis while others have none, and thus much also as to its use. It is a contrivance of nature to remedy the vicious position of the windpipe in front of the œsophagus. That position is the result of necessity. For it is in the front and centre of the body that

the heart is situated, in which we say is the principle of life and the source of all motion and sensation. (For sensation and motion are exercised in the direction which we term forwards, and it is on this very relation that the distinction of before and behind is founded.) But where the heart is, there and surrounding it is the lung. Now inspiration, which occurs for the sake of the lung and for the sake of the principle which has its seat in the heart, is effected through the windpipe. Since then the heart must of necessity lie in the very front place of all, it follows that the larynx also and the windpipe must of necessity lie in front of the œsophagus. For they lead to the lung and heart,⁹ whereas the œsophagus leads to the stomach. And it is an universal law that, as regards above and below, front and back, right and left, the nobler and more honourable part invariably is placed uppermost, in front, and on the right, rather than in the opposite positions, unless some more important object stands in the way.¹⁰

(*Ch.* 4.) We have now dealt with the neck, the œsophagus, and the windpipe, and have next to treat of the viscera. These are peculiar to sanguineous animals, some of which have all of them, others only a part, while no bloodless animals have any at all.¹ Democritus then seems to have been mistaken in the notion he formed of the viscera, if he fancied that the reason why none were discoverable in bloodless animals was that these animals were too small to allow them to be seen. For, in sanguineous animals, the heart and liver are visible enough when the body is only just formed, and while it is still extremely small. For these parts are to be seen in the egg sometimes as early as the third day, being then no bigger than a point;² and are visible also in aborted³ embryos, while still excessively minute. Moreover, as the external organs are not precisely alike in all animals, but each creature is provided with such as are suited to its special mode of life and motion, so is it with the internal parts, these also differing in different animals. Viscera, then, are peculiar to sanguineous animals; and this accords with the fact that the viscera are each and all formed from sanguineous material, as is plainly to be seen in the new-born young of these animals. For in such the viscera are more sanguineous, and of greater bulk in proportion to the body, than at any later period of life, because the nature of the material and its abundance are most distinctly marked in objects at the period of their first formation.⁴ There is a heart, then, in

all sanguineous animals, and the reason for this has already been given. For that sanguineous animals must necessarily have blood is self-evident. And, as the blood is fluid, it is also a matter of necessity that there shall be a receptacle for it ; and it is clearly to meet this requirement that nature has devised the blood-vessels. These, again, must necessarily have one primary source. For it is preferable that there shall be one such, when possible, rather than several. This primary source of the vessels is the heart.⁵ For the vessels clearly are *from* it, and not *through* it.⁶ Moreover, being as it is homogeneous, it has the character of a blood-vessel. Again its position is that of a primary or dominating part. For nature, when no other more important purpose stands in her way, places the more honourable part in the more honourable position ;⁷ and the heart lies about the centre of the body, but rather in its upper than its lower half, and also more in front than behind. This is most evident in the case of man, but even in other animals there is a tendency in the heart to assume a similar position, in the centre of the necessary part of the body, that is to say of the part which terminates in the vent for excrement. For the limbs vary in position in different animals, and are not to be counted with the parts which are necessary for life. For life can be maintained even when they are removed ; while it is self-evident that the addition of them to an animal is not destructive of it.

There are some⁸ who say that the vessels commence in the head. In this they are clearly mistaken. For in the first place, according to their representation, there would be many sources for the vessels, and these scattered ; and secondly these sources would be in a region that is manifestly cold, as is shown by its intolerance of chill, whereas the region of the heart is as manifestly hot.⁹ Again, as already said, the vessels extend through all the other viscera, but no vessel extends through the heart.¹⁰ From this it is quite evident that the heart is a part of the vessels and their origin ; and for this it is well suited by its structure. For its central¹¹ part consists of a thick and hollow body, and is moreover full of blood, as though the vessels took thence their origin. It is hollow to serve for the reception of the blood, while its wall is thick, that it may serve to protect the source of heat. For here, and here alone in all the viscera and in fact in all the body, there is blood without blood-vessels, the blood elsewhere being always contained within vessels.¹² Nor is this but con-

sistent with reason. For the blood is conveyed into the vessels from the heart, but none passes into the heart from without. For in itself it constitutes the origin and fountain, or primary receptacle, of the blood. It is, however, from dissections and from observations on the process of development that the truth of these statements receives its clearest demonstration. For the heart is the first of all the parts to be formed; and no sooner is it formed than it contains blood. Moreover the motions of pain and pleasure,¹³ and generally of all sensation, plainly start from the heart, and find in it their ultimate termination. This, indeed, reason would lead us to expect. For the starting-point must, whenever possible, be one; and, of all places, the best suited for a starting-point is the centre. For the centre is one, and is equally or almost equally within reach of every part. Again, as neither the blood itself, nor yet any part which is bloodless, is endowed with sensation, it is plain that that part which first has blood, and which holds it as it were in a receptacle, must be the primary source of sensation.¹⁴ And that this part is the heart is not only a rational inference, but is also evident to the senses. For no sooner is the embryo formed, than its heart is seen in motion like a living creature, and this before any of the other parts, as though it constituted the starting-point of life in all animals that have blood. A further evidence of the truth of what has been stated is the fact that no sanguineous animal is without a heart. For the primary source of blood must of necessity be present in them all. It is true that sanguineous animals not only have a heart but also invariably have a liver. But no one could ever deem the liver to be the primary organ either of the whole body or of the blood.¹⁵ For the position in which it is placed is far from being that of a primary or dominating part; and moreover in the most perfectly finished animals there is another part, the spleen, which as it were counterbalances it.¹⁶ Still further, the liver contains no spacious receptacle in its substance, as does the heart, but has its blood in a vessel like all the other viscera. The vessel moreover extends through it, and no vessel whatsoever originates in it; for it is from the heart that all the vessels take their rise. Since then one or other of these two parts must be the central source, and since it is not the liver which is such, it follows of necessity that it is the heart which is the source of the blood, as also the primary

organ in other respects. For the definitive characteristic of an animal is the possession of sensation; and the first sensory part is that which first has blood; that is to say is the heart, which is the source of blood and the first of the parts to contain it.

The apex of the heart is pointed and more solid than the rest of the organ. It lies against the breast, and entirely in the anterior part of the body, in order to prevent that region from getting chilled. For in all animals there is comparatively but little flesh over the breast, whereas there is a more abundant covering of that substance on the posterior surface, so that the heat has in the back a sufficient amount of protection. In all animals but man the heart is placed in the centre of the pectoral region; but in man¹⁷ it inclines a little towards the left, so that it may counterbalance the chilliness of that side. For the left side is colder in man, as compared with the right, than in any other animal.¹⁸ It has already been stated that even in fishes the heart holds the same position as in other animals; and the reasons have been given why it appears not to do so. The apex of the heart, it is true, is in them turned towards the head, but this in fishes is the front aspect, for it is the direction in which their motion occurs.¹⁹

The heart again is abundantly supplied with sinews,²⁰ as might reasonably be expected. For the motions of the body commence from the heart, and are brought about by traction and relaxation. The heart therefore, which, as already said, is as it were a living creature inside its possessor, requires some such subservient and strengthening parts.

In no animals does the heart contain a bone, certainly in none of those that we have ourselves inspected,²¹ with the exception of the horse and a certain kind of ox. In these exceptional cases the heart, owing to its large bulk, is provided with a bone as a support; just as the bones serve as supports for the body generally.²²

In animals of great size the heart²³ has three cavities; in smaller animals it has two; and in all has at least one. The reason for this, as already stated, is that there must be some place in the heart to serve as a receptacle for the first blood; which, as has been mentioned more than once, is formed in this organ. But inasmuch as the main blood-vessels are two in number,²⁴ namely the so-called great vessel and the aorta, each of

which is the origin of other vessels; inasmuch, moreover, as these two vessels present differences, hereafter to be discussed, when compared with each other, it is of advantage that they also shall themselves have distinct origins. This advantage will be obtained if each side have its own blood, and the blood of one side be kept separate from that of the other.²⁵ For this reason the heart, whenever it is possible, has two receptacles. And this possibility exists in the case of large animals, for in them the heart, as the body generally, is of large size. Again it is still better that there shall be three cavities, so that the middle and odd one may serve as a common origin for both sides.²⁶ But this requires the heart to be of greater magnitude, so that it is only in the largest animals that the heart has three cavities.

Of these three cavities it is the right which has the most abundant and the hottest blood,²⁷ and this explains why the limbs on the right side of the body are also warmer than those on the left.²⁸ The left cavity has the least blood of all, and the coldest; while in the middle cavity the blood, as regards quantity and heat, is intermediate to the other two, being however of purer quality than either. For it behoves the supreme part to be as tranquil as possible, and this tranquillity can be ensured by the blood being pure, and of moderate amount and warmth.²⁹

In the heart of animals there is also a kind of joint-like division, something like the sutures of the skull.³⁰ This is not however attributable to the heart being formed by the union of several parts into a compound whole, but is rather, as already said, the result of a joint-like division. These jointings are most distinct in animals of keen sensibility, and less so in those that are of duller feeling, in swine for instance. Different hearts differ also from each other in their sizes, and in their degrees of firmness; and these differences somehow extend their influence to the temperaments of the animals. For in animals of low sensibility the heart is hard and solid, while it is softer in such as are endowed with keener feeling. So also when the heart is of large size the animal is timorous, while it is more courageous if the organ be smaller and of moderate bulk. For in the former the bodily affection which results from terror already pre-exists;³¹ for the bulk of the heart is out of all proportion to the animal's heat, which being small is reduced to insignificance in the large space, and thus the blood is made colder than it would otherwise be.

The heart is of large size³² in the hare, the deer, the mouse, the hyæna, the ass, the leopard, the marten, and in pretty nearly all other animals that are either manifestly timorous, or that betray their cowardice by their spitefulness.

What has been said of the heart as a whole is no less true of its cavities and of the blood-vessels; these also if of large size being cold.³³ For just as a fire of equal size gives less heat in a large room than in a small one, so also does the heat in a large cavity or a large blood-vessel, that is in a large receptacle, have less influence than in a small one. The more spacious, moreover, these cavities and vessels are, the greater is the amount of spirit³⁴ which they contain, and the greater its effect, for all hot substances are cooled by the motion of external objects.³⁵ Thus it is that no animal that has large cavities in its heart, or large blood-vessels, is ever fat, the vessels being indistinct and the cavities small in all or most fat animals.³⁶

The heart again is the only one of the viscera, and indeed the only part of the body, that is unable to tolerate any serious affection.³⁷ This is but what might reasonably be expected. For, if the primary or dominant part be diseased, there is nothing from which the other parts which depend upon it can derive succour. A proof, that the heart is thus unable to tolerate any morbid affection, is furnished by the fact that never has it been seen diseased in a sacrificial victim, though this is not the case with the other viscera. For the kidneys are frequently found to be full of stones, and growths, and small abscesses, as also is the liver, the lung, and more than all the spleen.³⁸ There are also many other affections which are seen to occur in these parts, those which are least liable to such being the portion of the lung which is close to the windpipe, and the portion of the liver which lies about the junction with the great blood-vessel. This again admits of a rational explanation. For it is in these parts that the lung and liver are most closely in communion with the heart. On the other hand when animals die not by sacrifice but from disease, they are found on dissection to have morbid affections of the heart.

(*Ch.* 5.) Thus much of the heart, its nature, and the end and cause of its existence in such animals as have it. In due sequence we have next to discuss the blood-vessels, that is to say the great vessel and the aorta. For it is into these two¹ that the blood first passes when it quits the heart; and all the other vessels

are but offshoots from them. Now that these vessels exist on account of the blood has already been stated. For every fluid requires a receptacle, and in the case of the blood the vessels are that receptacle. Let us now explain why these vessels are two, and why they spring from one single source, and extend throughout the whole body.

The reason, then, why these two vessels coalesce into one centre, and spring from one source, is that the sensory soul is in all animals actually one; (in sanguineous animals one not only actually but potentially, but in some bloodless animals one actually only and not potentially;²) and this one-ness of the sensory soul determines a corresponding one-ness of the part in which it primarily abides. Where, however, the sensory soul is lodged, there also and in the self-same place must necessarily be the source of heat;³ and, again, where this is there also must be the source of the blood, seeing that it thence derives its warmth and fluidity. Thus, then, in the one-ness of the part in which is lodged the prime source of sensation and of heat is involved the one-ness of the source in which the blood originates; and this, again, explains why the blood-vessels have one common starting-point.

The vessels, again, are two, because the body of every sanguineous animal that is capable of locomotion⁴ is bilateral; for in all such animals there is a distinguishable before and behind, a right and left, an above and below. Now as the front is more honourable and of higher supremacy than the hinder aspect, so also and in like degree is the great vessel superior to the aorta. For the great vessel is placed in front, while the aorta is behind; the former again is plainly visible in all sanguineous animals, while the latter is in some indistinct and in some not discernible at all.

Lastly, the reason for the vessels being distributed throughout the entire body is that in them, or in parts analogous to them, is contained the blood, or the fluid which in ex-sanguineous animals takes the place of blood, and that the blood or analogous fluid is the material from which the whole body is made. Now as to the manner in which animals are nourished,⁵ and as to the ways and means by which they absorb nutriment from the stomach, these are matters which will be more suitably considered and explained in the treatise on Generation and Development. But inasmuch as the parts are, as already said, formed out of the blood,

it is but rational that the flow of the blood should extend, as it does, throughout the whole of the body. For since each part is formed of blood, each must have blood through and in its substance.

To give an illustration of this. The water-courses⁶ in gardens are so constructed as to distribute water from one single source or fount into numerous channels, which divide and subdivide so as to convey it to all parts; and, again, in house-building stones are thrown down along the whole ground-plan of the foundation walls; because the garden-plants in the one case grow at the expense of the water, and the foundation walls in the other are built out of the stones. Now just after the same fashion has nature laid down channels for the conveyance of the blood throughout the whole body, because this blood is the material out of which the whole fabric is made. This fact becomes very evident in bodies that have undergone great emaciation. For in such there is nothing to be seen but the blood-vessels; just as when fig-leaves or vine-leaves or the like have dried up, there is nothing left of them but their vessels. The explanation of these facts is that the blood, or fluid which takes its place, is potentially body and flesh, or substance analogous to flesh. Now just as in irrigation the larger dykes are permanent, while the smaller ones are soon filled up with mud and disappear, again to become visible when the deposit of mud ceases; so also do the largest blood-vessels remain permanently open, while the smallest ones are converted actually into flesh, though potentially they are no whit less vessels than before.⁷ This too explains why, so long as the flesh of an animal is in its integrity, blood will flow from any part of it whatsoever that is cut, though no vessel, however small, be visible in it. Yet there can be no blood, unless there be a blood-vessel. The vessels then are there, but are invisible owing to their being clogged up, just as the dykes for irrigation are invisible until they have been cleared of mud.

As the blood-vessels advance, they become gradually smaller and smaller, until at last their tubes are too fine to admit the thick blood. This fluid can therefore no longer find its way through them, though they still give passage to the moisture which we call sweat; and especially so when the body is heated, and the mouths of the small vessels are dilated. Instances, indeed, are not unknown of persons who in consequence of a cachectic state have secreted sweat that resembled blood,⁸ their

body having become loose and flabby, and their blood watery, owing to the heat in the small vessels⁹ having been too scanty for its concoction. For, as was before said, every mixture of earth and water—and both nutriment and blood are such—becomes thicker from concoction.¹⁰ The inability of the heat to effect concoction may be due either to its being absolutely small in amount, or to its being small in proportion to the quantity of food, when this has been taken in excess. This excess again may be of two kinds, either quantitative or qualitative; for all substances are not equally amenable to concoction.

The larger the channels, the more easily does the blood flow through them. Thus it is that when hæmorrhages occur from the nostrils and gums and fundament, and sometimes also from the mouth, they are of a passive kind, and not violent as are those from the windpipe.¹¹

The great vessel and the aorta, which above lie somewhat apart, lower down exchange positions, and by so doing give compactness to the body. For when they reach the point where the legs diverge, they each split into two, and the great vessel passes from the front to the rear, and the aorta from the rear to the front. By this they contribute to the unity of the whole fabric. For as in plaited work the parts hold more firmly together because of the interweaving, so also by the interchange of position between the blood-vessels are the anterior and posterior parts of the body more closely knit together. A similar exchange of position occurs also in the upper part of the body, between the vessels that have issued from the heart.¹² The details however of the mutual relations of the different vessels must be looked for in the treatises on Anatomy and the Researches concerning Animals.

So much, then, as concerns the heart and the blood-vessels. We must now pass on to the other viscera and apply the same method of enquiry to them.

(*Ch.* 6.) The lung,¹ then, is an organ found in all the animals of a certain class, because they live on land. For there must of necessity be some or other means of cooling down the heat of the body; and in sanguineous animals, as they are of an especially hot nature, the cooling agency must be an external one; whereas in the bloodless kinds the innate spirit is sufficient of itself for the purpose.² The external cooling agent must be

either air or water. In fishes the agent is water. Fishes therefore never have a lung, but have gills in its place, as was stated in the treatise on Respiration. But animals that breathe are cooled by air. These therefore are all provided with a lung.³

All land animals breathe, and even some water animals, such as the whale,⁴ the dolphin, and all the spouting Cetacea. For many animals lie halfway⁵ between terrestrial and aquatic; some that are terrestrial and that inspire air being nevertheless of such a bodily constitution that they abide for the most time in the water; and some that are aquatic partaking so largely of the land character, that respiration constitutes for them the main condition of life.

The organ of respiration is the lung. This derives its motion from the heart; but it is its own large size and spongy texture that affords amplitude of space for the entrance of the breath. For when the lung rises up the breath streams in, and is again expelled when the lung contracts.⁶ It has been stated, but incorrectly, that it is to the lung that the beating of the heart is due. That this is not so is shown by the phenomenon of palpitation occurring, so to speak, in man alone; inasmuch as man alone is influenced by hope and expectation.⁷ Again in most animals the heart is at a distance from the lung and placed above it;⁸ so that its beating can in no degree be brought about by this latter.

The lung differs much in different animals. For in some it is of large size and contains blood; while in others it is smaller and of spongy texture. In the vivipara it is large and rich in blood, because⁹ of their natural heat; while in the ovipara it is small and dry but capable of expanding to a vast extent when inflated. Among terrestrial animals, the oviparous quadrupeds, such as lizards, tortoises, and the like, have this kind of lung; and, among inhabitants of the air, the animals known as birds.¹⁰ For in all these the lung is spongy, and like foam. For it is bladdery and collapses from a large bulk to a small one, as does foam when it runs together. In this too lies the explanation of the fact that these animals are little liable to thirst and¹¹ drink but sparingly, and that they are able to remain for a considerable time under water.¹² For, inasmuch as they have but little heat,¹³ the very motion of the lung, airlike and void, suffices by itself to cool them for a considerable period.¹⁴

These animals, speaking generally, are also distinguished from others by their smaller bulk.¹⁵ For heat promotes growth, and abundance of blood is a sure indication of heat. Heat again tends to make the body erect; and thus it is that man is the most erect of animals, and the vivipara more erect than other quadrupeds.¹⁶ For no viviparous animal, be it apodous¹⁷ or be it possessed of feet, is so given to creep into holes as are the ovipara.

The lung then exists for respiration; and this is its universal office. But the amount of blood it contains, and its structure generally, varies with the requirements of different groups of animals.¹⁸ There is however no one term to denote all animals that have a lung, no designation, that is, like the term Bird, applicable to the whole of a certain class. Yet the possession of a lung is a part of their essence, just as much as the presence of certain characters constitutes the essence of a bird.

(*Ch. 7.*) Of the viscera some appear to be single, as the heart and lung; others to be double, as the kidneys; while of a third kind it is doubtful in which class they should be reckoned. For the liver and the spleen would seem to lie half-way between the single and the double organs. For they may be regarded either as constituting each a single organ, or as a pair of organs resembling each other in character.¹

In reality however all the organs are double.² The reason for this is that the body itself is double, consisting of two halves, which are however combined together under one supreme centre. For there is an upper and a lower half, a front and a rear, a right side and a left.

This explains why it is that even the brain and the several organs of sense tend in all animals to consist of two parts; and the same explanation applies to the heart with its cavities. The lung again in ovipara is divided to such an extent, that these animals look as though they had actually two lungs.³ As to the kidneys no one can overlook their double character. But when we come to the liver and the spleen, any one might fairly be in doubt. The reason of this is, that, in animals that necessarily have a spleen,⁴ this organ is such that it might be taken for a kind of bastard liver; while in those, in which a spleen is not an actual necessity but is merely present, as it were, by way of token, in an extremely minute form, the liver plainly consists of two parts; of which the larger tends to lie

on the right side and the smaller on the left. Not but what there are some ovipara in which this condition is comparatively indistinctly marked; while on the other hand there are some vivipara in which the liver is manifestly divided into two parts.⁵ Examples of such division are furnished by the hares of certain regions that have the appearance of having two livers, and by some fishes, especially the cartilaginous kinds.⁶

It is the position of the liver on the right side of the body that is the main cause for the formation of the spleen; the existence of which thus becomes to a certain extent a matter of necessity in all animals, though not of very stringent necessity.

The reason, then, why the viscera are bilateral is, as we have said, that there are two sides to the body, a right and a left.⁷ For each of these sides aims at similarity with the other, and so likewise do their several viscera; and as the sides, though dual, are knit together into unity, so also do the viscera tend to be bilateral and yet one by unity of constitution.

Those viscera which lie below the diaphragm exist one and all on account of the blood-vessels;⁸ serving as a bond, by which these vessels, while floating freely, are yet held in connection with the body. For the vessels give off branches which run to the body through the out-stretched structures,⁹ like so many anchor-lines thrown out from a ship. The great vessel sends such branches to the liver, the spleen, and the kidneys; and these viscera—the liver and spleen on either side and the kidneys behind—attach the great vessel¹⁰ immoveably to the body. The aorta sends similar branches to the kidneys, but none to the liver nor to the spleen.¹¹

These viscera, then, contribute in this manner to the compactness of the animal body. The liver and spleen assist moreover in the concoction of the food; for both are rich in blood,¹² and therefore of a hot character. The kidneys on the other hand take part in the separation of the excretion which flows into the bladder.¹³

The heart then and the liver are essential constituents of every animal; the liver that it may effect concoction, the heart that it may lodge the central source of heat. For some or other part there must be which like a hearth shall hold the kindling fire; and this part must be well protected, seeing that it is, as it were, the citadel of the body.

All sanguineous animals, then, need these two parts; and this explains why these two viscera, and these two alone, are invariably found in them all. In such of them, however, as breathe, there is also as invariably a third, namely the lung. The spleen on the other hand is not invariably present; and, in those animals that have it, is only present of necessity in the same sense as the excretions of the belly and of the bladder are necessary, in the sense that is of being an inevitable concomitant. Therefore it is that in some animals the spleen is but scantily developed as regards size. This for instance is the case in such feathered animals as have a hot stomach. Such are the pigeon, the hawk, and the kite.¹⁴ It is the case also in oviparous quadrupeds, where the spleen is excessively minute, and in many of the scaly fishes. These same animals are also without a bladder, because the loose texture of their flesh allows the residual fluid to pass through and to be converted into feathers and scales. For the spleen attracts the residual humours from the stomach, and owing to its blood-like character is enabled to assist in their concoction.¹⁵ Should however this residual fluid be too abundant, or the heat of the spleen be too scanty, the body becomes sickly from over-repletion with food. Often too, when the spleen is affected by disease, the belly becomes hard¹⁶ owing to the reflux into it of the fluid; just as happens to those who form too much urine, for they also are liable to a similar diversion of the fluids into the belly. But in those animals that have but little superfluous fluid to excrete, such as birds and fishes, the spleen is never large, and in some exists no more than by way of token. So also in the oviparous quadrupeds it is small, compact, and like a kidney. For their lung is spongy,¹⁷ and they drink but little, and such superfluous fluid as they have is diverted to the formation of feathers.

On the other hand in such animals as have a bladder, and whose lung contains blood, the spleen is watery, both for the reason already mentioned, and also because the left side of the body is more watery and colder than the right. For each of two contraries has been so placed as to go together with that which is akin to it in another pair of contraries. Thus right and left, hot and cold, are pairs of contraries; and right is conjoined with hot, after the manner described, and left with cold.

The kidneys when they are present exist not of actual necessity, but as matters of greater finish and perfection. For by their

special character they are suited to serve in the excretion of the fluid which collects in the bladder. In animals therefore where this fluid is very abundantly formed, their presence enables the bladder to perform its proper office with greater perfection.¹⁸

Since then both kidneys and bladder exist in animals for one and the same function, we must next treat of the bladder, though in so doing we disregard the due order of succession¹⁹ in which the parts should be enumerated. For not a word has yet been said of the midriff, which, though not one of the viscera, is yet one of the parts that environ them, and has to be considered with them.

(*Ch.* 8.) It is not every animal that has a bladder; those only being apparently intended by nature to have one, whose lung contains blood.¹ To such it was but reasonable that she should give this part. For the character of their lung with its abundant blood causes them to be the thirstiest of animals, and makes them require a more than ordinary quantity not merely of solid but also of liquid nutriment. This increased consumption necessarily entails the production of an increased amount of residue; which thus becomes too abundant to be concocted by the stomach and excreted with its own residual matter. The residual fluid must therefore of necessity have a receptacle of its own; and thus it comes to pass that all animals whose lung contains blood are provided with a bladder. Those animals on the other hand that are without a lung of this character, and that either drink but sparingly owing to their lung being of a spongy texture, or that never imbibe fluid at all for drinking's sake but only as nutriment, insects for instance and fishes, and that are moreover clad with feathers or scales or scaly plates²—all these animals, owing to the small amount of fluid which they imbibe, and owing also to such residue as there may be being converted into feathers and the like, are invariably without a bladder.³ The Tortoises,⁴ which are comprised among animals with scaly plates, form the only exception; and this is merely due to the imperfect development of their natural conformation; the explanation of the matter being that in the sea-tortoises the lung is flesh-like and contains blood, resembling the lung of the ox, and that in the land-tortoises it is of larger size in comparison with the bulk of the body than in other animals of the same class.⁵ Moreover, inasmuch

as the covering which invests them is dense and shell-like, so that the moisture cannot exhale through the porous flesh, as it does in birds and in snakes and other animals with scaly plates, such an amount of secretion is formed that some special part is required to receive and hold it. This then is the reason why these animals, alone of their kind, have a bladder, the sea-tortoises a large one, the land-tortoises an extremely small one.⁶

(*Ch. 9.*) What has been said of the bladder is equally true of the kidneys. For these also are wanting in all animals that are clad with feathers or with scales or with plates; the sea and land tortoises¹ forming the only exception. In some of the birds, however, there are flattened kidney-like bodies, as though the flesh allotted to the formation of the kidneys, unable to find one single place of sufficient size, had been distributed over several regions.²

The Emys³ has neither bladder nor kidneys. For the softness of its shell allows of the ready transpiration of fluid; and for this reason neither of the organs mentioned exists in this animal. All other animals, however, whose lung contains blood are, as before said, provided with kidneys. For nature uses these organs for two separate purposes, namely for the excretion of the residual fluid, and to subserve the blood-vessels,⁴ a channel leading to them from the great vessel.

In the centre of the kidney is a cavity of variable size. This is the case in all animals, excepting the seal.⁵ The kidneys of this animal are more solid than those of any other, and in form resemble the kidneys of the ox. The human kidneys are of similar shape; being as it were made up of numerous small kidneys,⁶ and not presenting one unbroken surface like the kidneys of sheep and other quadrupeds.⁷ For this reason, should the kidneys of a man be once attacked by disease, the malady is not easily expelled. For it is as though many kidneys were diseased and not merely one; which naturally enhances the difficulties of a cure.

The duct which runs to the kidney from the great vessel does not terminate in the central cavity, but is expended in the substance of the organ, so that there is no blood in the cavity, nor is any coagulum found there after death. A pair of stout ducts, void of blood, run, one from the cavity of each kidney, to the bladder; and other ducts, strong and continuous, lead into the kidneys from the aorta.⁸ The purpose of this arrangement is to

allow the superfluous fluid to pass from the blood-vessel into the kidney, and the resulting renal excretion to collect, by the percolation of the fluid through the solid substance of the organ, in its centre, where as a general rule there is a cavity. (This by the way explains why the kidney is the most ill-savoured of all the viscera.) From the central cavity the fluid is discharged into the bladder by the ducts just mentioned, having already assumed in great degree the character of residuum.⁹ The bladder is as it were moored¹⁰ to the kidneys; for, as already has been mentioned, it is attached to them by strong ducts. These then are the purposes for which the kidneys exist, and such the functions of these organs.

In all animals that have kidneys, that on the right is placed higher than that on the left.¹¹ For, inasmuch as motion commences from the right, the organs on this side become stronger than those on the left, and must all push forward in advance of their opposite fellows; as may be seen in the fact that men even raise the right eyebrow more than the left, and that the former is more arched than the latter.¹² The right kidney being thus drawn upwards is brought into contact with the liver; for the liver in all animals lies on the right side.

Of all the viscera the kidneys are those that have the most fat.¹³ This is in the first place the result of necessity, because the kidneys are the parts through which the residual matters percolate. For the blood which is left behind after this excretion, being of pure quality, is of easy concoction, and the final result of thorough blood-concoction is lard and suet. For just as a certain amount of heat is left in the ashes of solid substances, such as wood, after combustion, so also does a remnant of the heat that has been developed remain in fluids after concoction; and this is the reason why oily matter is light, and floats on the surface of other fluids.¹⁴ The fat is not formed in the kidneys themselves, the density of their substance forbidding this, but is deposited on their external surface. It consists of lard or of suet, according as the animal's fat is of the former or latter character. The difference between these two kinds of fat has already been set forth in other passages.¹⁵ The formation, then, of fat in the kidneys is the result of necessity; being, as explained, a consequence of the necessary conditions which accompany the possession of such organs. But at the same time the fat has a final cause, namely to ensure

the safety of the kidneys, and to maintain their natural heat. For placed, as these organs are, close to the surface, they require a greater supply of heat than other parts. For while the back is thickly covered with flesh, so as to form a shield for the heart and neighbouring viscera, the loins, in accordance with a rule that applies to all joints, are destitute of flesh;¹⁶ and fat is therefore formed as a substitute for it, so that the kidneys may not be without protection. The kidneys, moreover, by being fat are the better enabled to secrete and concoct their fluid; for fat is hot, and it is heat that effects concoction.

Such then are the reasons why the kidneys are fat. But in all animals the right kidney is less fat than its fellow.¹⁷ The reason for this is, that the parts on the right side are naturally more solid and more suited for motion than those on the left. But motion is antagonistic to fat, for it tends to melt it.

Animals then, as a general rule, derive advantage from their kidneys being fat; and the fat is often very abundant and extends over the whole of these organs. But, should the like occur in the sheep, death ensues. Be its kidneys however as fat as they may, they are never so fat but that some part, if not in both at any rate in the right one, is left free.¹⁸ The reason why sheep are the only animals that suffer in this manner, or suffer more than others, is that their fat is harder and more abundant than that of other animals. For the soft lard, of which the fat of some animals is composed, is of fluid consistence, so that there is not the same chance in their case of wind getting shut in and causing mischief. But it is to such an enclosure of wind that rot¹⁹ is due. And thus even in men, though it is beneficial to them to have fat kidneys, yet should these organs become over-fat and diseased, deadly pains ensue. As to those animals whose fat consists of suet, their suet is not so dense as that of sheep, neither is it nearly so abundant; for of all animals there is none in which the kidneys become so soon gorged with fat as in the sheep.²⁰ Rot then is produced by the moisture and the wind getting shut up in the kidneys, and is a malady that carries off sheep with great rapidity. For the disease forthwith reaches the heart, passing thither by the aorta and the great vessel, the ducts which connect these with the kidneys being of unbroken continuity.

(*Ch.* 10.) We have now dealt with the heart and the lung,

as also with the liver, spleen, and kidneys. The latter are separated from the former by the midriff or, as some call it, the *Phrenes*. This divides off the heart and lung, and as already said is called *Phrenes* in sanguineous animals, all¹ of which have a midriff, just as they all have a heart and a liver. For they require a midriff to divide the region of the heart from the region of the stomach; so that the centre wherein abides the sensory soul may be undisturbed, and not be overwhelmed, directly food is taken, by its up-steaming vapour² and by the abundance of heat then superinduced.³ For it was to guard against this that nature made a division, constructing the midriff as a kind of partition-wall and fence, and so separated the nobler from the less noble parts, in all cases where a separation of upper from lower was possible.⁴ For the upper part is the more⁵ honourable, and is that for the sake of which the rest exists; while the lower part exists for the sake of the upper and constitutes the necessary element in the body, inasmuch as it is the recipient of the food.

That part of the midriff which is near the ribs is somewhat fleshy and thick; but the central part has more of a membranous character, for this structure gives it strength and capability of extension. Now that the midriff is as it were a curtain or screen to prevent heat mounting up from below, is shown by what happens, should it, owing to its proximity to the stomach, attract thence the hot and residual fluid. For when this occurs there ensues forthwith a marked disturbance of intellect and of sensation. It is indeed because of this that the midriff is called⁶ *Phrenes*, as though it had some share in the process of thinking, for which the Greek term is *Phronein*. In reality however it has no part whatsoever itself in the matter, but, lying in close proximity to organs that have, it brings about the manifest changes of intelligence in question by acting upon them. This too explains why its central part is thin. For though this is in some measure the result of necessity, inasmuch as those portions of the fleshy whole which lie nearest to the ribs must necessarily be fleshier than the rest,⁷ yet besides this there is a final cause, namely to give it as small a proportion of moisture as possible; for, had it been made of flesh throughout, it would have been more likely to attract and hold a large amount of fluid. That rapid heating of it affects sensation in a notable manner is

shown by the phenomena of laughing. For when men are tickled they are quickly set a laughing, because the motion quickly reaches this region. And, even when the heating is more slowly applied, there is still a manifest affection and motion of the intellect in opposition to the will. That man alone is affected by tickling,⁸ is due firstly to the delicacy of his skin, and secondly to the fact that he is the only animal that laughs. For to be tickled is to be set in laughter, the laughter being produced by such a motion as mentioned of the region of the armpit.

It is said also that when men in battle are wounded anywhere near the midriff, they are seen to laugh, owing to the heat produced by the wound.⁹ This may possibly be the case. At any rate it is a statement made by much more credible persons than those who tell the story of the human head, how it speaks after it is cut off. For so some assert, and even call in Homer to support them, representing him as alluding to this when he wrote,¹⁰ "His head still speaking rolled into the dust," instead of "The head of the speaker." So fully was the possibility of such an occurrence accepted in Caria, that one of that country was actually brought to trial under the following circumstances. The priest of Zeus Hoplosmios¹¹ had been murdered; but as yet it had not been ascertained who was the assassin; when certain persons asserted that they had heard the murdered man's head, which had been severed from the body, repeat several times the words, "It was Cercidas that killed the man." Search was thereupon made and a man of those parts who bore the name of Cercidas hunted out and put upon his trial. But it is impossible that any one should utter a word when the windpipe is severed and no motion any longer derived from the lungs. Moreover among the Barbarians, where heads are chopped off with great rapidity, nothing of the kind has ever yet occurred. Why again does not the like occur in the case of other animals than man? For that none of them should laugh, when their midriff is wounded, is but what one would expect; for no animal but man ever laughs. So too there is nothing irrational in supposing that the trunk may run forwards to a certain distance after the head has been cut off; seeing that bloodless animals at any rate can live, and that for a considerable time, after decapitation, as has been set forth and explained in other passages.¹²

The purposes then for which the viscera severally exist have now been stated. It is of necessity upon the inner terminations of the vessels that they are developed; for moisture, and that of a bloody character, cannot but exude at these points, and it is of this, solidified and coagulated, that the substance of the viscera is formed.¹³ Thus they are of a bloody character, and in substance resemble each other while they differ from other parts.

(*Ch.* 11.) The viscera are enclosed each in a membrane. For they require some covering to protect them from injury, and require moreover that this covering shall be light. To such requirements membrane is well adapted; for it is close in texture so as to form a good protection, destitute of flesh so as neither to attract nor hold moisture, and thin so as to be light and not add to the weight of the body. Of the membranes those are the stoutest and strongest, which invest the heart and the brain;¹ as is but consistent with reason. For these are the parts which require most protection, seeing that they are the main governing powers of life,² and that it is to governing powers that guard is due.

(*Ch.* 12.) Some animals have all the viscera that have been enumerated; others have only some of them. In what kind of animals this latter is the case, and what is the explanation, has already been stated. Moreover the self-same viscera present differences in different possessors. For the heart is not precisely alike in all animals that have one; nor in fact is any viscus whatsoever. Thus the liver is in some animals split into several parts, while in others it is comparatively undivided.¹ Such differences in its form present themselves even among the viviparous quadrupeds, but are more marked in fishes and in the oviparous quadrupeds, and this whether we compare them with each other or with the vivipara. As for birds, their liver very nearly resembles that of the vivipara; for in them, as in these, it is of a pure and blood-like colour. The reason of this is that the body in both these classes of animals admits of the freest exhalation, so that the amount of foul residual matter within is but small. Hence it is that some of the vivipara are without any gall-bladder² at all. For the liver takes a large share in maintaining the purity of composition and the healthiness of the body. For these are conditions that depend finally and in the main upon the blood, and there is more blood in the liver than in any of the other viscera, the heart only excepted.³ On the other hand the liver

of oviparous quadrupeds and of fishes is, as a rule, of a pale yellow,⁴ and there are even some in which its colour is utterly foul,⁵ so as to match the foul composition of their bodies. Such, for instance, is the case in the toad, the tortoise, and other similar animals.

The spleen, again, varies in different animals. For in those that have horns and cloven hoofs, such as the goat, the sheep, and the like, it is of a rounded form;⁶ excepting when increased growth and size has caused some part of it to be lengthened out, as has happened in the case of the ox.⁷ In all polydactylous animals on the other hand it is elongated. Such for instance is the case in the pig,⁸ in man, and in the dog. While in animals with solid hoofs it is of a form intermediate to these two, being broad in one part, narrow in another. Such for example is its shape in the horse, the mule, and the ass.

(*Ch. 13.*) The viscera differ from the flesh not only in the turgid aspect of their substance, but also in position; for they lie within the body, whereas the flesh is placed on the outside. The explanation of this is that these parts partake of the character of blood-vessels, and that while the former exist for the sake of the vessels, the latter cannot exist without them.¹

(*Ch. 14.*) Below the mid-line of the body lies the stomach, placed at the end of the œsophagus, when there is one, and in immediate contiguity with the mouth, when the œsophagus is wanting.¹ Continuous with this stomach is what is called the gut. These parts are present in all animals, for reasons that are self-evident. For it is a matter of necessity that an animal shall receive the incoming food; and necessary also that it shall discharge the same when its goodness is exhausted. This residual matter, again, must not occupy the same place as the yet uncocted nutriment. For as the ingress of food and the discharge of the residue occur at distinct periods, so also must they necessarily occur in distinct places. Thus there must be one receptacle for the ingoing food and another for the useless residue, and between these, therefore, a part in which the change from one condition to the other may be effected. These however are matters which will be more suitably set forth, when we come to deal with Development and Nutrition.

We must now consider the variations presented by the stomach and its subsidiary parts. For neither in size nor in shape are

these parts uniformly alike in all animals. Thus the stomach is single in all such sanguineous and viviparous animals as have teeth in front of both jaws. It is single therefore in all the polydactylous kinds, such as man, dog, lion, and the rest; in all the solid-hoofed animals also, such as horse, mule, ass; and in all those which like the pig, though their hoof is cloven, yet have front teeth in both jaws.² When, however, an animal is of large size, and feeds on substances of so thorny and ligneous a character as to be difficult of concoction, it may in consequence have several stomachs, as for instance is the case with the camel. A similar multiplicity of stomachs exists also in the horned animals; the reason being that horn-bearing animals have no front teeth in the upper jaw. The camel also, though it has no horns, is yet without upper front teeth.³ The explanation of this is that it is more essential for the camel to have a multiple stomach than to have these teeth. Its stomach, then, is shaped like that of animals without upper front teeth, and, its dental arrangements being such as to match its stomach, the teeth in question are wanting.⁴ They would indeed be of no service. Its food, moreover, being of a thorny character, and its tongue necessarily made of a fleshy substance,⁵ nature uses the earthy matter which is saved from the teeth to give hardness to the palate.⁶ The camel ruminates like the horned animals, because its multiple stomach resembles theirs. For all animals that have horns, the sheep for instance, the ox, the goat, the deer, and the like, have several stomachs. For since the mouth, owing to its lack of teeth, fails to perform its due office as regards the food, this multiplicity of stomachs is intended to supply its place; the several cavities receiving the food one from the other in succession; the first taking the unreduced substances, the second the same when somewhat reduced, the third when reduction is complete, and the fourth when the whole has become a smooth pulp. Such is the reason why there is this multiplicity of parts and cavities in animals with such dentition. The names given to the several cavities are the paunch, the honey-comb, the manyplies and the reed. How these parts are related to each other, in position and in shape, must be looked for in the treatises on Anatomy and the Researches concerning Animals.⁷

Birds also present variations in the part which acts as a recipient

of the food; and the reason for these variations is the same as in the animals just mentioned. For here again it is because the mouth fails to perform its office and fails even more completely;—for birds have no teeth at all, nor any instrument whatsoever with which to comminute or grind down their food—it is, I say, because of this, that in some of them⁸ what is called the crop precedes the stomach and does the work of the mouth; while in others the œsophagus either is dilated throughout,⁹ or expands just before it enters the stomach, so as to form a preparatory store-house for the un-reduced food;¹⁰ or the stomach itself has a protuberance in some part,¹¹ or is strong and fleshy,¹² so as to be able to store up the food for a considerable period and to concoct it, in spite of its not having been ground into a pulp. For nature retrieves the inefficiency of the mouth by increasing the efficiency and heat of the stomach. Other birds there are, such, namely, as have long legs and live in marshes, that have none of these provisions, but merely an elongated œsophagus.¹³ The explanation of this is to be found in the moist character of their food. For all these birds feed on substances easy of reduction, and their food being moist and not requiring much concoction, their digestive cavities are of a corresponding character.¹⁴

Fish are provided with teeth, which are almost invariably of the serrated kind. For there is but one small section in which it is otherwise. Of these the fish called *Scarus* is an example. And this is¹⁵ probably the reason why this fish apparently ruminates, though no other fishes do so. For those horned animals that have no front teeth in the upper jaw also ruminate.

In all fishes the teeth are sharp;¹⁶ so that these animals can divide their food, though imperfectly. For it is impossible for a fish to linger or spend time in the act of mastication,¹⁷ and therefore they have no teeth that are flat or suitable for grinding; for such teeth would be to no purpose. The œsophagus again in some fishes is entirely wanting, and in the rest is but short.¹⁸ In order however to facilitate the concoction of the food, some of them, as the *Cestreus*, have a fleshy stomach resembling that of a bird;¹⁹ while most of them have numerous processes close against the stomach, to serve as a sort of antechamber in which the food may be stored up and undergo putrefaction²⁰ and concoction. There is a contrast between fishes and birds in the position of these processes. For in fishes they are placed

close to the stomach; while in birds, if present at all, they are lower down, near the end of the gut.²¹ Some of the vivipara also have processes connected with the lower part of the gut which serve the same purpose as that stated above.²²

The whole tribe of fishes is of gluttonous appetite, owing to the arrangements for the reduction of their food being very imperfect, and much of it consequently passing through them without undergoing concoction; and, of all, those are the most gluttonous that have a straight intestine. For as the passage of food in such cases is rapid, and the enjoyment derived from it in consequence but brief, it follows of necessity that the return of appetite is also speedy.²³

It has already been mentioned that in animals with front teeth in both jaws the stomach is of small size.²⁴ It may be classed pretty nearly always under one or other of two headings, namely as resembling the stomach of the dog, or as resembling the stomach of the pig. In the pig the stomach is larger than in the dog, and presents certain flat projections of moderate size, the purpose of which is to lengthen out the period of concoction; while the stomach of the dog is of small size, not much larger in calibre than the gut, and smooth on the internal surface.²⁵

Not much larger, I say, than the gut; for in all animals after the stomach comes the gut. This, like the stomach, presents numerous modifications. For in some animals it is uniform, when uncoiled, and alike throughout, while in others it differs in different portions. Thus in some cases it is wider in the neighbourhood of the stomach, and narrower at the other end. This is the case in dogs, and explains why they have to strain so much in discharging their excrement. While in other animals, and these the majority, it is the upper portion that is the narrower and the lower that is of greater width.²⁶

Of larger size than in any of these animals, and much convoluted, are the intestines of those that have horns.²⁷ The bulgings moreover both of their stomach and of their intestines are more prominent, in accordance with the larger bulk of their bodies generally. For the horned animals are, as a rule, of large bulk, because of the thorough elaboration which their food undergoes.²⁸ The gut, excepting in those animals where it runs in a straight line, invariably widens out as we get farther from the stomach and come to what is called the colon and to a kind of

cæcal dilatation. After this it again becomes narrower and convoluted.²⁹ Then succeeds a straight portion which runs right on to the vent. This vent is known as the fundament, and is in some animals surrounded by fat, in others not so. All these varying parts have been so contrived by nature as to harmonize with the various operations that concern the food and the residue. For, as the residue gets farther on and lower down, the space to contain it becomes ampler. This is suited to the wants of those animals that, owing either to their large size or to the heat of their digestive cavities,³⁰ require more nutriment and consume more fodder than the rest; for it allows the food to remain stationary and undergo conversion.

Neither is it without a purpose that, just as a narrower gut succeeds to the upper stomach, so also does the residual food, when its goodness is thoroughly exhausted, pass from the colon and the ample space of the lower stomach³¹ into a narrower channel and into the spiral coil. For so nature can regulate her expenditure and prevent the residual substances from being discharged all at once.³²

Now in all such animals as it behoves to be more temperate³³ in the consumption of food than those we have been considering the lower stomach presents no wide and roomy spaces, neither is their gut ever straight, but has numerous convolutions. For amplitude of space causes desire for ample food, and straightness of the intestine causes quick return of appetite.³⁴ And thus it is that all animals whose food receptacles are either simple or spacious are of gluttonous habits, the latter eating enormously at a meal, the former making many meals at short intervals.

Again, since the food in the upper stomach, having just been swallowed, must of necessity be quite fresh, while that which has reached the lower stomach must have had its juices exhausted and resemble dung, it follows of necessity that there must also be some intermediate part, in which the change may be effected, and where the food will be neither perfectly fresh nor yet dung. And thus it is that, in all such animals as we are now considering, there is found what is called the jejunum;³⁵ which is a part of the small gut, of the gut, that is, which comes next to the stomach. For this jejunum lies between the upper cavity which contains the yet unconcocted food and the lower cavity which holds the residual matter, which by the time it has got here is

quite worthless. . There is a jejunum in all animals, but it is only plainly visible in those of larger bulk, and in these only when they have abstained from food for a certain time. For thus alone can one hit on the exact period when the food lies half-way between the upper and lower cavities; a period which is very short, for the time occupied in the transition of food is but brief. In females this jejunum may occupy any part whatsoever of the upper intestine, but in males it comes just before the cæcum and the lower stomach.³⁶

(*Ch.* 15.) The substance called rennet¹ is found in all animals that have a multiple stomach, but only in the hare² among animals whose stomach is single. In the former the rennet neither occupies the large paunch, nor yet the reticulum, nor the terminal abomasus, but is found in the cavity which separates this terminal one from the two first, namely in the so-called psalterium.³ It is the thick character of their milk which causes all these animals to have rennet; whereas in animals with a single stomach the milk is thin, and consequently no rennet is formed. It is this difference in thickness⁴ which makes the milk of horned animals coagulate, while that of animals without horns does not. Rennet forms in the hare because it feeds on⁵ herbage that has juice like that of the fig. For juice of this kind coagulates the milk in the stomach of the young animal. Why it is in the psalterium that rennet is formed in animals with multiple stomachs has been stated in the Problems.⁶

BOOK IV.

(*Ch.* I.) The account which has now been given of the viscera,¹ the stomach, and the other several parts of the vivipara, holds equally good for the oviparous quadrupeds, and also for such apodous animals as the Serpents. These two classes of animals are indeed nearly akin, a serpent resembling a lizard which has been lengthened out and deprived of its feet. Fishes again resemble these two groups in all their parts, excepting that, while these, being land animals, have a lung, fishes have no lung, but gills in its place. Of all these animals, including the Fishes, none excepting the tortoise has an urinary bladder.² For, owing to the bloodlessness of their lung, they drink but little; and such moisture as they have is converted into scales, as in birds it is converted into feathers;³ and so they come to have the same white matter on the surface of their excretions as we see on those of birds. For even in animals that have a bladder, if the excretion when voided be placed in a vessel, it will throw down a deposit of earthy brine.⁴ For the sweet and fresh elements, being light, are expended on the flesh.

Among the Serpents, the same peculiarity attaches to vipers, as among the fishes attaches to the Selachia. For both these and vipers are externally viviparous, but previously produce ova internally.⁵

The stomach in all these animals is single, just as it is single in all other animals⁶ that have teeth in front of both jaws; and their viscera are excessively small, as always happens when there is no bladder. In serpents these viscera are, moreover, differently shaped from what they are in other animals. For, a serpent's

body being long and narrow, its contents are as it were moulded into a similar form, and thus come to be themselves elongated.⁷

All animals that have blood possess an omentum, a mesentery,⁸ an intestine with its appendages, and, moreover, a diaphragm and a heart; and all, excepting fishes, a lung and a trachea. The relative positions, moreover, of the windpipe and the œsophagus are precisely similar in them all; and the reason is the same as has already been given.⁹

(*Ch. 2.*) Almost all sanguineous animals have a gall-bladder. In some this is attached to the liver, in others separated from that organ¹ and attached to the intestines, being apparently in the latter case no less than in the former an appendage of the lower stomach.² It is in fishes that this is most clearly seen. For all fishes³ have a gall-bladder; and in most of them it is attached to the intestine, being in some, as in the⁴ *Amia*, united with this, like a border, along its whole length. It is similarly placed in most serpents. There are therefore⁵ no good grounds for the view entertained by some writers, that the use of the gall-bladder is to give rise to certain sensations. They say that its final cause is to affect that part of the soul which is lodged in the neighbourhood of the liver, and that it induces a gloomy or a cheerful disposition, according as it irritates this or leaves it alone. But this cannot be. For in some animals there is absolutely no gall-bladder at all, in the horse for instance, the mule, the ass, the deer, and the roe; and in others, as the camel, there is no distinct bladder, but merely small vessels that bear some resemblance to one.⁶ Again there is no such organ in the seal, nor, of sea-animals, in the dolphin.⁷ Even within the limits of the same genus, some animals appear to have and others to be without it.⁸ Such for instance is the case with mice; such also with man. For in some individuals there is apparently a gall-bladder attached to the liver, and in others none at all. This explains how the existence of this part in the whole genus has been a matter of dispute. For each observer, according as he has found it present or absent in the individual cases he has examined, has supposed it to be present or absent in the whole genus. The same has occurred in the case of sheep and of goats. For these animals usually have a gall-bladder; but, while in some it is so enormously big as to appear a monstrosity, as is the case in *Naxos*, in other instances

it is altogether wanting, as is the case in a certain district of Chalcis in Eubœa.⁹ Another fact, which shows that the opinion of these writers is erroneous, is that the gall-bladder in fishes is separated, as already mentioned, by a considerable interval from the liver. No less mistaken seems to be the opinion of Anaxagoras and his followers, that the gall-bladder is the cause of acute diseases, inasmuch as it becomes over-full, and spirts out its excess into the lung, the blood-vessels, and the chest. For, almost invariably, those who suffer from these acute forms of disease are persons who have no gall-bladder at all, and, were they to be dissected, this fact would be quite evident.¹⁰ Moreover there is no kind of correspondence between the amount of bile which is present in these diseases and the amount which is exuded.¹¹ The most probable opinion is that, as the bile when it exists in any other part of the body is a mere residuum or a product of decay, so also when it exists in the region of the liver it is equally excremental and has no further use; just as is the case with the dejections of the stomach and intestines. For though even the residua are occasionally used¹² by nature for some useful purpose, yet we must not in all cases expect to find such a final cause; for granted the existence in the body of this or that constituent, with such and such properties, many results must ensue merely as necessary consequences of these properties.¹³ All animals, then, whose liver is healthy in composition and supplied with none but sweet blood, are either entirely without a gall-bladder, or have merely¹⁴ bile-containing ducts; or are some with and some without such parts. Thus it is that the liver in animals that have no gall-bladder is, as a rule, of good colour and sweet; and that, when there is a gall-bladder, that part of the liver is sweetest, which lies immediately underneath it. But, when animals are formed of blood less pure in composition, the bile serves for the excretion of its impure residue. For the very meaning of residuum is that it is the opposite of nutriment, and of bitter that it is the opposite of sweet; and sweet blood it is which alone is nutritious.¹⁵ So that it is evident that the bile, which is bitter, cannot have any useful end, but must simply be a purifying excretion.¹⁶ It was therefore no bad saying of old writers that the absence of a gall-bladder gave long life. In so saying they had in mind deer and animals with uncloven hoofs. For such have no

gall-bladder and live long. But besides these there are other animals that have no gall-bladder, though those old writers were not aware of the fact, such as the camel and the dolphin;¹⁷ and these also are, as it happens, long-lived. Seeing, indeed, that the liver is not only useful, but absolutely necessary, to all animals that have blood, it is but reasonable that on its character should depend the length or the shortness of life. Nor less reasonable is it that this organ and none other should have such an excretion as the bile. For the heart, unable as it is to stand any violent affection,¹⁸ would be utterly intolerant of the proximity of such a fluid; and, as to the rest of the viscera, none excepting the liver are necessary parts of an animal.¹⁹ In conclusion, wherever we see bile we must take it to be residual or excremental. For to suppose that it has one character in this part, another in that, would be as great an absurdity as to suppose mucus or the dejections of the stomach to vary in character according to locality and not to be excremental wherever found.

(*Ch. 3.*) So much then of the gall-bladder, and of the reasons why some animals have one, while others have not.¹ We have next to speak of the mesentery and the omentum; for these are contained in the same cavity as the parts already described. The omentum, then, is a membrane containing fat; the fat being suet or lard according as the fat of the animal generally is of the former or latter description. What kinds of animals are so distinguished has been already set forth in an earlier part of this treatise.² This membrane, alike in animals that have a single and in those that have a multiple stomach, grows from the middle of that organ, along a line which is marked on it like a seam. Thus attached, it covers the rest of the stomach and the greater part of the bowels. It is found in all sanguineous animals whether they live on land or in water.³ Now the development of this part into such a form as has been described is the result of necessity. For, whenever solid and fluid are mixed together and heated, the surface invariably becomes membranous and skin-like.⁴ But the region in which the omentum lies is full of nutriment of such a mixed character. Moreover, in consequence of the thickness of the membrane, that portion of the sanguineous nutriment will alone filter into it which is of a greasy character; for this portion is composed of the finest particles; and when it has so

filtered in, it will be concocted by the heat of the part, and will be converted into suet or lard, and will not acquire a flesh-like or sanguineous constitution. The development, then, of the omentum is simply the result of necessity. But when once formed, it is used by nature for an end, namely, to facilitate and to hasten the concoction of food.⁵ For all that is hot aids concoction; and fat is hot, and the omentum is fat. This too explains why it hangs from the middle of the stomach; for the upper part of the stomach has no need of it, being assisted in concoction by the neighbouring liver. Thus much as concerns the omentum.

(*Ch. 4.*) The so-called mesentery is also a membrane; and stretches from the intestines, in their whole extent, to the great vessel and the aorta. In it are numerous and close-packed vessels, which run from the intestines into the great vessel and into the aorta. The formation of this membrane we shall find to be the result of necessity, as is that of the other parts.¹ What however is the final cause of its existence in sanguineous animals is manifest on reflection. For it is necessary that animals shall get nutriment from without; and, again, that this shall be converted into the ultimate nutriment, which is then distributed as sustenance to the various parts; this ultimate nutriment being, in sanguineous animals, what we call blood, and having, in exsanguineous animals, no definite name. This being so, there must be channels through which the nutriment shall pass, as it were through roots,² from the stomach into the blood-vessels. Now the roots of plants are in the ground; for thence their nutriment is derived. But in animals the stomach and intestines represent the ground, from which the nutriment is to be taken. The mesentery, then, is an organ to contain the roots; and these roots are the vessels that traverse it. This then is the final cause of its existence. But how it absorbs nutriment, and how that portion of the food which enters into the vessels is distributed by them to the various parts of the body, are questions which will be considered when we come to deal with the generation and nutrition of animals.³

The constitution of sanguineous animals, so far as the parts as yet mentioned are concerned, and the reasons for such constitution, have now been set forth. In natural sequence we should next go on to the organs of generation, as yet undescribed, on which depend the distinctions of male and female. But, inasmuch as

we shall have to deal specially with generation and development hereafter, it will be more convenient to defer the consideration of these parts to that occasion.

(*Ch. 5.*) Very different from the animals we have as yet considered are the Cephalopods, and the Crustacea. For these have absolutely no viscera¹ whatsoever; as indeed is the case with all ex-sanguineous animals, in which are included two other genera, namely the Testacea and the Insects. For in none of them does the material out of which viscera are formed exist. None of them, that is, have blood. The cause of this lies in their essential constitution. For the presence of blood in some animals, its absence from others, will be included in the conception which determines their respective essences.² Moreover, in the animals we are now considering, none of those final causes will be found to exist, which, in sanguineous animals, determine the presence of viscera. For³ they have no blood-vessels nor urinary bladder, nor do they breathe. All, in fact, that it is necessary for them to have is some organ which shall be analogous to a heart; for in all animals there must be some central and commanding part of the body, to lodge the sensory portion of the soul and the source of life. They must of course also all have the organs of nutrition; and a diversity of character exists among them, depending on differences of these receptacles of food.

In the Cephalopods there are two teeth,⁴ enclosing what is called the mouth; and inside this mouth is a fleshy substance, which represents a tongue and serves for the discrimination of pleasant and unpleasant food. The Crustacea have teeth corresponding to those of the Cephalopods, namely their anterior teeth,⁵ and also have the fleshy representative of a tongue. This latter part is found, moreover, in all Testacea,⁶ and serves, as in sanguineous animals, for gustatory sensations. Similarly provided also are the Insects. For some of these, such as the Bees and the Flies, have, as already described, a proboscis protruding from the mouth;⁷ while others, that have no such piercing instrument in front, have a part which acts as a tongue inside the mouth.⁸ Such for instance is the case in the Ants and the like. As for teeth, some insects have them, the Bees and the Flies for instance, though in a somewhat modified form, while others, that live on fluid nutriment, are without them. For in many insects the teeth are not organs of mastication, but weapons of defence.

In some Testacea, as was said in the⁹ first treatise, the organ which is called the tongue is of considerable strength; and in one kind, namely the Sea-snails, in addition to the tongue there are two teeth,¹⁰ resembling those of the Crustacea. The mouth in the Cephalopods¹¹ is succeeded by a long gullet. This leads to a crop, like that of a bird. Directly continuous with this is the stomach, from which a gut runs without windings to the vent. The Sepias and the Poulps resemble each other completely, so far as regards the shape and consistency of these parts. But not so the Calamaries. Here, as in the other groups, there are two stomach-like receptacles; but the first of these cavities has less resemblance to a crop, and in neither is the form or the consistency the same as in the other kinds, the whole animal indeed being made of a softer kind of flesh.

The object of this arrangement of the parts in question is the same in the Cephalopods as in Birds. For in neither of these groups is the mouth suited for mastication; and therefore it is that a crop precedes their stomach.

For purposes of defence, and to enable them to escape from their foes, the Cephalopods have what is called an ink-bag attached to their body. This is a membranous pouch, which is provided with a terminal outlet just at the point where what is termed the funnel gives issue to the residua of the stomach. This funnel is placed on the ventral surface of the animal. All Cephalopods alike have this characteristic ink-bag, but chief of all the sepia, where it is of greater size than in the rest.¹² When the animal is disturbed and frightened it uses the ink to make the surrounding water black and turbid, and so, as it were, puts a shield in front of its body.

In the Calamaries and the Poulps the ink-bag is placed in the upper part of the body, in close proximity to the *myris*,¹³ whereas in the sepia it is much lower down, against the stomach. For the sepia has a larger ink-bag than the rest, owing to its making more use of it. The reasons for this are, firstly, that it lives near the shore, and, secondly, that it has no other means of protection; whereas the poulp has twining feet¹⁴ that it can use in its defence, and is moreover endowed with the power of changing its colour.¹⁵ This changing of colour, like the discharge of ink, occurs as the result of fright. As to the calamary, it lives far out at sea, being the only one of the Cephalopods that

does so ; and this gives it protection.¹⁶ These then are the reasons why the ink-bag is much larger in the sepia than in the other Cephalopods ; and its larger size again explains its lower position. For though this position removes it farther from the funnel, yet the larger size of the bag gives increased force, and so allows the ink to be ejected with ease even from a distance. The ink itself is of an earthy character, in this resembling the white deposit on the surface of a bird's excrement. And the explanation in both cases is the same, namely the absence of an urinary bladder.¹⁷ For, in default of this, it is the ink that serves for the excretion of the earthy matter. And this is more especially the case in the sepia ; because there is a greater proportion of earth in its composition than in that of the other Cephalopods. The earthy character of its bone is a clear proof of this. For in the poulp there is no bone at all, and in the calamary it is small and merely cartilaginous.¹⁸ Why this bone should be present in some Cephalopods, and wanting in others, and how its character varies in those that have it, has now been set forth.

These animals, having no blood, are in consequence cold and of a timid character.¹⁹ Now, in some animals, fear causes an evacuation from the bowels, and, in others, a flow of urine from the bladder.²⁰ Similarly in these it produces a discharge of ink, and though this ink, like the urine, is residual matter, and though its ejection in fright is merely the result of necessity, yet it is used by nature for a purpose,²¹ namely the protection and safety of the animal that excretes it.

The Crustacea also, both the Caraboid forms and the Crabs,²² are provided with teeth, namely the two so-called anterior teeth ; and between these they also present the tongue-like piece of flesh, as has indeed been already mentioned.²³ Directly after their mouth comes a gullet, which is very small considering the size of the whole body ; and then a stomach, which in the Carabi and some of the Crabs is furnished with a second set of teeth, the anterior set being insufficient for mastication. From the stomach an unconvoluted gut runs in a direct line to the vent.²⁴

The parts described are to be found also in all the various Testacea.²⁵ The degree of distinctness, however, with which they are formed varies much in the different kinds, and the larger the size of the animal the more easily distinguishable are all these parts

severally. In the Sea-snails, for example, we find teeth, hard and sharp, as before mentioned,²⁶ and between them a fleshy substance like the tongue of Crustacea and Cephalopods, and again a proboscis which has already been described as something between a piercing instrument and a tongue.²⁷ Directly after the mouth comes a kind of bird-like crop,²⁸ then a gullet, and continuous with this a stomach, in which is the *mecon*, as it is styled ; and this *mecon* in turn gives rise to an intestine, starting directly from it.²⁹ It is the excretion of this *mecon*, which appears in all the Testacea to form the most palatable morsel. *Purpuræ*,³⁰ and whelks, and all other Testacea that have turbinated shells, in structure resemble the sea-snail.

The genera and species of Testacea are very numerous. There are those with turbinated shells, of which some have just been mentioned ; and, besides these, there are bivalves and univalves. Those with turbinated shells may indeed after a certain fashion be said to resemble bivalves. For they all, from their very birth, have a covering to protect that part of their body which would otherwise be exposed.³¹ This is the case with the *Purpuræ*, with whelks, with the *Nerites*,³² and all the like. Were it not for this, the part which is undefended by the shell would be very liable to injury from without. The Univalves also are not without protection. For on their dorsal surface they have a shell, and by the under surface they attach themselves to the rocks, and so after a manner may be said to become bivalved, the rock representing the second valve. Of these the animals known as limpets are an example. The Bivalves, scallops and mussels for instance, are protected by the power they have of closing their valves ; and the turbinated genera by the operculum just mentioned, which transforms them, as it were, from univalves into bivalves. But of all there is none so perfectly protected as the *Echinus*. For here there is a globular shell which encloses the body completely, and which is moreover set with sharp spines. This peculiarity distinguishes the *Echinus* from all other Testacea, as has already been mentioned.

The structure of the Testacea and of the Crustacea is exactly the reverse of that of the Cephalopods. For in the latter the fleshy substance is on the outside and the earthy substance within, whereas in the former the soft parts are inside and the hard part without. In the *Echinus*, however, there is no fleshy part whatsoever.

All the Testacea, then, those that have not been mentioned as well as those that have, agree as stated in possessing a mouth³³ with the tongue-like body, a stomach, and a vent; but they differ from each other in the positions and proportions of these parts. The details, however, of these differences must be looked for in the Researches concerning Animals and the treatises on Anatomy. For while there are some points which can be made clear by verbal description there are others which are more suited for ocular demonstration.³⁴

Peculiar among the Testacea are the Echini and the animals known as Ascidians.³⁵ The Echini have five teeth,³⁶ and in the centre of these the fleshy body,³⁷ which is common to all the animals we have been discussing. Immediately after this comes a gullet, and then a stomach, divided into a number of separate compartments, which look like so many distinct stomachs; for the cavities are separate, and each contains residual matter. They are all however connected with one and the same œsophagus, and they all end in one and the same vent.³⁸ There is nothing, with the exception of the stomach, of a fleshy character, as has already been stated. All that can be seen are the so-called ova,³⁹ of which there are several, contained each in a separate membrane, and certain black bodies which have got no name, and which, beginning at the animal's mouth, are scattered round its body here and there profusely.⁴⁰ Of these Echini there are many different species, and in all of them the parts mentioned are to be found. It is not however in every kind that the so-called ova are edible. Neither do these attain to any size in any other species than those which are found in shallow water.⁴¹ A similar distinction may be made generally in the case of all Testacea. For there is a great difference in the edible qualities of the flesh of different kinds; and in some, moreover, the residual substance known as the *mecon*⁴² is good for food, while in others it is uneatable. This *mecon* in the turbinated genera is lodged in the spiral part of the shell, while in univalves, such as limpets, it occupies the apex, and in bivalves is placed near the hinge. In these bivalves the so-called ovum lies on the right; while on the opposite side is the vent.⁴³ The former is incorrectly termed ovum,⁴⁴ for it merely corresponds to what in well-fed sanguineous animals is fat; and thus it is that it only makes its appearance in Testacea at those seasons of the year when they are in good condition, namely spring

and autumn. For no Testacea can abide extremes of temperature, and they are therefore in evil plight in winter and in summer. This is clearly shown by what occurs in the case of the Echini. For though the ova are to be found in these animals even directly they are born, yet they acquire a much greater size than usual at the time of full moon; not, as some think, because the Echini eat more at that season, but because the nights are then warmer, owing to the moonlight.⁴⁵ For these creatures are bloodless, and so are unable to stand cold and require warmth. Therefore it is that they are found in better condition in summer than at any other season; and this all over the world excepting in the Pyrrhean tidal strait. There the Echini flourish as well in winter as in summer. But the reason for this is that they have a greater abundance of food in the winter, because the fish desert the strait at that season.⁴⁶

The number of the ova is the same in all Echini, and is an odd one. There are in fact five ova, just as there are also five teeth and five stomachs; and the explanation of this is to be found in the fact that the so-called ova are not really ova, but merely, as was said before, the result of the animal's well-fed condition. Oysters also have an ovum, corresponding in character to that of the Echini, but existing only on one side of their body.⁴⁷ Now inasmuch as the Echinus is of a spherical form, and not merely a single disk like the oyster, and in virtue of its spherical shape is the same from whatever side it be examined, it follows as a necessary consequence that its ovum must be of a corresponding symmetry. For the spherical shape has not the asymmetry of the disk-shaped body of the Oysters. For though the head in all of these is central, yet their ovum is on one side, namely the upper.⁴⁸ Now the necessary symmetry would be observed, were the ovum to form a continuous ring. But this may not be. For it would be in opposition to what prevails in the whole tribe of Testacea;⁴⁹ for in all the ovum is discontinuous, and in all, excepting the Echini, asymmetrical, being placed only on one side of the body. Owing, then, to this necessary discontinuity of the ovum, which belongs to the Echinus as a member of the class, and owing to the spherical shape of the body, which is its individual peculiarity, this animal cannot possibly have an even number of ova. For were they in even number, they would have to be arranged exactly opposite to each other, in pairs, so

as to keep the necessary symmetry ; one ovum of each pair being placed at one end, the other ovum at the other end, of a transverse diameter. This again would violate the universal provision in Testacea. For both in the Oysters and in the Scallops we find the ovum only on one side of the circumference. The number then of the ova must be an uneven one, three for instance or five. But if there were only three, they would be much too far apart ; while if there were more than five, they would come to form a continuous mass. The former arrangement would be disadvantageous to the animal, the latter an impossibility. There can, therefore, be neither more nor less than five. For the same reason the stomach is divided into five parts, and there is a corresponding number of teeth. For seeing that the ova represent each of them a kind of body for the animal, from which the materials for growth are derived, there must be a conformity between them and the general configuration of the vital organs. Now, if there were only one stomach, this conformity would be wanting. For either the ova would be too far off from the stomach, or the stomach would be so big as to fill up the whole cavity, and the Echinus would have great difficulty in moving about, and finding due nourishment for its repletion. As then there are five intervals between the five ova, so are there of necessity five divisions of the stomach, one for each interval. So also and on like grounds there are five teeth. For by this arrangement nature is enabled to allot to each stomachal compartment and ovum its separate and similar tooth. These then are the reasons why the number of ova in the Echinus is an odd one, and why that odd number is five. In some Echini the ova are excessively small, in others of considerable size, the explanation being that the latter are of a warmer constitution, and so are able to concoct their food more thoroughly ; while in the former concoction is less perfect, so that the stomach is found full of residual matter, while the ova are small and uneatable.⁵⁰ Those of a warmer constitution are, moreover, in virtue of their warmth more given to motion, so that they make expeditions in search of food, instead of remaining stationary like the rest. In proof of this, it will be found that they always have something or other sticking to their spines, which they use as feet in their frequent ramblings.⁵¹

The Ascidians differ but slightly from plants and yet have more of an animal nature than the Sponges, which are in

fact virtually plants and nothing more. For nature passes from lifeless objects to animals in such unbroken sequence, interposing between them beings which live and yet are not animals, that scarcely any difference seems to exist between two neighbouring classes owing to their close proximity.⁵²

A sponge⁵³ then, as already said, in these respects completely resembles a plant, that throughout its life it is attached to a rock, and that when separated from this it dies. Slightly different from the Sponges are the so-called *Holothuriæ* and the *Sea-lungs*,⁵⁴ as also sundry other sea-animals that resemble them. For these are free and unattached. Yet have they no feeling,⁵⁵ and their life is simply that of a plant, separated from the ground. For even among land-plants there are some that are independent of the soil, and that spring up and grow, either parasitically upon other plants,⁵⁶ or even entirely free. Such for example is the plant which is found on Parnassus, and which some call the *Epipetrum*.⁵⁷ This you may take up and hang from the rafters, and it will yet live for a considerable time. Sometimes it is a matter of doubt whether a given organism should be classed with plants or with animals. The *Ascidians*, for instance, and the like so far resemble plants as that they never live free and unattached,⁵⁸ but, on the other hand, inasmuch as they have a certain flesh-like substance, they must be supposed to possess some degree of sensibility.⁵⁹

In the *Ascidians* there are two orifices and a single septum, which latter separates the part into which the animal takes the fluid that ministers to its nutrition from the part by which it again discharges the superfluity of moisture. For it appears to have no distinct residual matter, such as have the other *Testacea*. This is itself a very strong justification for considering an *Ascidian*, and any thing else there may be among animals that resembles it, to be of a vegetable character; for plants also never have any residuum.⁶⁰ Finally there runs across the middle of the body of these *Ascidians* a thin partition, and here it is that we may reasonably suppose the part on which life depends to be situated.

The animals which some call *Sea-nettles* and others *Acalephæ*⁶¹ are not *Testacea* at all nor included in their divisions. Their constitution approximates them on the one side to plants, on the other to animals. For seeing that some of them can detach

themselves and can seize hold of their food, and that they are sensible of objects which come in contact with them, they must be considered to have an animal nature. The like conclusion follows from their using the roughness of their bodies⁶² as a protection against their enemies. But on the other hand they are closely allied to plants, firstly by the imperfection of their structure, secondly by their being able to attach themselves to the rocks with great rapidity, and lastly by their having no visible residuum, notwithstanding that they possess a mouth.

Very similar again to the Acalephæ, are the Starfishes. For these also seize hold of their prey, and suck out its juices, and thus destroy a vast number of oysters.⁶³ At the same time they present a certain resemblance to such animals as the Cephalopods and Crustacea, inasmuch as they are free and unattached. The same may also be said of the Testacea.

Such then is the structure of the parts that minister to nutrition, and which every animal must necessarily possess. But besides these organs it is quite plain that in every animal there must be some part or other which shall be analogous to what in sanguineous animals is the presiding seat of sensation. Whether an animal has or has not blood, it cannot possibly be without this. In the Cephalopods this part consists of a membrane, containing fluid, through which runs the gullet on its way to the stomach. It lies rather towards the dorsal surface of the animal, and is by some called the *mytis*.⁶⁴ Just such another organ is found in the Crustacea, and is known in them also by the same name. This part is formed by a combination of fluid and solid, and is, as before said, traversed by the gullet. For had the gullet been placed farther back, between the *mytis* and the dorsal surface of the animal, the hardness of the back would have interfered with its due dilatation in the act of deglutition. On the outer surface of the *mytis* runs the intestine; and in contact with this latter is the ink-bag, which is thus removed as far as possible from the mouth, while a considerable interval divides its irritating fluid from the nobler and sovereign parts.⁶⁵ The position of the *mytis* shows that it corresponds to the heart of sanguineous animals; for it occupies the self-same place. The same is proved by the sweetness of its fluid, which has the character of concocted matter and resembles blood.⁶⁶

In the Testacea the presiding seat of sensation is in a cor-

responding position, but is less easily made out.⁶⁷ It should, however, always be looked for in the following situations. In such Testacea as are stationary, between the gullet and the channel through which either the excrement or the spermatic fluid⁶⁸ is voided; but, in those species which are capable of locomotion, invariably in the centre, midway that is between the right and left sides.

In Insects this important organ lies, as was stated in the first treatise,⁶⁹ between the head and the cavity which contains the stomach. In most of them it consists of a single part; but in others, for instance in such as have long bodies and resemble the Juli,⁷⁰ it is made up of several parts, so that such insects continue to live after they have been cut into pieces.⁷¹ For the aim of nature is to give each animal only one such governing part; and when she is unable to carry out this intention she causes the parts, though potentially many, to work together actually as one.⁷² The phenomenon referred to is much more clearly marked in some insects than in others.

The parts concerned in nutrition are not alike in all insects, but show a considerable diversity. Thus some have what is called a piercer in their mouths, which is a kind of compound instrument that combines in itself the character of a tongue and of lips.⁷³ In others, that have not got this anterior piercer, there is an organ inside the mouth that answers the same sensory purposes. After the mouth comes the intestine, which is never wanting in any insect. This runs in a straight line and without further complication to the vent; occasionally, however, it has a spiral coil. There are, moreover, some insects in which a stomach succeeds to the mouth, and is itself succeeded by a convoluted intestine.⁷⁴ By this arrangement the larger and more voracious insects are enabled to take in a more abundant supply of food.⁷⁵ More curious than any are the Cicadæ. For here the mouth and the tongue are united so as to form a single part, through which, as through a root, the fluids on which the creature lives are sucked up.⁷⁶ Insects are always small eaters, not so much because of their diminutive size as because of their cold temperament. For it is heat which requires sustenance; just as it is heat which speedily concocts it.⁷⁷ But cold neither requires sustenance nor concocts it. In no insects is this more evident than in these Cicadæ. For they find enough to live on in the moisture which

is deposited from the air.⁷⁸ So also do the Ephemera found about the Black Sea.⁷⁹ But while these latter only live for a single day, the Cicadæ subsist on such food for several days, though still not many.

We have now done with the internal parts of animals, and must return to the consideration of such external parts as have not yet been described. It will be better to begin with the animals we have just been describing, that is with the bloodless animals, so that we may not be hampered with them hereafter, but may be free to deal leisurely with the more perfect kinds of animals, those namely that have blood.

(*Ch. 6.*) We will begin with Insects.¹ These animals, though they consist of but few parts, are yet not without diversities when compared with each other. They are all many-footed; the object of this being to compensate their natural slowness and frigidity, and give greater activity to their motions. Accordingly we find that those which, as the² Juli, have long bodies, and are therefore the most liable to refrigeration, have also the greatest number of feet. Again in all insects the body is made up of segments—the reason for this being that in these animals there is no one supreme and sovereign part³ but several—and the number of feet corresponds to the number of segments. Should the feet fall short of this, their deficiency is supplied by the presence of feathers. Of such feathered insects some live a wandering life and are forced to make long expeditions in search of food. These have a body of light weight, and four feathers, two on either side, to support it. Such are bees and other insects akin to them. When, however, such insects are of very small bulk, their feathers are reduced to two, as is the case with flies.⁴ Even insects of squarer build and of stationary habits of life yet come to have as many feathers in all as have bees; for they have shards, which protect the acting feathers from injury. Such are the Melalonthæ⁵ and the like. For their stationary habits expose them to much greater risks than are run by those insects that are more constantly in flight, and on this account they are provided with this protecting shield. The feather of an insect has neither barbs nor shaft.⁶ For, though it is called a feather, it is no feather at all, but merely a skin-like membrane that, owing to its dryness, necessarily becomes detached from the surface of the fleshy body.

The body of an insect is made of segments, not only for the reasons already assigned, but also to enable it to bend in such a manner as may protect it from injury. For such insects as have long bodies can roll themselves up,⁷ which would have been impossible, had they not been formed of segments; and even those which cannot do this can yet draw their segments closer together, and so increase the hardness of their bodies. This can be felt quite plainly by putting the finger on any of the insects known as *Canthari*.⁸ The touch frightens the insect, and it remains perfectly motionless, while its body is felt to become harder than before. The division, then, of the body into segments has this final cause; but it is also a necessary result of there being several supreme organs in place of one; and this again is a part of the essential constitution of insects, and is a character which approximates them to plants. For as plants, though cut into pieces, still live, so also do insects. There is however this difference between the two cases, that the portions of the divided insect live for a very short space, whereas the portions of the plant live on and attain the perfect form of the whole, so that from one single plant you may obtain two or more.⁹

Some insects are also provided with another means of protection against their enemies, namely a piercer or sting. In some this is in front, connected with the tongue, in others behind and connected with the tail. For just as the organ of smell in elephants¹⁰ answers several uses, serving alike for purposes of nutrition and for purposes of defence, so also does the lingual arrangement in some insects answer more than one end. For it is the instrument through which they derive their sensations of food, as well as that with which they suck it up and bring it to their mouths; and, when no such anterior piercer or sting exists, the mouth is furnished with teeth, which so far supply its place as to serve either for the mastication of food or for its prehension and conveyance to the mouth. They serve this latter use for instance in ants and in all the various kinds of bees.¹¹

As for a tail-sting, nature has given it to such insects as are of a fierce disposition, and to no others. Sometimes this instrument is lodged inside the body, as in bees and wasps. This is a necessary consequence of their being made for flight. For, were their piercer or sting external and of delicate make, it would very easily get spoiled. If on the other hand, still being

external, it were of stouter build, as in scorpions, its weight would interfere with flight. As for scorpions, they never rise from the ground, and their piercer or sting must therefore be arranged in the way it is, as otherwise it would be of no use as a weapon.¹² The Diptera never have a tail-sting.¹³ For the very reason of their being dipterous is that they are small and weak, and therefore require no more than two feathers to support their light weight. And the same reason which reduces their feathers to two causes their piercer or sting to be in front; for their strength is not sufficient to allow them to strike efficiently with the hinder part¹⁴ of their body. Polypterous insects on the other hand are of greater bulk—indeed it is this which causes them to have so many feathers—and their greater size makes them much stronger in their hinder parts. These insects therefore have tail-stings. For it is better, when possible, that one and the same instrument shall not be made to serve several dissimilar uses; but that there shall be one organ to serve as a weapon, which can then be very sharp, and a second distinct one to serve as a tongue, which can then be of spongy texture and fit to absorb nutriment. Whenever, therefore, nature is able to provide two separate instruments for two separate uses, without the one hampering the other, she does so, instead of acting like a coppersmith, who for cheapness makes a spit and lampholder in one. It is only when this is impossible, that she uses one organ for several functions.¹⁵

The anterior legs are in some cases longer¹⁶ than the others, that they may serve to clean off the dust or other matter which may fall into the insect's eyes and obstruct its sight, which already is not very distinct owing to the eyes being made of a hard substance. Flies and bees and the like may be constantly seen thus dressing themselves with crossed fore-legs. Of the other legs, the hinder are bigger than the middle pair, both to aid in running and also that the insect, when it takes flight, may spring more easily from the ground. This difference is still more marked in such insects as leap, in grasshoppers for instance and in the various kinds of fleas.¹⁷ For these first bend and then extend the legs, and, by so doing, are necessarily shot up from the ground. It is only the hind legs of grasshoppers, and never the front ones, that resemble the two long stern oars by which a ship is steered.¹⁸ For it is essential that the joint shall be

bent inwards, and this never occurs in the anterior limbs.¹⁹ The whole number of legs, including those used in leaping, is six, in all these insects.

(*Ch.* 7.) In the Testacea¹ the body consists of but few parts, the reason being that these animals lead a stationary life. For such animals as move much about must of necessity have more numerous parts than such as remain quiet; for, the more diversified the movements, the greater the number of organs required to effect them.² Some species of Testacea are absolutely motionless, and others very nearly so. They would thus fall an easy prey to their enemies, were it not for the hardness of the shell with which nature has invested their body as a means of protection. This shell, as already has been said, may have one valve, or two valves, or be turbinated. In the latter case it may either be spiral as in whelks, or merely globular, as in the Echini.³ When it has two valves, these may be gaping, as in scallops and mussels, where the valves are united together on one side only, so as to open and shut on the other, or they may be united together on both sides, as in razor-fishes.⁴ In all cases alike these Testacea have, like plants, their head downwards. The reason for this is that they imbibe their nourishment from below, just as do plants with their roots.⁵ Thus the under parts come in them to be above, and the upper parts to be below. The body is enclosed in a membrane, and through this the animal filters suitable fluid and absorbs nutriment. In all there is a head;⁶ but none of the other parts, excepting the receptacle for food, has any distinctive name.

(*Ch.* 8.) All the Crustacea¹ can crawl as well as swim, and accordingly they are provided with numerous feet. There are four main genera, viz. the Carabi, as they are called, the Astaci, the Carides, and the Carcini. In each of these genera, again, there are numerous species, which differ from each other not only as regards shape, but also very considerably as regards size. For, while in some species the individuals are large, in others they are excessively minute. The Carcinoid and Caraboid Crustacea resemble each other in possessing claws. These claws are not for locomotion, but to serve as hands for seizing and holding objects; and they are therefore bent in exactly the opposite direction to the feet, being so twisted as to turn their convexity towards the body, while the feet turn towards it their concavity

For by this position the claws are best suited for laying hold of food and carrying it to the mouth. The distinction between the Carabi and Carcini consists in the former having a tail while the latter have none. For the Carabi swim about, and a tail is therefore of use to them, serving for their propulsion like the blade of an oar. But it would be of no good to the Carcini; for these animals as a rule live close to shore, and creep into holes and corners. In such of them as live out at sea, the feet are much less adapted for crawling than in the rest, because they make but little use of them for this purpose and depend for protection on their shell-like covering. The Maiaæ and the crabs known as Heracleotic are examples of this; the legs in the former being very thin, in the latter very short.²

The very minute crabs, that are found at the bottom of the net mixed with small fishes, have their posterior legs expanded into the resemblance of fins or oar-blades, so as to help the animal in swimming.³ The Carides are distinguished from the Carcinoid species by the presence of a tail; and from the Caraboids by the absence of claws. This is explained by their large number of feet, on which has been expended the material for the growth of claws.⁴ Their feet again are numerous to suit their mode of progression, which is mainly by swimming.⁵

The parts on the ventral surface and near the head are in some of these animals formed like gills, for the admission and discharge of water. The parts lower down differ in the different sexes. For in the female Carabi the parts are more laminar than in the males,⁶ and in the female Crabs the flap is furnished with hairier appendages. This gives ampler space for the disposal of the ova, which the females retain in these parts instead of letting them go free, like fishes and other animals, as soon as they are brought forth.

In the Carabi and in the Carcini the right claw is invariably the larger and the stronger.⁷ For it is natural⁸ to every animal to use its right side in preference to its left; and nature, in distributing the organs, invariably assigns each, either exclusively or in a more perfect condition, to such animals as can use it. So is it with tusks, and teeth, and horns, and spurs, and all such offensive and defensive weapons.⁹

In the Astaci alone it is a matter of chance which claw is the larger, and this in either sex.¹⁰ Claws they must have, because they belong to a genus in which this is a constant character,¹¹

but they have them in this irregular way, owing to imperfect formation and to their not using them for their natural purpose, but for locomotion.

For a detailed account of the several parts of these animals, of their position, and their differences, those parts being also included which distinguish the sexes, reference must be made to the treatises on Anatomy and to the Researches concerning Animals.

(*Ch. 9.*) We come now to the Cephalopods.¹ Their internal organs have been already described with those of other animals.² Externally there is an uniform sac inclosing the body; and in front of this sac a head surrounded by feet, which form a circle about the mouth and teeth, and are set between these and the eyes. Now in all other animals the feet, if there are any, are placed in one of two ways; either before and behind, or along the sides, the latter being the method adopted in those bloodless animals whose feet are numerous. But in the Cephalopods there is a peculiar arrangement, distinct from either of these. For their feet are all placed at what may be called the fore end. The reason for this is that the hind and fore parts of their body have been drawn up close to each other,³ as is also the case in the turbinated Testacea. For these latter, while in some points they resemble the Crustacea, in others resemble the Cephalopods. Their earthy matter is on the outside, and their fleshy substance within. So far they are like the Crustacea. But the general plan of their body is that of the Cephalopods; and, though this is true in a certain degree of all the Testacea, it is more especially true of those turbinated species, that have a spiral shell.⁴ This general plan, common to the two, is as follows. Let us first consider the case of quadrupeds and of man, where the arrangement is that of a straight line. At the upper end of this line, *A* represents the mouth, then *B* the gullet, and *C* the stomach. After this comes the intestine reaching to the vent, which is marked by *D*.⁵ Such is the plan in sanguineous animals; and round this straight line as an axis are disposed the head and trunk; the remaining parts, such as the anterior and posterior limbs, having been super-added by nature, merely to minister to these and for locomotion.

Now in Crustacea and in Insects there is a tendency to a similar arrangement of the internal parts in a straight line; the distinctions between these groups and the sanguineous animals depending on differences of the external organs which minister to locomotion.

But the Cephalopods and the turbinated Testacea have in common an arrangement which stands in contrast with this. For here the two extremities are brought together by a curve, as if one were to bend the straight line marked *E* until *D* came close to *A*. Such, then, is the disposition of the internal parts; and round these, in the Cephalopods, is placed the sac (in the Poulps alone called a head),⁶ and, in the Testacea, the turbinated shell which corresponds to the sac. There is, in fact, only this difference between them, that the sac of the Cephalopods is soft while the shell of the Testacea is hard, nature having invested the fleshy parts with this hard coating as a protection to the animal, which from its limited power of locomotion is exposed to considerable risk. In both classes, owing to this arrangement of the internal organs, the excrement is voided near the mouth; at a point below this orifice in the Cephalopods, and in the Turbinata somewhat on one side of it.⁷

Such then is the explanation of the position of the feet in the Cephalopods, and of the contrast they present to other animals in this matter. The arrangement, however, in the Sepias and the Calamaries is not precisely the same as in the Poulps, owing to the former having no other mode of progression than by swimming, while the latter not only swim but crawl.⁸ Thus the former have six of their feet above the teeth, and of these six the two outer ones are the biggest; while the remaining two, which make up the total eight, are below the mouth and are much the biggest of all, just as the hind limbs in quadrupeds are stronger than the fore limbs. For it is these lower feet and these⁹ hind legs that have to support the weight, and to take the main part in locomotion. And the outer pair of the upper six are bigger than the pair which intervene between them and the uppermost of all, because they have to assist the lowermost pair in their office. In the Poulps, on the other hand, the four central feet are the biggest.¹⁰ Again, though the number of feet is the same in all the Cephalopods, namely eight,¹¹ their length varies in different kinds, being short in the Sepias and the Calamaries, but greater in the Poulps. For in these latter the body with its sac is of small bulk, while in the former it is of considerable size; and so in the one case nature has used the materials subtracted from the body to give length to the feet, while in the other she has acted in precisely the opposite way, and has given to the growth

of the body what she has first taken from the feet.¹² The Poulps, then, owing to the length of their feet, can not only swim but crawl, whereas in the other genera the feet are useless for the latter mode of progression, being small while the bulk of the body is considerable. These short feet would not enable their possessors to cling to the rocks and keep themselves from being torn off by the waves in times of storm; neither would they serve to lay hold of objects at all remote and bring them in; but, to supply these defects, the animal is furnished with two¹³ proboscides, by which it can moor itself and ride at anchor like a ship in rough weather.¹⁴ The same processes serve also to catch prey even at some distance and to bring it to the mouth. They are so used both by the Sepias and the Calamaries. In the Poulps the feet are themselves able to perform these offices, and there are consequently no proboscides. In all the Cephalopods that have suckers on their feet and¹⁵ twining tentacles, these act in the same way, and have indeed the same structure, as those plaited instruments which were used of old by physicians to reduce dislocations of the fingers.¹⁶ The tentacles, like these, consist of plaited fibres, which act by pulling upon portions of flesh or any substance of a yielding nature. The physician places his instrument in a slackened condition round the finger; and, when it is put on the stretch, it grasps and clings tightly to whatever may be in contact with its inner surface. The tentacles act in the same manner, serving in place of hands for offensive or defensive purposes, the Cephalopods having indeed no other instrument than either feet or proboscides with which to lay hold on anything and bring it to the mouth.

The suckers are set in double line in all the Cephalopods excepting in one kind of poulp, where there is but a single row.¹⁷ The length and the slimness which is part of the nature of this kind of poulp explains the exception. For a narrow space cannot possibly admit of more than a single line of suckers. This exceptional character, then, belongs to them, not because it is the most advantageous arrangement, but because it is the necessary consequence of their essential specific constitution.¹⁸

In all these animals there is a fin, encircling the sac. In the Poulps and the Sepias this fin is unbroken and continuous, as is also the case in the larger calamaries. But in the smaller kind, called Teuthides, the fin is not only broader than in the Sepias

and the Poulps, where it is very narrow, but moreover only begins in the middle of the side, and does not encircle the entire sac. The use of this fin is to enable the animal to swim, and also to direct its course. It acts, that is, like the rump-feathers in birds, or the tail-fin in fishes. In none is it so small as in the Poulps; where it is difficult to make it out.¹⁹ For in these the body is of small bulk and can be steered by the feet sufficiently well without other assistance.

All the bloodless animals, namely the Insects, the Crustacea, the Testacea, and the Cephalopoda, have now been dealt with in turn; and the parts of all have been described, whether internal or external.

(Ch. 10.) We must now go back to the animals that have blood, and consider such of their parts, already enumerated, as were before passed over. We will take the¹ viviparous animals first, and, when we have done with these, will pass on to the ovipara, and treat of them in like manner.

The parts that border on the head, and on what is known as the neck and throat, have been already taken into consideration. As for the head itself, such a part is found in all animals that have blood; whereas in some bloodless animals, in crabs for instance, there is no head distinctly separable from the trunk. As to the neck, it is present in all the vivipara, but only in some of the ovipara; for while those that have a lung also have a neck, those that do not inhale the outer air have none.²

The head exists mainly for the sake of the brain. For every animal that has blood must of necessity have a brain; and must, moreover, for reasons already given, have it placed as far as possible from the heart.³ But the head has also been chosen by nature as the part in which to set some of the senses; because its blood is mixed in such suitable proportions as to ensure quiet and precision to the sensations, while at the same time it can supply the brain with such warmth as it requires.⁴ There is yet a third constituent superadded to the head, namely the part which ministers to the ingestion of food. This has been placed here by nature, because such a situation accords best with the general configuration of the body. For the stomach could not possibly be placed above the heart, seeing that this is the sovereign organ; and if placed below, as in fact it is, then the mouth could not possibly be placed there also. For this would have neces-

sitated a great increase in the length of the body ; and the stomach moreover would have been removed too far from the source of motion and of concoction.⁵ The head, then, exists for the sake of these three parts.

The neck, again, exists for the sake of the windpipe. For it acts as a defence to this and to the œsophagus, encircling them and keeping them from injury. In all other animals this neck is flexible and contains several vertebræ ; but in wolves and lions it contains only a single bone.⁶ For the object of nature was to give these animals an organ which should be serviceable in the way of strength, rather than one that should be useful for any of the other purposes to which necks are subservient.⁷

Continuous with the head and neck is the trunk with the anterior limbs. In man the fore-legs and fore-feet are replaced by arms and by what we call hands. For of all animals man alone stands erect, in accordance with his god-like nature and essence. For it is the function of the god-like to think and to be wise ; and no easy task were this under the burden of a heavy body, pressing down from above and obstructing by its weight the motions of the intellect and of the common sense.⁸ When, moreover, the weight and corporeal substance become excessive, the body must of necessity incline towards the ground. In such cases therefore nature, in order to give support to the body, has replaced the arms and hands by fore-feet, and has thus converted the animal into a quadruped. For, as every animal that walks must of necessity have the two hinder feet, such an animal becomes a quadruped, its body inclining downwards in front from the weight which its soul cannot sustain. For all animals, man alone excepted, are dwarf-like in form. For, as in a dwarf, their upper part is large, while that which bears the weight and is used in walking is comparatively small.⁹ This upper part is what we call the trunk, and reaches from the mouth to the vent. In man it is duly proportionate to the part below, and diminishes much in its comparative size as the man attains to full growth. But in his infancy the contrary obtains, and the upper parts are large, the lower small ; so that the infant can only crawl, and is unable to walk ; nay, at first cannot even crawl, but remains without motion. In fact all children are dwarfs in shape, but cease to be so as they become men, from the growth of their lower portion ; whereas in quadrupeds the reverse occurs, their lower parts being largest

in youth, and advance of years bringing increased growth above, that is in the trunk, which extends from the buttocks to the head. Thus it is that colts are scarcely, if at all, below full-grown horses in height; and that while quite young they can touch their heads with the hind legs, though this is no longer possible when they are older. Such then is the form of animals that have either a solid or a cloven hoof. But such as are polydactylous and without horns; though they too are of dwarf-like shape, are so in a less degree; nor is the original disproportion between their upper and lower parts increased by aftergrowth.¹⁰

Dwarf-like again is the race of birds and of fishes; and so in fact, as already has been said, is every animal that has blood. This is the reason that no other animal is so intelligent as man. For even among men themselves if we compare children with adults, or such adults as are of dwarf-like shape with such as are not, we find that, whatever other superiority the former may possess, they are at any rate deficient as compared with the latter in intelligence. The explanation, as already stated, is that their psychical principle is corporeal, and much impeded in its motions. Let now a still further decrease occur in the¹¹ elevating heat, and a still further increase in the earthy matter, and the animals become smaller in bulk, and their feet more numerous, until at a later stage they become apodous, and extended full length on the ground. Thus, by gradual small successions of change, they come to have their principal organ below; and at last their cephalic part becomes motionless and destitute of sensation. Thus the animal becomes a plant, that has its upper parts downwards and its lower parts above.¹² For in plants the roots are the equivalents of mouth and head, while the seed, which is produced above at the extremities of the twigs, has an opposite significance.¹³

The reasons have now been stated why some animals have many feet, some only two, and others none; why, also, some living things are plants and others animals; and, lastly, why man alone of all animals stands erect. Standing thus erect, man has no need of legs in front, and in their stead has been endowed by nature with arms and hands. Now it is the opinion of Anaxagoras, that the possession of these hands is the cause of man being of all animals the most intelligent. But it is rational to suppose that the possession of hands is the consequence rather than the cause

of his superior intelligence. For the hands are instruments or organs, and the invariable plan of nature in distributing the organs is to give each to such animal as can make use of it; nature acting in this matter as any prudent man would do. For to such an one it would seem much more appropriate to take a person who was already a flute-player and give him a flute, than to take one who possessed a flute and teach him the art of flute-playing. For by the former plan something comparatively insignificant would be added to something of much greater importance; while by the latter the more valuable and the more important element would be superadded to the less valuable one. Seeing then that it is a better plan to assign an instrument to a workman than to assign a workman to an instrument, and seeing also that of all available plans nature invariably adopts the best, we must conclude that man does not owe his superior intelligence to his hands, but his hands to his superior intelligence.¹⁴ For the most intelligent of animals is the one which would put the most organs to the best use; and the hand is apparently not a single organ but many in one; for it is an organ that can serve in the place of many. This instrument, then,—the hand—of all instruments the most variously serviceable; has been given by nature to man, the animal of all animals the most capable of acquiring the most varied handicrafts.

Much in error then are they, who say that the construction of man is not only faulty, but inferior to that of all other animals; seeing that he is, as they point out, barefooted, naked, and without weapon of which to avail himself. For other animals have each but one mode of defence, and this they can never change; so that they must perform all the offices of life and even, so to speak, sleep with sandals on, never laying aside whatever serves as a protection to their bodies, nor changing such single weapon as they may chance to possess. But to man numerous modes of defence are open, and these moreover he may change at will; as also he may adopt such weapon as he pleases, and at such times as suit him. For the hand is talon, hoof, and horn, at will. So too is it spear, and sword, and whatsoever other weapon or instrument you please; for all these can it be from its power of grasping and holding them all.¹⁵ In harmony with this varied office is the form which nature has contrived for it. For it is split into several divisions, and these are capable of divergence.

Such capacity of divergence does not prevent their again converging so as to form a single compact body, whereas had the hand been an undivided mass, divergence would have been impossible. Again these parts may be used singly or together, and in various combinations. The joints, moreover, of the fingers are well constructed for prehension and for pressure. One of the digits also, and this not long like the rest but short and thick, is placed laterally. For were it not so placed all prehension would be as impossible, as were there no hand at all. For the pressure of this lateral digit is applied from below upwards, while the rest act from above downwards; an arrangement which is essential, if the grasp is to be firm and hold like a tight clamp. As for the shortness of this lateral digit, the object is to increase its strength, so that it may be able, though but one, to counterbalance its more numerous opponents. Indeed were it long it would be of no use. This is the explanation of its being sometimes called the great digit, in spite of its small size; for without it all the rest would be practically useless. The finger which stands at the other end of the row is small, while the central one of all is long, like the centre oar in a ship. This is rightly so; for in grasping an object, as a workman grasps his tool, it is the central part of the encircling hold which is of the most importance.

No less skilfully contrived are the nails. For, while in man these serve simply as coverings to protect the ends of the fingers, in other animals they are also used for active purposes; [and their form in each case is suited to their office.]¹⁶

The arms in man and the fore limbs in quadrupeds bend in precisely¹⁷ contrary directions, this difference having reference to the ingestion of food¹⁸ and to the other offices which belong to these parts. For quadrupeds use their anterior limbs as feet in progression, and must therefore have them bent inwards. In such of the quadrupeds indeed as are polydactylous, these fore limbs are at any rate intended not only to serve in locomotion, but also to act as hands. And they are in fact so used, as any one may see. For these animals seize hold of objects, and also repel assailants, with their anterior limbs; whereas quadrupeds with solid hoofs use their hind legs for this latter purpose.¹⁹ For their fore limbs are not analogous to the arms and hands of man.²⁰

It is this hand-like office of the anterior limbs which explains

why in some of the polydactylous quadrupeds, such as wolves, lions, dogs, and leopards, there are five digits on each fore-foot, though there are only four on each hind one. For the fifth digit of the foot corresponds to the fifth digit of the hand, and like it is called the big one. It is true that in the smaller polydactylous quadrupeds the hind feet also have each five toes. But this is because these animals are creepers; and the increased number of nails serves to give them a tighter grip of the ground, and so enables them to creep up steep places, or even to run overhead, with greater facility.²¹

In man between the arms, and in other animals between the fore legs, lies what is called the breast. This in man is broad, as one might expect. For as the arms are set laterally on the body, they offer no impediment to the expansion of this part. But in quadrupeds the breast is narrow, owing to the legs having to be extended in a forward direction in progression and locomotion.

Owing to this narrowness the mammæ of quadrupeds are never placed on the breast. But in the human body there is ample space in this part; moreover the heart and neighbouring organs require such protection as would be afforded by a fleshy covering; and for these reasons the mammæ in man are placed on the breast, side by side. In the male these mammæ are of a fleshy substance and are therefore of use in the way just stated; but, in the female, nature, in accordance²² with what we say is her frequent practice, makes these organs minister to an additional function, employing them as a store-place of nutriment for the offspring. The human mammæ are two in number, in accordance with the division of the body into two halves, a right and a left. They are somewhat firmer than they would otherwise be, because the ribs²³ in this region are joined together; and they form two distinct masses, because their presence is in no wise burdensome.²⁴ In other animals²⁵ than man, it is impossible for the mammæ to be placed on the breast between the fore legs, for they would interfere with locomotion; they are therefore disposed of otherwise, and in a variety of ways. Thus in such animals as produce but few at a birth, whether horned quadrupeds or those with solid hoofs, the mammæ are placed in the groins, and are two in number,²⁶ while in such as produce litters, or such as are polydactylous, the dugs are

either numerous and placed laterally on the belly, as in swine and dogs, or are only two in number and placed in the centre of the abdomen, as is the case in the lion.²⁷ The explanation of this latter condition is not that the lion produces few at a birth, for sometimes it has more than two cubs at a time, but is to be found in the fact that this animal has no plentiful supply of milk. For, being a flesh-eater, it gets food at but rare intervals, and such nourishment as it obtains is all expended on the growth of its body.

In the elephant also there are but two mammæ, which are placed under the axillæ of the fore limbs. The mammæ are not more than two, because this animal has only a single young one at a birth; and they are not placed in the groins, because they never occupy that position in any polydactylous animal such as this.²⁸ Lastly they are placed above close to the axillæ, because this is the position of the foremost dugs in all animals whose dugs are numerous, and the dugs, so placed give the most milk. A proof of this is furnished by the sow. For she always presents these foremost dugs to the first-born of her litter. A single young one is of course a first-born; and so such animals as only produce a single young one must have these anterior dugs to present to it; that is they must have dugs close to the axillæ. This, then, is the reason why the elephant has but two mammæ, and why they are so placed. But, in such animals as have litters of young, the dugs are disposed about the belly; the reason being that more dugs are required by those that have more young to nourish. Now as there are but two sides to the body, the right and the left, it is impossible that these dugs should be set transversely in rows of more than two. They must therefore be placed lengthways, and the only place where there is sufficient length for this is the region between the front and hind legs. As to the animals that are not polydactylous but produce few at a birth, or have horns, their dugs are placed in the groins. The horse, the ass, the camel are examples; all of which bear but a single young one at a time, and of which the two former have solid hoofs, while in the last the hoof is cloven. As still further examples, may be mentioned the deer, the ox, the goat, and all other similar animals.

The explanation is that in these animals growth takes place

in an upward direction;²⁹ so that there must be an abundant collection of residual matter and of blood in the lower region, that is to say in the neighbourhood of the excremental orifices. Here therefore nature has placed the *mammæ*. For the part whence they can most easily derive nutriment will clearly be that part in which the nutriment is set in motion. In man there are *mammæ* in the male as well as in the female; but some of the males of other animals are without them. Such for instance is the case with horses, some stallions being destitute of these parts, while others that resemble their dams have them.³⁰ Thus much then concerning the *mammæ*.

Next after the breast comes the region of the belly, which is left unenclosed by the ribs for a reason which has already been given;³¹ namely that there may be no impediment to the swelling which necessarily occurs in the food as it gets heated, nor to the expansion of the womb in pregnancy.

At the extreme end of what is called the trunk are the parts concerned in the evacuation of the solid and also of the fluid residues. In all sanguineous animals with some few exceptions,³² and in all vivipara without any exception at all, the same part which serves for the evacuation of the fluid residue is also made by nature to serve in sexual congress, and this alike in male and female. For the semen is a kind of fluid and a residue. The proof of this will be given hereafter, but for the present let it be taken for granted. The like holds good of the menstrual fluid in women, and of the part by which they give issue to it. This also, however, is a matter of which a more accurate account will be given hereafter. For the present let it be simply stated as a fact, that the catamenia of the female as also the semen of the male are residual matter.³³ The catamenia, then, and the semen are both fluid, and thus it is only reasonable that the same parts which serve for voidance of the urine should give issue to these residues, which have identical or similar characters. Of the internal structure of these parts, and of the differences which exist between the parts concerned with semen and the parts concerned with conception, a clear account is given in the book of Researches concerning Animals and in the treatises on Anatomy. Moreover I shall have to speak of them again when I come to deal with Generation and Development. As regards, however, the external shapes of the parts, it is plain enough that they are adapted to their

operations, as indeed of necessity they must be. There are, however, differences in the male organ corresponding to differences in the body generally. For it is not of an equally sinewy character in all animals. This organ, again, is the only one that, independently of any morbid change, admits of augmentation and of diminution of bulk. The former change is of service in copulation, while the other is required for the advantage of the body at large. For, were the organ constantly in a state of erection, it would be an incumbrance. The organ therefore has been formed of such constituents, as will admit of either condition. For it is partly sinewy, partly cartilaginous,³⁴ and thus is enabled either to contract or to become extended, and is capable of admitting air.³⁵

All female quadrupeds void their urine backwards, because the position of the parts which this implies is useful to them in the act of copulation. This is the case with very few males, though there are some exceptions, as the lynx, the lion, the camel, and the hare.³⁶ No quadruped with a solid hoof is retromingent.

The posterior portion of the body and the parts about the legs are different in man from what they are in quadrupeds. Nearly all these latter have a tail, and this whether they are viviparous or oviparous. For, even if the tail be of no great size, yet they have a kind of scut; as at any rate a small representative of it. But man is tail-less. He has however buttocks, which exist in none of the quadrupeds. His legs also, calves and thighs alike, are fleshy; while in all other animals that have legs, viviparous or not, they are fleshless, being made of sinew and bone and a substance resembling fish-spine.³⁷ For all these differences there is, so to say, one common explanation, and this is that of all animals man alone stands erect. It was to facilitate the maintenance of this position that nature made his upper parts light, taking away some of their corporeal substance, and using it to increase the weight of the parts below, so that the buttocks, the thighs, and the calves of the legs, all became fleshy. The character which she thus gave to the buttocks renders them at the same time useful in resting the body. For standing causes no fatigue to quadrupeds, and even the long continuance of this posture produces in them no weariness; for they are supported the whole time by four props,

which is much as though they were lying down.³⁸ But to man it is no easy task to remain for any length of time on his feet, his body demanding rest in a sitting position. This, then, is the reason why man has buttocks and fleshy legs; and the presence of these fleshy parts explains why he has no tail. For the nutriment which would otherwise go to the tail is used up in the production of these parts,³⁹ while at the same time the existence of buttocks does away with the necessity of a tail. But in quadrupeds and other animals the reverse obtains. For they are of dwarf-like form, so that all the pressure of their weight and corporeal substance is on their upper part; and is withdrawn from the parts below.⁴⁰ On this account they are without buttocks and have hard legs. In order however to cover and protect that part which serves for the evacuation of excrement, nature has given them a tail or brush, subtracting for the purpose some of the nutriment which would otherwise go to the legs. Half-way in shape between man and quadrupeds is the ape,⁴¹ belonging therefore to neither or to both, and having on this account neither tail nor buttocks; no tail in its character of biped, no buttocks in its character of quadruped. There is a great diversity of so-called tails; and this organ like others⁴² is sometimes used by nature for bye-purposes, being made to serve not only as a covering and protection to the fundament, but also for other uses and advantages of its possessor.⁴³

There are differences in the feet of quadrupeds. For in some of these animals there is a solid hoof, and in others a hoof cloven into two, and again in others a foot divided into many parts.

The hoof is solid when the body is large and the earthy matter present in great abundance; in which case the earth, instead of forming teeth and horns, is separated in the character of a nail, and being very abundant forms one continuous nail, that is a hoof, in place of several. This consumption of the earthy matter on the hoof explains why these animals, as a rule, have no⁴⁴ huckle-bones; a second reason, however, being that the presence of such a bone in the joint of the hind leg somewhat impedes its free motion. For extension and flexion can be made more rapidly in parts that have but one angle than in parts that have many. But the presence of a huckle-bone, as a connecting bolt, is the introduction as it were of a new limb-segment between the two ordinary ones. Such an addition adds to the weight of

the foot, but renders the act of progression more secure. Thus it is that even in such animals as have a huckle-bone, it is only in the posterior limbs and never in the anterior ones that this bone is found. For the anterior limbs, moving as they do in advance of the others, require to be light and capable of ready flexion, whereas firmness and extension are what is wanted in the hind limbs. Moreover a huckle-bone adds weight to the blow of a limb, and so renders it a suitable weapon of defence; and these animals all use their hind legs to protect themselves, kicking out with their heels against anything which annoys them. In the cloven-hoofed quadrupeds the lighter character of the hind legs admits of there being a huckle-bone; and the presence of the huckle-bone prevents them from having a solid hoof, the bony substance remaining in the joint, and therefore being deficient in the foot. As to the polydactylous quadrupeds, none of them have huckle-bones. For if they had they would not be polydactylous, but the divisions of the foot would only extend to that amount of its breadth which was covered by the huckle-bone.⁴⁵ Thus it is that most of the animals that have huckle-bones are cloven-hoofed.

Of all animals man has the largest foot in proportion to the size of the body.⁴⁶ This is only what might be expected. For seeing that he is the only animal that stands erect, the two feet which have to bear all the weight of the body must be both long and broad. Equally intelligible is it that the proportion between the size of the fingers and that of the whole hand should be inverted in the case of the toes and feet. For the function of the hands is to take hold of objects and retain them by pressure; so that the fingers require to be long. For it is by its flexed portion that the hand grasps an object. But the function of the feet is to enable us to walk with security; so that here the undivided part is to be looked on as of most importance. However it is better for an extremity to be divided than to be undivided. For in an undivided foot disease of any one part would affect by sympathy the whole; whereas, if the foot be divided into separate digits, there is not an equal liability to such an occurrence.⁴⁷ The digits, again, by being short would be less liable to injury. For these reasons the feet in man are many-toed, while the separate digits are of no great length. The toes, finally, are furnished with nails for the same reason as are the

fingers, namely because such projecting parts are weak and therefore require especial protection.

(*Ch.* 11.) We have now done with such sanguineous animals as live on land and bring forth their young alive;¹ and, having dealt with all their main kinds, we may pass on to such sanguineous animals as are oviparous. Of these some have four feet, while others have none. The latter form a single genus, namely the Serpents; and why these are apodous has been already explained in the dissertation on Animal Progression.² Irrespective of this, absence of feet, serpents resemble the oviparous quadrupeds in their conformation.³

In all these animals there is a head with its component parts; its presence being determined by the same causes⁴ as obtain in the case of other sanguineous animals; and in all, with the single exception of the river crocodile, there is a tongue inside the mouth.⁵ In this one exception there would seem to be no actual tongue, but merely a space left vacant for it. The reason is that a crocodile is in a way a land-animal and a water-animal combined. In its character of land-animal it has a space for a tongue; but in its character of water-animal it is without the tongue itself. For in fishes, as has already been mentioned, there is either no appearance of a tongue at all, unless the mouth be stretched open very widely indeed; or, if there be a tongue, it is indistinctly separated from the rest of the mouth.⁶ The reason for this is that a tongue would be of but little service to such animals, seeing that they are unable to chew their food or to taste it before swallowing, the pleasurable sensations they derive from it being limited to the act of deglutition.⁷ For it is in their passage down the gullet that solid edibles cause enjoyment, while it is by the tongue that the savour of fluids is perceived. Thus it is during deglutition that the oiliness, the heat, and other such qualities of food, are recognised; and in fact the satisfaction from solid edibles and dainties in general is derived almost exclusively from the dilatation of the œsophagus during deglutition.⁸ This pleasurable sensation, then, belongs to all sanguineous animals, viviparous or not, alike; but, while the rest have in addition the sensations of taste, tongueless animals have no other satisfaction than it. What has now been said explains why intemperance as regards drinks and savoury fluids does not go hand in hand with intemperance as regards eating and solid relishes.⁹

In some oviparous quadrupeds, namely in lizards, the tongue is bifid, as also it is in serpents, and its terminal divisions are of hair-like fineness, as has already been described.¹⁰ (Seals also have a forked tongue.) This it is which accounts for all these animals being so fond of dainty food.¹¹ The teeth in these oviparous quadrupeds are of the sharp interfitting kind, as also are the teeth of fishes.¹² The organs of all the senses are present; and these resemble those of other animals. Thus there are nostrils for smell, eyes for vision, and ears for hearing. The latter organs, however, do not project from the sides of the head, but consist simply of the duct, as also is the case in birds. This is due in both cases to the hardness¹³ of the integument; birds having their bodies covered with feathers, and these oviparous quadrupeds with horny plates. These plates are equivalent to scales, but of a harder character. This is very manifest in tortoises and river crocodiles, and also in the large serpents. For so hard are they in these animals, as to become even stronger than the bones.¹⁴

These animals have no upper eyelid, but close the eye with the lower lid.¹⁵ In this they resemble birds, and the reason of the peculiarity is the same as was assigned in their case.¹⁶ Among birds there are some that can not only thus close the eye, but can also blink by means of a membrane which comes from its inner corner. But none of the oviparous quadrupeds blink in this manner.¹⁷ For their eyes are harder than those of birds.¹⁸ The reason for this is that sharp-sightedness is of very considerable service to birds, flying as they do in the air, whereas it would be of comparatively small use to the oviparous quadrupeds, seeing that they all live in holes.

Of the two separate portions which constitute the head, namely the upper part and the lower jaw, the latter in man and in the viviparous quadrupeds moves not only upwards and downwards, but also from side to side;¹⁹ while in fishes, and birds, and oviparous quadrupeds, the only movement is up and down. The reason is that this latter movement is the one required in biting and dividing food, while the lateral movement serves to reduce substances to a pulp. To such animals, therefore, as have grinder-teeth this lateral motion is of service; but to those animals that have no grinders it would be quite useless, and they are therefore invariably without it. For nature never makes anything that is

superfluous. While in all other animals it is the lower jaw that is moveable, in the river crocodile it is exceptionally the upper one.²⁰ This is because the feet in this creature are so excessively small as to be perfectly useless for seizing and holding prey; on which account nature has given it a mouth that can serve for these purposes in their stead. For clearly that direction of motion, which will give the greater force to a blow, will be the more serviceable one in holding or in seizing prey; and a blow from above is always more forcible than one from below. Seeing, then, that both the prehension and the mastication of food are offices of the mouth, and that the former of these two is the more essential in an animal that has neither hands nor suitably formed feet, these crocodiles will derive greater benefit from a motion of the upper jaw downwards than from a motion of the lower jaw upwards. The same considerations explain why crabs move only the upper division of each claw and not the lower. For their claws are substitutes for hands, and so require to be suitable for the prehension of food, and not for its comminution; for such comminution and mastication is the office of teeth. In crabs, then, and in such other animals as are able to seize their food in a leisurely manner, and are not forced by being constantly in the water to perform this office with the mouth,²¹ the two functions are assigned to different parts, prehension to the hands or feet, the division or mastication of food to the mouth. But in crocodiles the mouth has been so framed by nature as to serve both purposes, the jaws being made to move in the manner just described.

All these animals have a neck, which is the necessary consequence of their having a lung. For the windpipe by which the air is admitted to the lung is of some length.²² If however the definition of a neck be correct, which calls it the portion between the head and the shoulders, a serpent can scarcely be said with the same right as the rest of these animals to have a neck, but only to have something analogous to that part of the body. It is a peculiarity of serpents, as compared with other animals allied to them, that they are able to turn their head backwards without stirring the rest of the body. The reason of this is that a serpent, like an insect, has a body that admits of being curled up, its vertebræ being cartilaginous and easily bent.²³ The faculty in question belongs then to serpents simply as a necessary consequence of this character of their vertebræ; but at the same

time it has a final cause, for it tells to their advantage by enabling them to guard against attacks from behind. For their body, owing to its length and its want of feet, is ill-suited for turning round and protecting the hinder parts; and merely to lift the head, without the power of turning it round, would be of no use whatsoever.

The ovipara with which we are dealing have, moreover, a part which corresponds to the breast; but neither here nor elsewhere in their body have they any mammæ, as neither has any bird or fish. This is a consequence of their having no milk; for a mamma is a receptacle for milk and, as it were, a vessel to contain it. This absence of milk is not peculiar to these animals, but is common to all such as are not internally viviparous.²⁴ For all such produce eggs, and the nutriment which corresponds to the milk is in them formed in the egg. Of all this, however, a clearer account will be given in the treatise on Generation and Development.²⁵ As to the mode in which the joints bend, a general account, in which all animals are considered, has already been given in the dissertation on Progression. So also the reasons for the presence or absence of a tail have been already stated at large.²⁶ It suffices therefore to say that these animals always have a tail of some sort, though the size which it attains varies considerably.

Of all the ovipara that live on land there is none so lean as the Chamæleon.²⁷ For there is none that has so little blood. The explanation of this is to be found in the psychical temperament of the creature. For it is of a timid nature, as the frequent changes it undergoes in its outward aspect testify.²⁸ But fear is a refrigeration, and results from deficiency of natural heat and scantiness of blood.²⁹

We have now done with the sanguineous animals, both such as are quadrupedous and such as are apodous, and have stated with sufficient completeness what external parts they possess, and for what reasons they have them.

(Ch. 12.) The differences of birds¹ compared one with another are differences of magnitude, and of the greater or smaller development of parts. Thus some have long legs, others short legs; some have a broad tongue, others a narrow tongue; and so on with the other parts. There are very few of their parts that present any greater differences than these, taking birds by

themselves.² But, when we compare birds with other animals, we find the parts differing not merely in relative size but in form. Thus birds are invariably feathered, and this is a peculiarity which is characteristic of them. For while other animals are some hairy, some scaly, some covered with scaly plates, birds alone are feathered. Insects it is true are also feathered, but the feathers of a bird are split and different in kind from the undivided feathers of an insect; for the bird's feather is barbed, while the insect's is not; the bird's feather has a shaft, the insect's has none.³

A second strange peculiarity which distinguishes birds from all other animals is their beak. For as in elephants⁴ the nostril serves in place of hands, and as in some insects⁵ the tongue serves in place of mouth, so in birds there is a bony⁶ beak which serves in place of teeth and lips. Their organs of sense have already been considered.

All birds have a neck extending from the body; and the purpose of this neck is the same as in such other animals as have one.⁷ This neck in some birds is long, in others short; its length as a general rule being pretty nearly determined by that of the legs. For long-legged birds have a long neck, short-legged birds a short one, to which rule, however, the web-footed birds form an exception. For to a bird perched up on long legs a short neck would be of no use whatsoever in collecting food from the ground; and equally useless would be a long neck, if the legs were short. Such birds, again, as are carnivorous would find length in this part interfere greatly with their habits of life. For a long neck is a weak one, and it is on their superior strength that carnivorous birds depend for their subsistence. No bird, therefore, that has talons ever has an elongated neck. In web-footed birds, however, and in those other birds that seem to belong to the same genus, inasmuch as their toes though actually separate are yet flattened and expanded into lobes,⁸ the neck is elongated, so as to be suitable for collecting food from the water; while at the same time the legs are short, so as to serve in swimming.

The beaks of birds, as their feet, vary with their modes of life. For in some the beak is straight, in others crooked; straight, in those whose food requires that form; crooked, in those that live on flesh. For a crooked beak is an advantage in fighting;

and carnivorous birds must of course get their food from the bodies of other animals, and in most cases by violence. In such birds, again, as live in marshes and are herbivorous the beak is broad, this form being best suited for digging and cropping, and for pulling up plants. In some of these marsh birds, however, the beak is elongated, as also is the neck, the reason for this being that the bird gets its food from some depth below the surface. For most birds that have this conformation, and most of those that are either actually web-footed or web-footed in the partial way already mentioned, live by preying on some of the small animals that are to be found in water. In these the neck acts the part of a fishing-rod, the beak representing the line and hook.

The upper and under sides of the body, that is of what in quadrupeds is called the trunk, present in birds one unbroken surface. For there are no arms nor fore legs attached to it, but in their stead wings, which are a distinctive peculiarity of birds; and, as these wings are substitutes for arms, their terminal segments lie on the back in the place of a shoulder-blade.⁹

The legs are two in number, as in man; not however, as in man, bent outwards, but bent inwards like the hind legs of a quadruped.¹⁰ The wings are bent like the fore legs of a quadruped, having their convexity turned outwards. That the feet should be two in number is a matter of necessity. For a bird is essentially a sanguineous animal, and at the same time essentially a winged animal; and no sanguineous animal has more than four points for motion.¹¹ In birds, then, as in those other sanguineous animals that live and move upon the ground,¹² the limbs attached to the trunk are four in number. But, while in all the rest these four limbs consist of a pair of arms and a pair of legs, or of four legs as in quadrupeds, in birds alone the arms or fore legs are replaced by a pair of wings, and this is their distinctive character. For it is of the essence of a bird that it shall be able to fly; and it is by the extension of wings that this is made possible. Of all arrangements, then, the only possible, and so the necessary, one is that birds shall have two feet; for this with the wings will give them four points for motion. The breast in all birds is sharp-edged, and fleshy.¹³ The sharp edge is of advantage in flight; for broad surfaces move with considerable difficulty, owing to the large quantity of air which they have to displace. The fleshy

character acts as a protection to the breast, which owing to its form would be very weak, were it not amply covered.¹⁴

Below the breast lies the belly, extending, as in quadrupeds and in man, to the vent and to the place where the legs are jointed to the trunk.

Such, then, are the parts which lie between the wings and the legs. Birds like all other animals, whether produced viviparously or from eggs, have an umbilical cord during their development, but, when the bird has attained to fuller growth, no signs of this remain visible. The explanation of this will be set forth in the treatise on Generation and Development, but amounts to this, that in birds the umbilical cord unites with the intestine, and is not a portion of the vascular system, as is the case in the vivipara.¹⁵

Some birds, again, are well adapted for flight, their wings being large and strong. Such, for instance, are those that have talons and live on flesh. For their mode of life renders the power of flight a necessity, and it is on this account that their feathers are so abundant and their wings so large. Besides these, however, there are also other genera of birds that can fly well; all those, for example, that depend on speed for security, and all those again that are of migratory habits. On the other hand some kinds of birds have heavy bodies and are not constructed for flight. These are birds that are frugivorous and live on the ground, or that are able to swim and gain their livelihood in watery places. In those that have talons the body, when stripped of [its feathers and] wings, is small; for the nutriment is consumed¹⁶ in the production of these weapons and defensive appliances; whereas in birds that do not fly the contrary obtains, and the body is bulky and so of heavy weight. In some of these heavy birds the legs are furnished with what are called spurs, which replace the wings as a means of defence. Spurs and talons never co-exist in the same bird. For nature never makes anything superfluous; and if a bird can fly, and has talons, it has no use for spurs; for these are weapons for fighting on the ground, and on this account are an appanage of certain heavy-bodied birds. These latter, again, would find the possession of talons not only useless but actually injurious; for the claws would stick into the ground and interfere with progression. This is the reason why all birds with talons walk so badly, and why

they never settle upon rocks.¹⁷ For the character of their claws is ill-suited for either action.

All this is the necessary consequence of the process of development. For the earthy matter in the body issues from it and is converted into some or other useful kind of weapon. That which flows upwards gives hardness or size to the beak; and, should any flow downwards, it either forms spurs upon the legs or gives size and strength to the talons. But it does not at one and the same time produce both these results, one in the legs, the other in the claws; for such a dispersion of this residual matter would destroy all its efficacy. In other birds this earthy residue furnishes the legs with the material for their elongation; or sometimes, in place of this, fills up the interspaces between the toes. Thus it is simply a matter of necessity,¹⁸ that such birds as swim shall either be actually web-footed, or, if not so, shall at any rate have a kind of broad blade-like margin running along the whole length of each distinct toe.¹⁹ The forms, then, of these feet are simply the necessary results of the causes that have been mentioned. Yet at the same time they are intended for the animal's advantage. For they are in harmony with the mode of life of these birds, who, living on the water, where their wings are useless, require that their feet shall be such as to serve in swimming. For these feet are so developed as to resemble the broad-bladed oars of a boat, or the fins of a fish; and the destruction of the foot-web has precisely the same effect as the destruction of the fin; that is to say, it puts an end to all power of swimming.²⁰

In some birds the legs are very long, the cause of this being that they inhabit marshes. I say the cause, because nature makes the organs for the function, and not the function for the organs.²¹ It is, then, because these birds are not meant for swimming that their feet are unwebbed, and it is because they live on ground that gives way under the foot that their legs and toes are elongated, and that these latter in most of them have an extra number of joints.²² Again, seeing that all birds are made of the same substance,²³ and yet are not all constructed alike to fly, the materials which in some are expended on organs of flight can in others be diverted to different purposes; and in these birds the nutriment which would otherwise go to form the tail-feathers is used in increasing the dimensions of the legs. This is the reason why these birds when they fly make use of their legs

as a tail, stretching them out behind, and so rendering them serviceable, whereas in any other position they would be simply in the way.²⁴

In other birds, where the legs are short, these are held close against the belly during flight. In some cases this is merely to keep the feet out of the way, but in birds that have talons the position has a further purpose, being the one best suited for rapine.²⁵ Birds that have a long and a thick neck, keep it stretched out during flight ; but those whose neck though long is slender fly with it coiled up. For in this position it is protected, and less likely to get broken, should the bird fly against any obstacle.²⁶

In all birds there is an ischium, but of such length that it would scarcely be taken for an ischium, but rather for a second thigh-bone ; for it extends as far as to the middle of the belly.²⁷ The reason for this is that the bird is a biped, and yet is unable to stand erect. For if its ischium extended but a short way from the fundament, and then immediately came the leg, as is the case in man and in quadrupeds, the bird would be unable to stand upright at all. For while man stands erect, and while quadrupeds have their heavy bodies propped up in front by the fore legs, birds can neither stand erect owing to their dwarf-like²⁸ shape, nor have anterior legs to prop them up, these legs being replaced by wings. As a remedy for this nature has given them a long ischium, and brought it to the centre of the body, fixing it firmly ; and under this central point she has placed the legs, that the weight on either side may be equally balanced, and standing or progression rendered possible. Such then is the reason why a bird, though it is a biped, does not stand erect. Why its legs are destitute of flesh has also already been stated ; for the reasons are the same as in the case of quadrupeds.²⁹

In all birds alike, whether web-footed or not, the number of toes in each foot is four.³⁰ For the Libyan ostrich may be disregarded for the present, and its cloven hoof and other discrepancies of structure as compared with the tribe of birds will be considered farther on. Of these four toes three are in front, while the fourth points backwards, serving, as a heel, to give steadiness. In the long-legged birds this fourth toe is much shorter³¹ than the others, as is the case also with the *Crex*.³² In no bird is the number of toes more than four. The arrangement of the toes is such

as has been described, in all birds with one exception. This exception is the wryneck.³³ Here two of the toes are in front, the other two behind; and the reason for this is that the body of the wryneck is not inclined forwards so much as that of other birds. All birds have testicles; but they are inside the body. The reason for this will be given in the treatise on the Generation and Development of Animals.³⁴

(*Ch.* 13.) Thus then are fashioned the parts of birds. But in fishes¹ a still further stunting has occurred in the external parts. For here, for reasons already given,² there are neither arms nor legs nor wings, the whole body from head to tail presenting one unbroken surface. This tail differs in different fishes, in some having a consistency similar to that of the body, while in others, namely in some of the flat kinds, it is spinous and elongated, because the material which should have gone to the tail has been diverted thence and used to increase the breadth of the body. Such for instance is the case with the Torpedos,³ the Trygons,⁴ and whatever other Selachia there may be of like nature. In such fishes, then, the tail is spinous and long; while in some others it is short and fleshy, for the very same reason which makes it spinous and long in the Torpedo. For to be short and fleshy comes to exactly the same thing as to be long and less amply furnished with flesh.

What has occurred in the Fishing-frog,⁵ is exactly the reverse of what has occurred in the other instances just given. For here the anterior and broad part of the body is destitute of flesh, and so all the fleshy substance which has been thence diverted has been placed by nature in the tail and hinder portion of the body.

In fishes there are no limbs attached to the body. For in accordance with their essential constitution they are swimming animals; and nature never makes anything superfluous or void of use. Now, seeing that fishes are essentially sanguineous animals, they must have four points of motion; and, seeing that they are meant for swimming, these must be fins and not feet; for feet are attached to the body that they may be of use in walking on dry ground.⁶ Moreover it is impossible for fishes, that are sanguineous animals, to have four fins and also at the same time to have feet or in fact any other kind of limb. Tadpoles,⁷ it is true, though they have gills, have feet; but then they have no fins, but merely have their tail flattened out and loose in texture.⁸

Fishes, unless, like the Batos⁹ and the Trygon, they are broad and flat, have four fins, two on the upper and two on the under aspect of the body; and no fish ever has more than these. For, if it had, it would be an ex-sanguineous animal.

The upper pair of fins is present in nearly all fishes, but not so the under pair; for these are wanting in some of those fishes that have long thick bodies, such as the eel, the conger, and the Cestreus which is found in the lake at Siphæ.¹⁰ When the body is still more elongated, and resembles that of a serpent rather than that of a fish, as is the case in the Smyræna,¹¹ there are absolutely no fins at all; and locomotion¹² is effected by the flexures of the body, the water being put to the same use by these fishes as is the ground by serpents. For in fact serpents swim in water exactly in the same way as they glide on the ground. The reason for these serpent-like fishes being without fins is precisely the same as that which causes serpents to be without feet; and what this is has been already stated in the dissertation on the Progression and the Motion of Animals.¹³ The reason was this. If the points of motion were four,¹⁴ motion would be effected under difficulties. For either the two pairs of fins would be near each other, in which case motion would scarcely be possible, or they would be at a very considerable distance apart, in which case the long interval between them would be just as great an evil. On the other hand, to have more than four such motor points is out of the question. For that would convert the fishes into ex-sanguineous animals.¹⁵ A similar explanation applies to the case of those fishes that have only two fins. For here again the body is of great length and like that of a serpent, and its undulations do the office of the two missing fins. It is owing to this that such fishes can even crawl on dry ground, and can live there for a considerable time; and do not, like other fishes, begin to gasp the moment they are taken out of the water; and the less so, the nearer their nature conforms to that of land-animals. In such fishes as have but two fins it is the upper pair that is present,¹⁶ excepting when the flat broad shape of the body prevents this. These fins, moreover, in such cases are placed at the head, because in this region there is no elongation, which might serve in the absence of fins as a means of locomotion; whereas in the direction of the tail there is a considerable lengthening out in fishes of this conformation. As for the Bati and the like, they

use the marginal parts of their flattened bodies in place of fins for swimming.¹⁷

In the Torpedo and the Fishing-frog the breadth and flatness of the anterior part of the body is not so great as to render locomotion by fins impossible. It necessitates, however, the displacement of the upper pair to a point further back, and the advancement of the under pair to the head. At the same time to compensate for this advancement these lower fins are reduced in size so as to be smaller than the upper ones.¹⁸ In the Torpedo the two upper fins are placed on the tail, and the fish uses the broad expansion of its body to supply their place, each lateral half of its circumference serving the office of a fin.¹⁹

The head, with its several parts, as also the organs of sense, have already come under consideration.²⁰

There is one peculiarity which distinguishes fishes from all other sanguineous animals, namely, the possession of gills. Why they have these organs has been explained in the treatise on Respiration.²¹ These gills are in most fishes covered by opercula, but in the Selachia, owing to the skeleton being cartilaginous, there are no such coverings. For an operculum requires fish-bone for its formation, and in other fishes the skeleton is made of this substance, whereas in the Selachia it is invariably formed of cartilage. Again, while the motions of bony fishes are rapid, those of the Selachia are but sluggish, owing to the absence of bone and of sinew. But an operculum requires rapidity of motion, seeing that the office of the gills is to minister as it were to expiration.²² For this reason in Selachia the branchial orifices themselves effect their own closure, and thus there is no need for an operculum to ensure its taking place with due rapidity.²³ In some fishes the gills are numerous, in others few in number; in some again they are double, in others single. The last gill in most cases is single.²⁴ For a detailed account of all this, reference must be made to that part of the treatises on Anatomy which relates to fishes, and to the book of Researches concerning Animals.²⁵

It is the abundance or the deficiency of the cardiac heat which determines the numerical abundance or deficiency of the gills. For, the greater an animal's heat, the more rapid and the more forcible does it require the branchial movement to be;²⁶ and numerous and double gills act with more force and rapidity than few and single ones. Thus too it is that some fishes that have

but few gills, and those of comparatively small efficacy, can live out of water for a considerable time; for in them there is no great demand for refrigeration. Such, for example, are the eel and all other fishes of serpent-like form.²⁷

Fishes also present diversities as regards the mouth. In some this is placed in front, at the very extremity of the body, while in others, as the dolphin²⁸ and such fishes as resemble the Selachia, it is placed on the under surface; so that these fishes have to turn on their backs in order to take their food. The purpose of nature in this was apparently not merely to provide a means of salvation for other animals, by allowing them opportunity of escape during the time lost in the act of turning—for all the fishes with this kind of mouth prey on living animals—but also to prevent these fishes from giving way too much to their gluttonous ravening after food.²⁹ For had they been able to seize their prey more easily than they do, they would soon have perished from over-repletion. An additional reason for placing their mouth on the under surface is that the projecting extremity of the head is round and small, so that a mouth, if placed there, could not possibly open widely. In such fishes as have the mouth placed at the anterior extremity there are differences in the extent to which that orifice can open. In those that are carnivorous, such as the fishes³⁰ with sharp inter-fitting teeth, whose strength lies in their mouth, this orifice can gape widely, whereas it is placed at the point of a small tapering snout in all such as are not carnivorous.

The skin is in some fishes covered with scales, a scale being a shiny film which owing to its thinness becomes detached from the surface of the body. In others it is rough, as for instance in the Rhine,³¹ the Batos, and the like. In others again, but in very few, the skin is smooth.³² The Selachia have no scales, but a rough skin. This is explained by their cartilaginous skeleton. For the earthy material which has been thence diverted is expended by nature upon the skin.³³

No fish has testicles³⁴ either externally or internally; as indeed have no apodous animals, among which of course are included the serpents.³⁵ One and the same orifice serves both for the excrement and for the generative secretions,³⁶ as is the case also in all other oviparous animals, the quadrupeds as well as the rest, inasmuch as they have no urinary bladder and form no fluid excretion.³⁷

Such then are the characters which distinguish fishes from all other animals. But dolphins and whales and all such Cetacea³⁸ are without gills; and, having a lung, are provided with a blow-hole. By this they discharge the sea-water which has been taken into the mouth.³⁹ For, feeding as they do in the water, they cannot but let this fluid enter into their mouth, and, having let it in, they must of necessity let it out again. This they cannot do by the aid of gills; for the use of gills, as has been explained in the treatise on Respiration, is limited to such animals as do not breathe; so that no animal can possibly possess gills and at the same time be a respiratory animal.⁴⁰ In order, therefore, that these Cetacea may discharge the water, they are provided with a blow-hole. This is placed in front of the brain; for, if placed behind, it would have cut off the brain from the spine.⁴¹ The reason for these animals having a lung and breathing, is that their large size demands an excess of heat, to render motion possible.⁴² A lung, therefore, is placed within their body, and is fully supplied with blood-heat. These creatures are after a fashion land and water animals in one. For so far as they are inhalers of air they resemble land-animals, while they resemble water-animals in having no feet and in deriving their food from the sea. So also seals⁴³ lie half-way between land and water animals, and bats half-way between animals that live on the ground and animals that fly; and so may be said to belong to both kinds or to neither. For seals, if looked on as water-animals, are yet found to have feet; and, if looked on as land-animals, are yet found to have fins. For their hind feet are exactly like the fins of fishes. Their teeth also are all of the sharp and interfitting kind, as in fishes.⁴⁴ Bats again, if regarded as winged animals, have feet; and, if regarded as quadrupeds, are without them.⁴⁵ So also they have neither the tail of a quadruped nor the tail of a bird; no quadruped's tail, because they are winged animals; no bird's tail, because they live on the ground. This absence of tail is the result of necessity. For bats fly by means of a membrane, not by means of barbed feathers; and no animal unless it has barbed feathers has the tail of a bird; for a bird's tail is composed of such barbed feathers. As for a quadruped's tail, it would be an impediment to an animal that flies.

(*Ch.* 14.) Much the same may be said also of the Libyan ostrich. For it has some of the characters of a bird, some of the characters

of a quadruped. It differs from a quadruped, in possessing wings; from a bird in being unable to soar aloft, and in having feathers that resemble hair and are useless for flight.¹ On the other hand it agrees with quadrupeds in having upper eyelashes,² which are the more richly supplied with hairs because the parts about the head and the upper portion of the neck are bare;³ and it agrees with birds in being feathered in all the parts posterior to these. Again it resembles a bird in being a biped, and a quadruped in having a cloven hoof. For it has hoofs and not toes.⁴ The explanation of these peculiarities is to be found in its bulk, which is that of a quadruped rather than of a bird. For, speaking generally, a bird must necessarily be of very small size. For a body of heavy bulk can with difficulty be raised into the air.

Thus much then as regards the parts of animals. We have discussed them all, and set forth the cause why each exists; and in so doing we have severally considered each group of animals. We must now pass on, and in due sequence must next deal with the question of their generation and development.